

Actividad - Proyecto práctico

La actividad se desarrollará en grupos pre-definidos de 4 alumnos. Se debe indicar los nombres en orden alfabético (de apellidos). Recordad que esta actividad se corresponde con un 30% de la nota final de la asignatura. Se debe entregar el trabajo en la presente notebook.

- Alumno 1: Rosa Ines Chamorro Nieto
- Alumno 2: Lucia Sanjuan Pérez
- Alumno 3: Ander Amatriain Amiano
- Alumno 4:

PARTE 1 - Instalación y requisitos previos

Las prácticas han sido preparadas para poder realizarse en el entorno de trabajo de Google Colab. Sin embargo, esta plataforma presenta ciertas incompatibilidades a la hora de visualizar la renderización en gym. Por ello, para obtener estas visualizaciones, se deberá trasladar el entorno de trabajo a local. Por ello, el presente dossier presenta instrucciones para poder trabajar en ambos entornos. Siga los siguientes pasos para un correcto funcionamiento:

1. **LOCAL:** Preparar el enviroment, siguiendo las intrucciones detalladas en la sección *1.1.Preparar enviroment*.
2. **AMBOS:** Modificar las variables "mount" y "drive_mount" a la carpeta de trabajo en drive en el caso de estar en Colab, y ejecturar la celda *1.2.Localizar entorno de trabajo*.
3. **COLAB:** se deberá ejecutar las celdas correspondientes al montaje de la carpeta de trabajo en Drive. Esta corresponde a la sección *1.3.Montar carpeta de datos local*.
4. **AMBOS:** Instalar las librerías necesarias, siguiendo la sección *1.4.Instalar librerías necesarias*.

1.1. Preparar enviroment (solo local)

1. En Windows, puede ser necesario instalar las C++ Build Tools. Para ello, siga los siguientes pasos. Alternativamente puedes utilizar WSL2: <https://towardsdatascience.com/how-to-install-openai-gym-in-a-windows-environment-338969e24d30>.
2. Instalar Anaconda
3. Siguiendo el código que se presenta comentado en la próxima celda: Crear un environment, cambiar la ruta de trabajo, e instalar librerías básicas.

Para preparar el entorno de trabajo en local, se han seguido los siguientes pasos:

```
conda create --name miar_rl python=3.8
conda activate miar_rl
cd "PATH_TO_FOLDER"
conda install git
pip install jupyter
```

4. Abrir la notebook con *jupyter-notebook*.

```
jupyter-notebook
```

1.2. Localizar entorno de trabajo: Google colab o local

```
In [ ]: # ATENCIÓN!! Modificar ruta relativa a La práctica si es distinta (drive_root)
mount='/content/gdrive'
drive_root = mount + "/My Drive/Estudios/Master Inteligencia Artificial/08MIAR - Aprendizaje por refuerzo/Proyecto Practico"

try:
    from google.colab import drive
    IN_COLAB=True
except:
    IN_COLAB=False
```

1.3. Montar carpeta de datos local (solo Colab)

```
In [ ]: # Switch to the directory on the Google Drive that you want to use
import os
if IN_COLAB:
    print("We're running Colab")

    if IN_COLAB:
        # Mount the Google Drive at mount
        print("Colab: mounting Google drive on ", mount)

        drive.mount(mount)

        # Create drive_root if it doesn't exist
        create_drive_root = True
        if create_drive_root:
            print("\nColab: making sure ", drive_root, " exists.")
            os.makedirs(drive_root, exist_ok=True)

        # Change to the directory
        print("\nColab: Changing directory to ", drive_root)
        %cd $drive_root
    # Verify we're in the correct working directory
    %pwd
    print("Archivos en el directorio: ")
    print(os.listdir())
```

Archivos en el directorio:
['Proyecto_práctico_2.ipynb']

1.4. Instalar librerías necesarias

```
In [ ]: if IN_COLAB:
        %pip uninstall -y jax jaxlib
        %pip install tensorflow==2.18.0
        %pip install tf-keras==2.18.0
        %pip install gym==0.17.3
        %pip install git+https://github.com/Kojoley/atari-py.git
        %pip install keras-rl2==1.0.5
        %pip install "numpy<2.0"
```

```
else:
    %pip install gym==0.17.3
    %pip install git+https://github.com/Kojoley/atari-py.git
    %pip install pygame==1.5.0
    %pip install h5py==3.1.0
    %pip install Pillow==9.5.0
    %pip install keras-rl2==1.0.5
    %pip install Keras==2.2.4
    %pip install tensorflow==2.5.3
    %pip install torch==2.0.1
    %pip install agents==1.4.0
    %pip install "numpy<2.0"
```

```
Collecting gym==0.17.3
  Downloading gym-0.17.3.tar.gz (1.6 MB)
Collecting scipy
  Downloading scipy-1.10.1-cp38-cp38-win_amd64.whl (42.2 MB)
Collecting numpy>=1.10.4
  Downloading numpy-1.24.4-cp38-cp38-win_amd64.whl (14.9 MB)
Collecting pygame<=1.5.0,>=1.4.0
  Downloading pygame-1.5.0-py2.py3-none-any.whl (1.0 MB)
Collecting cloudpickle<1.7.0,>=1.2.0
  Downloading cloudpickle-1.6.0-py3-none-any.whl (23 kB)
Collecting future
  Downloading future-1.0.0-py3-none-any.whl (491 kB)
Using legacy 'setup.py install' for gym, since package 'wheel' is not installed.
Installing collected packages: numpy, future, scipy, pygame, cloudpickle, gym
  Running setup.py install for gym: started
  Running setup.py install for gym: finished with status 'done'
Successfully installed cloudpickle-1.6.0 future-1.0.0 gym-0.17.3 numpy-1.24.4 pygame-1.5.0 scipy-1.10.1
Note: you may need to restart the kernel to use updated packages.
Collecting git+https://github.com/Kojoley/atari-py.git
  Cloning https://github.com/Kojoley/atari-py.git to c:\users\rosai\appdata\local\temp\pip-req-build-y22qlqeb
Requirement already satisfied: numpy in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por re
fuerzo\proyecto\gym_env\lib\site-packages (from atari-py==1.2.2) (1.24.4)
Using legacy 'setup.py install' for atari-py, since package 'wheel' is not installed.
Installing collected packages: atari-py
  Running setup.py install for atari-py: started
  Running setup.py install for atari-py: still running...
  Running setup.py install for atari-py: still running...
  Running setup.py install for atari-py: finished with status 'done'
Successfully installed atari-py-1.2.2
Note: you may need to restart the kernel to use updated packages.

Running command git clone -q https://github.com/Kojoley/atari-py.git 'C:\Users\rosai\AppData\Local\Temp\pip-req-build-y22qlqeb'
```

Requirement already satisfied: pygame==1.5.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (1.5.0)Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: future in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from pygame==1.5.0) (1.0.0)

Collecting h5py==3.1.0

Downloading h5py-3.1.0-cp38-cp38-win_amd64.whl (2.7 MB)

Requirement already satisfied: numpy>=1.17.5 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from h5py==3.1.0) (1.24.4)

Installing collected packages: h5py

Successfully installed h5py-3.1.0

Note: you may need to restart the kernel to use updated packages.

Collecting Pillow==9.5.0

Downloading Pillow-9.5.0-cp38-cp38-win_amd64.whl (2.5 MB)

Installing collected packages: Pillow

Successfully installed Pillow-9.5.0

Note: you may need to restart the kernel to use updated packages.

Collecting keras-rl2==1.0.5

Downloading keras_rl2-1.0.5-py3-none-any.whl (52 kB)

Collecting tensorflow

Downloading tensorflow-2.13.1-cp38-cp38-win_amd64.whl (1.9 kB)

Downloading tensorflow-2.13.0-cp38-cp38-win_amd64.whl (1.9 kB)

Collecting tensorflow-intel==2.13.0

Downloading tensorflow_intel-2.13.0-cp38-cp38-win_amd64.whl (276.5 MB)

Collecting libclang>=13.0.0

Downloading libclang-18.1.1-py2.py3-none-win_amd64.whl (26.4 MB)

Collecting tensorflow-estimator<2.14,>=2.13.0

Downloading tensorflow_estimator-2.13.0-py2.py3-none-any.whl (440 kB)

Collecting numpy<=1.24.3,>=1.22

Downloading numpy-1.24.3-cp38-cp38-win_amd64.whl (14.9 MB)

Collecting flatbuffers>=23.1.21

Downloading flatbuffers-25.12.19-py2.py3-none-any.whl (26 kB)

Collecting protobuf!=4.21.0,!<4.21.1,!<4.21.2,!<4.21.3,!<4.21.4,!<4.21.5,<5.0.0dev,>=3.20.3

Downloading protobuf-4.25.9-cp38-cp38-win_amd64.whl (413 kB)

Requirement already satisfied: h5py>=2.9.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow-intel==2.13.0->tensorflow->keras-rl2==1.0.5) (3.1.0)

Collecting astunparse>=1.6.0

Downloading astunparse-1.6.3-py2.py3-none-any.whl (12 kB)

Collecting google-pasta>=0.1.1

Downloading google_pasta-0.2.0-py3-none-any.whl (57 kB)

Collecting tensorboard<2.14,>=2.13

Downloading tensorboard-2.13.0-py3-none-any.whl (5.6 MB)
Requirement already satisfied: packaging in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow-intel==2.13.0->tensorflow->keras-rl2==1.0.5) (26.2)
Collecting typing-extensions<4.6.0,>=3.6.6
Downloading typing_extensions-4.5.0-py3-none-any.whl (27 kB)
Collecting gast<=0.4.0,>=0.2.1
Downloading gast-0.4.0-py3-none-any.whl (9.8 kB)
Collecting tensorflow-io-gcs-filesystem>=0.23.1
Downloading tensorflow_io_gcs_filesystem-0.31.0-cp38-cp38-win_amd64.whl (1.5 MB)
Collecting opt-einsum>=2.3.2
Downloading opt_einsum-3.4.0-py3-none-any.whl (71 kB)
Collecting wrapt>=1.11.0
Downloading wrapt-2.0.1-cp38-cp38-win_amd64.whl (60 kB)
Requirement already satisfied: setuptools in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow-intel==2.13.0->tensorflow->keras-rl2==1.0.5) (56.0.0)
Collecting termcolor>=1.1.0
Downloading termcolor-2.4.0-py3-none-any.whl (7.7 kB)
Collecting keras<2.14,>=2.13.1
Downloading keras-2.13.1-py3-none-any.whl (1.7 MB)
Collecting absl-py>=1.0.0
Downloading absl_py-2.3.1-py3-none-any.whl (135 kB)
Collecting grpcio<2.0,>=1.24.3
Downloading grpcio-1.70.0-cp38-cp38-win_amd64.whl (4.3 MB)
Requirement already satisfied: six>=1.12.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow-intel==2.13.0->tensorflow->keras-rl2==1.0.5) (1.17.0)
Collecting wheel<1.0,>=0.23.0
Downloading wheel-0.45.1-py3-none-any.whl (72 kB)
Collecting requests<3,>=2.21.0
Downloading requests-2.32.4-py3-none-any.whl (64 kB)
Collecting tensorboard-data-server<0.8.0,>=0.7.0
Downloading tensorboard_data_server-0.7.2-py3-none-any.whl (2.4 kB)
Collecting google-auth-oauthlib<1.1,>=0.5
Downloading google_auth_oauthlib-1.0.0-py2.py3-none-any.whl (18 kB)
Collecting markdown>=2.6.8
Downloading Markdown-3.7-py3-none-any.whl (106 kB)
Collecting werkzeug>=1.0.1
Downloading werkzeug-3.0.6-py3-none-any.whl (227 kB)
Collecting google-auth<3,>=1.6.3
Downloading google_auth-2.50.0-py3-none-any.whl (246 kB)
Collecting cryptography>=38.0.3
Downloading cryptography-47.0.0-cp38-abi3-win_amd64.whl (3.8 MB)

```
Collecting pyasn1-modules>=0.2.1
  Downloading pyasn1_modules-0.4.2-py3-none-any.whl (181 kB)
Collecting cffi>=1.14
  Downloading cffi-1.17.1-cp38-cp38-win_amd64.whl (181 kB)
Collecting cryptography>=38.0.3
  Downloading cryptography-46.0.7-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-46.0.6-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-46.0.5-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-46.0.4-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-46.0.3-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-46.0.2-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-46.0.1-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-46.0.0-cp38-abi3-win_amd64.whl (3.5 MB)
  Downloading cryptography-45.0.7-cp37-abi3-win_amd64.whl (3.4 MB)
Collecting pycparser
  Downloading pycparser-2.23-py3-none-any.whl (118 kB)
Collecting requests-oauthlib>=0.7.0
  Downloading requests_oauthlib-2.0.0-py2.py3-none-any.whl (24 kB)
Requirement already satisfied: importlib-metadata>=4.4 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from markdown>=2.6.8->tensorboard<2.14,>=2.13->tensorflow-intel==2.13.0->tensorflow->keras-rl2==1.0.5) (8.5.0)
Requirement already satisfied: zipp>=3.20 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard<2.14,>=2.13->tensorflow-intel==2.13.0->tensorflow->keras-rl2==1.0.5) (3.20.2)
Collecting pyasn1<0.7.0,>=0.6.1
  Downloading pyasn1-0.6.4-py3-none-any.whl (84 kB)
Collecting certifi>=2017.4.17
  Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)
Collecting idna<4,>=2.5
  Downloading idna-3.15-py3-none-any.whl (72 kB)
Collecting charset_normalizer<4,>=2
  Downloading charset_normalizer-3.4.9-py3-none-any.whl (64 kB)
Collecting urllib3<3,>=1.21.1
  Downloading urllib3-2.2.3-py3-none-any.whl (126 kB)
Collecting oauthlib>=3.0.0
  Downloading oauthlib-3.3.1-py3-none-any.whl (160 kB)
Collecting MarkupSafe>=2.1.1
  Downloading MarkupSafe-2.1.5-cp38-cp38-win_amd64.whl (17 kB)
Installing collected packages: pycparser, urllib3, pyasn1, idna, charset-normalizer, cffi, certifi, requests, pyasn1-modules, oauthlib, cryptography, requests-oauthlib, MarkupSafe, google-auth, wheel, werkzeug, tensorboard-data-server, protobuf, numpy, markdown, grpcio, google-auth-oauthlib, absl-py, wrapt, typing-extensions, termcolor, tensorflow-io-gcs-filesystem, tensorflow-e
```


stimator, tensorboard, opt-einsum, libclang, keras, google-pasta, gast, flatbuffers, astunparse, tensorflow-intel, tensorflow, keras-rl2

Attempting uninstall: numpy

Found existing installation: numpy 1.24.4

Uninstalling numpy-1.24.4:

Successfully uninstalled numpy-1.24.4

Attempting uninstall: typing-extensions

Found existing installation: typing-extensions 4.13.2

Uninstalling typing-extensions-4.13.2:

Successfully uninstalled typing-extensions-4.13.2

Successfully installed MarkupSafe-2.1.5 absl-py-2.3.1 astunparse-1.6.3 certifi-2026.7.22 cffi-1.17.1 charset-normalizer-3.4.9 cryptography-45.0.7 flatbuffers-25.12.19 gast-0.4.0 google-auth-2.50.0 google-auth-oauthlib-1.0.0 google-pasta-0.2.0 grpcio-1.70.0 idna-3.15 keras-2.13.1 keras-rl2-1.0.5 libclang-18.1.1 markdown-3.7 numpy-1.24.3 oauthlib-3.3.1 opt-einsum-3.4.0 protobuf-4.25.9 pyasn1-0.6.4 pyasn1-modules-0.4.2 pycparser-2.23 requests-2.32.4 requests-oauthlib-2.0.0 tensorboard-2.13.0 tensorboard-data-server-0.7.2 tensorflow-2.13.0 tensorflow-estimator-2.13.0 tensorflow-intel-2.13.0 tensorflow-io-gcs-filesystem-0.31.0 termcolor-2.4.0 typing-extensions-4.5.0 urllib3-2.2.3 werkzeug-3.0.6 wheel-0.45.1 wrapt-2.0.1

Note: you may need to restart the kernel to use updated packages.

Collecting Keras==2.2.4

Downloading Keras-2.2.4-py2.py3-none-any.whl (312 kB)

Collecting pyyaml

Downloading PyYAML-6.0.3-cp38-cp38-win_amd64.whl (160 kB)

Requirement already satisfied: six>=1.9.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from Keras==2.2.4) (1.17.0)

Requirement already satisfied: h5py in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from Keras==2.2.4) (3.1.0)

Requirement already satisfied: scipy>=0.14 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from Keras==2.2.4) (1.10.1)

Collecting keras-preprocessing>=1.0.5

Downloading Keras_Preprocessing-1.1.2-py2.py3-none-any.whl (42 kB)

Requirement already satisfied: numpy>=1.9.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from Keras==2.2.4) (1.24.3)

Collecting keras-applications>=1.0.6

Downloading Keras_Applications-1.0.8-py3-none-any.whl (50 kB)

Installing collected packages: pyyaml, keras-preprocessing, keras-applications, Keras

Attempting uninstall: Keras

Found existing installation: keras 2.13.1

Uninstalling keras-2.13.1:

Successfully uninstalled keras-2.13.1

Successfully installed Keras-2.2.4 keras-applications-1.0.8 keras-preprocessing-1.1.2 pyyaml-6.0.3

Note: you may need to restart the kernel to use updated packages.

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
tensorflow-intel 2.13.0 requires keras<2.14,>=2.13.1, but you have keras 2.2.4 which is incompatible.

Collecting tensorflow==2.5.3
 Downloading tensorflow-2.5.3-cp38-cp38-win_amd64.whl (428.2 MB)
Collecting termcolor~=1.1.0
 Downloading termcolor-1.1.0.tar.gz (3.9 kB)
Collecting numpy~=1.19.2
 Downloading numpy-1.19.5-cp38-cp38-win_amd64.whl (13.3 MB)
Collecting opt-einsum~=3.3.0
 Downloading opt_einsum-3.3.0-py3-none-any.whl (65 kB)
Requirement already satisfied: astunparse~=1.6.3 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (1.6.3)
Collecting flatbuffers~=1.12.0
 Downloading flatbuffers-1.12-py2.py3-none-any.whl (15 kB)
Requirement already satisfied: gast==0.4.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (0.4.0)
Requirement already satisfied: keras-preprocessing~=1.1.2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (1.1.2)
Requirement already satisfied: tensorboard~=2.5 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (2.13.0)
Requirement already satisfied: protobuf>=3.9.2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (4.25.9)
Requirement already satisfied: google-pasta~=0.2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (0.2.0)
Collecting grpcio~=1.34.0
 Downloading grpcio-1.34.1-cp38-cp38-win_amd64.whl (2.9 MB)
Requirement already satisfied: h5py~=3.1.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (3.1.0)
Collecting wrapt~=1.12.1
 Downloading wrapt-1.12.1.tar.gz (27 kB)
Collecting keras-nightly~=2.5.0.dev
 Downloading keras_nightly-2.5.0.dev2021032900-py2.py3-none-any.whl (1.2 MB)
Collecting tensorflow-estimator<2.6.0,>=2.5.0
 Downloading tensorflow_estimator-2.5.0-py2.py3-none-any.whl (462 kB)
Collecting absl-py~=0.10
 Downloading absl_py-0.15.0-py3-none-any.whl (132 kB)
Requirement already satisfied: wheel~=0.35 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow==2.5.3) (0.45.1)
Collecting typing-extensions~=3.7.4
 Downloading typing_extensions-3.7.4.3-py3-none-any.whl (22 kB)
Collecting six~=1.15.0
 Downloading six-1.15.0-py2.py3-none-any.whl (10 kB)
Requirement already satisfied: google-auth-oauthlib<1.1,>=0.5 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08

miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow==2.5.3) (1.0.0)
Requirement already satisfied: markdown>=2.6.8 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow==2.5.3) (3.7)
Requirement already satisfied: setuptools>=41.0.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow==2.5.3) (56.0.0)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow==2.5.3) (2.32.4)
Requirement already satisfied: google-auth<3,>=1.6.3 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow==2.5.3) (2.50.0)
Collecting tensorboard~=2.5
 Downloading tensorboard-2.14.0-py3-none-any.whl (5.5 MB)
 Downloading tensorboard-2.12.3-py3-none-any.whl (5.6 MB)
 Downloading tensorboard-2.12.2-py3-none-any.whl (5.6 MB)
 Downloading tensorboard-2.12.1-py3-none-any.whl (5.6 MB)
 Downloading tensorboard-2.12.0-py3-none-any.whl (5.6 MB)
 Downloading tensorboard-2.11.2-py3-none-any.whl (6.0 MB)
Collecting protobuf>=3.9.2
 Downloading protobuf-3.20.3-cp38-cp38-win_amd64.whl (904 kB)
Collecting tensorboard-data-server<0.7.0,>=0.6.0
 Downloading tensorboard_data_server-0.6.1-py3-none-any.whl (2.4 kB)
Collecting tensorboard-plugin-wit>=1.6.0
 Downloading tensorboard_plugin_wit-1.8.1-py3-none-any.whl (781 kB)
Collecting google-auth-oauthlib<0.5,>=0.4.1
 Downloading google_auth_oauthlib-0.4.6-py2.py3-none-any.whl (18 kB)
Requirement already satisfied: werkzeug>=1.0.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow==2.5.3) (3.0.6)
Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow==2.5.3) (0.4.2)
Requirement already satisfied: cryptography>=38.0.3 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow==2.5.3) (45.0.7)
Requirement already satisfied: cffi>=1.14 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from cryptography>=38.0.3->google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow==2.5.3) (1.17.1)
Requirement already satisfied: pycparser in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from cffi>=1.14->cryptography>=38.0.3->google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow==2.5.3) (2.23)
Requirement already satisfied: requests-oauthlib>=0.7.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from google-auth-oauthlib<0.5,>=0.4.1->tensorboard~=2.5->tensorflow==2.5.3) (2.0.0)

Requirement already satisfied: importlib-metadata>=4.4 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from markdown>=2.6.8->tensorboard~=2.5->tensorflow==2.5.3) (8.5.0)

Requirement already satisfied: zipp>=3.20 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard~=2.5->tensorflow==2.5.3) (3.20.2)

Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from pyasn1-modules>=0.2.1->google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow==2.5.3) (0.6.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow==2.5.3) (2.2.3)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow==2.5.3) (3.4.9)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow==2.5.3) (2026.7.22)

Requirement already satisfied: idna<4,>=2.5 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow==2.5.3) (3.15)

Requirement already satisfied: oauthlib>=3.0.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests-oauthlib>=0.7.0->google-auth-oauthlib<0.5,>=0.4.1->tensorboard~=2.5->tensorflow==2.5.3) (3.3.1)

Requirement already satisfied: MarkupSafe>=2.1.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from werkzeug>=1.0.1->tensorboard~=2.5->tensorflow==2.5.3) (2.1.5)

Building wheels for collected packages: termcolor, wrapt

Building wheel for termcolor (setup.py): started

Building wheel for termcolor (setup.py): finished with status 'done'

Created wheel for termcolor: filename=termcolor-1.1.0-py3-none-any.whl size=4853 sha256=593c4b773d6218ea1a0aab22952acc7732d73acc88daf1806e895abcae20e7ef

Stored in directory: c:\users\rosai\appdata\local\pip\cache\wheels\16\9c\5473df82468f958445479c59e784896fa24f4a5fc024b0f501

Building wheel for wrapt (setup.py): started

Building wheel for wrapt (setup.py): finished with status 'done'

Created wheel for wrapt: filename=wrapt-1.12.1-cp38-cp38-win_amd64.whl size=33367 sha256=057b1333747adbc55c94c09c12c53fe8948c587146905aa8758e66b4088a6601

Stored in directory: c:\users\rosai\appdata\local\pip\cache\wheels\5f\fd\9e\b6cf5890494cb8ef0b5eaff72e5d55a70fb56316007d6dfe73

Successfully built termcolor wrapt

Installing collected packages: six, tensorboard-plugin-wit, tensorboard-data-server, protobuf, numpy, grpcio, google-auth-oauthlib, absl-py, wrapt, typing-extensions, termcolor, tensorflow-estimator, tensorboard, opt-einsum, keras-nightly, flatbuffers, tensorflow

Attempting uninstall: six

Found existing installation: six 1.17.0
Uninstalling six-1.17.0:
 Successfully uninstalled six-1.17.0
Attempting uninstall: tensorboard-data-server
 Found existing installation: tensorboard-data-server 0.7.2
 Uninstalling tensorboard-data-server-0.7.2:
 Successfully uninstalled tensorboard-data-server-0.7.2
Attempting uninstall: protobuf
 Found existing installation: protobuf 4.25.9
 Uninstalling protobuf-4.25.9:
 Successfully uninstalled protobuf-4.25.9
Attempting uninstall: numpy
 Found existing installation: numpy 1.24.3
 Uninstalling numpy-1.24.3:
 Successfully uninstalled numpy-1.24.3
Attempting uninstall: grpcio
 Found existing installation: grpcio 1.70.0
 Uninstalling grpcio-1.70.0:
 Successfully uninstalled grpcio-1.70.0
Attempting uninstall: google-auth-oauthlib
 Found existing installation: google-auth-oauthlib 1.0.0
 Uninstalling google-auth-oauthlib-1.0.0:
 Successfully uninstalled google-auth-oauthlib-1.0.0
Attempting uninstall: absl-py
 Found existing installation: absl-py 2.3.1
 Uninstalling absl-py-2.3.1:
 Successfully uninstalled absl-py-2.3.1
Attempting uninstall: wrapt
 Found existing installation: wrapt 2.0.1
 Uninstalling wrapt-2.0.1:
 Successfully uninstalled wrapt-2.0.1
Attempting uninstall: typing-extensions
 Found existing installation: typing-extensions 4.5.0
 Uninstalling typing-extensions-4.5.0:
 Successfully uninstalled typing-extensions-4.5.0
Attempting uninstall: termcolor
 Found existing installation: termcolor 2.4.0
 Uninstalling termcolor-2.4.0:
 Successfully uninstalled termcolor-2.4.0
Attempting uninstall: tensorflow-estimator
 Found existing installation: tensorflow-estimator 2.13.0

```

Uninstalling tensorflow-estimator-2.13.0:
  Successfully uninstalled tensorflow-estimator-2.13.0
Attempting uninstall: tensorboard
  Found existing installation: tensorboard 2.13.0
  Uninstalling tensorboard-2.13.0:
    Successfully uninstalled tensorboard-2.13.0
Attempting uninstall: opt-einsum
  Found existing installation: opt-einsum 3.4.0
  Uninstalling opt-einsum-3.4.0:
    Successfully uninstalled opt-einsum-3.4.0
Attempting uninstall: flatbuffers
  Found existing installation: flatbuffers 25.12.19
  Uninstalling flatbuffers-25.12.19:
    Successfully uninstalled flatbuffers-25.12.19
Attempting uninstall: tensorflow
  Found existing installation: tensorflow 2.13.0
  Uninstalling tensorflow-2.13.0:
    Successfully uninstalled tensorflow-2.13.0
Successfully installed absl-py-0.15.0 flatbuffers-1.12 google-auth-oauthlib-0.4.6 grpcio-1.34.1 keras-nightly-2.5.0.dev20210329
00 numpy-1.19.5 opt-einsum-3.3.0 protobuf-3.20.3 six-1.15.0 tensorboard-2.11.2 tensorboard-data-server-0.6.1 tensorboard-plugin
-wit-1.8.1 tensorflow-2.5.3 tensorflow-estimator-2.5.0 termcolor-1.1.0 typing-extensions-3.7.4.3 wrapt-1.12.1
Note: you may need to restart the kernel to use updated packages.

```

```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is th
e source of the following dependency conflicts.
tensorflow-intel 2.13.0 requires absl-py>=1.0.0, but you have absl-py 0.15.0 which is incompatible.
tensorflow-intel 2.13.0 requires flatbuffers>=23.1.21, but you have flatbuffers 1.12 which is incompatible.
tensorflow-intel 2.13.0 requires keras<2.14,>=2.13.1, but you have keras 2.2.4 which is incompatible.
tensorflow-intel 2.13.0 requires numpy<=1.24.3,>=1.22, but you have numpy 1.19.5 which is incompatible.
tensorflow-intel 2.13.0 requires tensorboard<2.14,>=2.13, but you have tensorboard 2.11.2 which is incompatible.
tensorflow-intel 2.13.0 requires tensorflow-estimator<2.14,>=2.13.0, but you have tensorflow-estimator 2.5.0 which is incompati
ble.

```

```
Collecting torch==2.0.1
  Downloading torch-2.0.1-cp38-cp38-win_amd64.whl (172.4 MB)
Collecting networkx
  Downloading networkx-3.1-py3-none-any.whl (2.1 MB)
Collecting jinja2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Collecting filelock
  Downloading filelock-3.16.1-py3-none-any.whl (16 kB)
Collecting sympy
  Downloading sympy-1.13.3-py3-none-any.whl (6.2 MB)
Requirement already satisfied: typing-extensions in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprend
izaje por refuerzo\proyecto\gym_env\lib\site-packages (from torch==2.0.1) (3.7.4.3)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendiz
aje por refuerzo\proyecto\gym_env\lib\site-packages (from jinja2->torch==2.0.1) (2.1.5)
Collecting mpmath<1.4,>=1.1.0
  Downloading mpmath-1.3.0-py3-none-any.whl (536 kB)
Installing collected packages: mpmath, sympy, networkx, jinja2, filelock, torch
Successfully installed filelock-3.16.1 jinja2-3.1.6 mpmath-1.3.0 networkx-3.1 sympy-1.13.3 torch-2.0.1
Note: you may need to restart the kernel to use updated packages.
Collecting agents==1.4.0Note: you may need to restart the kernel to use updated packages.

  Downloading agents-1.4.0.tar.gz (37 kB)
Requirement already satisfied: tensorflow in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje p
or refuerzo\proyecto\gym_env\lib\site-packages (from agents==1.4.0) (2.5.3)
Requirement already satisfied: gym in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refu
erzo\proyecto\gym_env\lib\site-packages (from agents==1.4.0) (0.17.3)
Collecting ruamel.yaml
  Downloading ruamel.yaml-0.18.16-py3-none-any.whl (119 kB)
Requirement already satisfied: scipy in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por re
fuerzo\proyecto\gym_env\lib\site-packages (from gym->agents==1.4.0) (1.10.1)
Requirement already satisfied: numpy>=1.10.4 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaj
e por refuerzo\proyecto\gym_env\lib\site-packages (from gym->agents==1.4.0) (1.19.5)
Requirement already satisfied: pygame<=1.5.0,>=1.4.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - ap
rendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from gym->agents==1.4.0) (1.5.0)
Requirement already satisfied: cloudpickle<1.7.0,>=1.2.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar
 - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from gym->agents==1.4.0) (1.6.0)
Requirement already satisfied: future in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por r
efuerzo\proyecto\gym_env\lib\site-packages (from pygame<=1.5.0,>=1.4.0->gym->agents==1.4.0) (1.0.0)
Collecting ruamel.yaml.clib>=0.2.7
  Downloading ruamel.yaml.clib-0.2.8-cp38-cp38-win_amd64.whl (118 kB)
Requirement already satisfied: termcolor~=1.1.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendi
```


zaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (1.1.0)
Requirement already satisfied: typing-extensions~=3.7.4 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (3.7.4.3)
Requirement already satisfied: flatbuffers~=1.12.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (1.12)
Requirement already satisfied: google-pasta~=0.2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (0.2.0)
Requirement already satisfied: wrapt~=1.12.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (1.12.1)
Requirement already satisfied: six~=1.15.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (1.15.0)
Requirement already satisfied: opt-einsum~=3.3.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (3.3.0)
Requirement already satisfied: wheel~=0.35 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (0.45.1)
Requirement already satisfied: grpcio~=1.34.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (1.34.1)
Requirement already satisfied: tensorboard~=2.5 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (2.11.2)
Requirement already satisfied: gast==0.4.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (0.4.0)
Requirement already satisfied: h5py~=3.1.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (3.1.0)
Requirement already satisfied: keras-preprocessing~=1.1.2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (1.1.2)
Requirement already satisfied: astunparse~=1.6.3 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (1.6.3)
Requirement already satisfied: tensorflow-estimator<2.6.0,>=2.5.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (2.5.0)
Requirement already satisfied: absl-py~=0.10 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (0.15.0)
Requirement already satisfied: protobuf>=3.9.2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (3.20.3)
Requirement already satisfied: keras-nightly~=2.5.0.dev in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorflow->agents==1.4.0) (2.5.0.dev2021032900)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (2.32.4)
Requirement already satisfied: tensorboard-plugin-wit>=1.6.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (1.8.1)
Requirement already satisfied: werkzeug>=1.0.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (3.0.6)

Requirement already satisfied: google-auth<3,>=1.6.3 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (2.50.0)

Requirement already satisfied: tensorboard-data-server<0.7.0,>=0.6.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (0.6.1)

Requirement already satisfied: google-auth-oauthlib<0.5,>=0.4.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (0.4.6)

Requirement already satisfied: setuptools>=41.0.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (56.0.0)

Requirement already satisfied: markdown>=2.6.8 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from tensorboard~=2.5->tensorflow->agents==1.4.0) (3.7)

Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow->agents==1.4.0) (0.4.2)

Requirement already satisfied: cryptography>=38.0.3 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow->agents==1.4.0) (45.0.7)

Requirement already satisfied: cffi>=1.14 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from cryptography>=38.0.3->google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow->agents==1.4.0) (1.17.1)

Requirement already satisfied: pycparser in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from cffi>=1.14->cryptography>=38.0.3->google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow->agents==1.4.0) (2.23)

Requirement already satisfied: requests-oauthlib>=0.7.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from google-auth-oauthlib<0.5,>=0.4.1->tensorboard~=2.5->tensorflow->agents==1.4.0) (2.0.0)

Requirement already satisfied: importlib-metadata>=4.4 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from markdown>=2.6.8->tensorboard~=2.5->tensorflow->agents==1.4.0) (8.5.0)

Requirement already satisfied: zipp>=3.20 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard~=2.5->tensorflow->agents==1.4.0) (3.20.2)

Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from pyasn1-modules>=0.2.1->google-auth<3,>=1.6.3->tensorboard~=2.5->tensorflow->agents==1.4.0) (0.6.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow->agents==1.4.0) (2.2.3)

Requirement already satisfied: idna<4,>=2.5 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow->agents==1.4.0) (3.15)

Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow->agents==1.4.0) (3.4.9)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests<3,>=2.21.0->tensorboard~=2.5->tensorflow->agents==1.4.0) (2026.7.22)

Requirement already satisfied: oauthlib>=3.0.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from requests-oauthlib>=0.7.0->google-auth-oauthlib<0.5,>=0.4.1->tensorboard~=2.5->tensorflow->agents==1.4.0) (3.3.1)

Requirement already satisfied: MarkupSafe>=2.1.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from werkzeug>=1.0.1->tensorboard~=2.5->tensorflow->agents==1.4.0) (2.1.5)

Building wheels for collected packages: agents

Building wheel for agents (setup.py): started

Building wheel for agents (setup.py): finished with status 'done'

Created wheel for agents: filename=agents-1.4.0-py3-none-any.whl size=62757 sha256=250b855d22a2ede6f70e99350b9d7a40483500597a02d9156a749cfc3f706585

Stored in directory: c:\users\rosai\appdata\local\pip\cache\wheels\87\bb\49\8e14e51995c7fa61cb6cd10934be577d01a51d4236c62c80f2

Successfully built agents

Installing collected packages: ruamel.yaml.clib, ruamel.yaml, agents

Successfully installed agents-1.4.0 ruamel.yaml-0.18.16 ruamel.yaml.clib-0.2.8

Requirement already satisfied: numpy<2.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (1.19.5)

Note: you may need to restart the kernel to use updated packages.

PARTE 2. Enunciado

Consideraciones a tener en cuenta:

- El entorno sobre el que trabajaremos será el indicado en el listado correspondiente de cada grupo y el algoritmo que usaremos será *DQN*.
- Para nuestro ejercicio, el requisito mínimo será alcanzado cuando el agente consiga una **media de recompensa por encima de los puntos indicados en el listado por grupos en modo test**. Por ello, esta media de la recompensa se calculará a partir del código de test en la última celda del notebook.

Este proyecto práctico consta de tres partes:

1. Implementar la red neuronal que se usará en la solución
2. Implementar las distintas piezas de la solución DQN y probar al menos 3 propuestas diferentes de mejora.
3. Justificar la respuesta en relación a los resultados obtenidos e incluir al menos 3 gráficas relevantes comparando las 3 propuestas.

Rúbrica: Se valorará la originalidad en la solución aportada, así como la capacidad de discutir los resultados de forma detallada. El requisito mínimo servirá para aprobar la actividad, bajo premisa de que la discusión del resultado sera apropiada.

IMPORTANTE:

- Si no se consigue una puntuación óptima, responder sobre la mejor puntuación obtenida.
- Para entrenamientos largos, recordad que podéis usar checkpoints de vuestros modelos para retomar los entrenamientos. En este caso, recordad cambiar los parámetros adecuadamente (sobre todo los relacionados con el proceso de exploración).
- Se deberá entregar unicamente el notebook y los pesos del mejor modelo en un fichero .zip, de forma organizada.
- Cada alumno deberá de subir la solución de forma individual.

PARTE 3. Desarrollo y preguntas

Importar librerías

```
In [ ]: #Comprobamos que se carga la versión esperada de TF y detecta la GPU
import os
import tensorflow as tf1

if IN_COLAB:
    os.environ['TF_USE_LEGACY_KERAS'] = "1"
    from keras import __version__
    import tensorflow.keras as tf
    tf.__version__ = __version__
else:
    import tensorflow.keras as tf # usa el Keras integrado en TF 2.5.3, sin tocar el paquete standalone
```

```
print(tf1.__version__)
print(tf1.config.list_physical_devices('GPU'))
print(tf1.config.list_logical_devices('GPU'))
```

2.5.3

[]

[]

```
In [ ]: from __future__ import division

from PIL import Image
import numpy as np
import gym

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Activation, Flatten, Convolution2D, Permute

if IN_COLAB:
    from tensorflow.keras.optimizers.legacy import Adam
else:
    from tensorflow.keras.optimizers import Adam

import tensorflow.keras.backend as K

from rl.agents.dqn import DQNAgent
from rl.policy import LinearAnnealedPolicy, BoltzmannQPolicy, EpsGreedyQPolicy
from rl.memory import SequentialMemory
from rl.core import Processor
from rl.callbacks import FileLogger, ModelIntervalCheckpoint
```

Configuración base

```
In [ ]: # Descomentar y adaptar a cada entorno según corresponda
INPUT_SHAPE = (84, 84)
WINDOW_LENGTH = 4

env_name = 'Enduro-v0'
env = gym.make(env_name)

np.random.seed(123)
```

```

env.seed(123)
nb_actions = env.action_space.n

# Umbral objetivo: superar el primer día de carrera (200 coches)
# de forma sostenida durante los 100 episodios de test
REWARD_THRESHOLD = 200

# --- Definición de las 3 propuestas a comparar ---
experiments = {
    'baseline_dqn': dict(double_dqn=False, dueling=False),
    'double_dqn': dict(double_dqn=True, dueling=False),
    'dueling_dqn': dict(double_dqn=False, dueling=True),
}

NB_STEPS_TRAIN = 1750000 # ajustar según tiempo/recursos disponibles

```

```

In [ ]: class AtariProcessor(Processor):
        def process_observation(self, observation):
            assert observation.ndim == 3 # (height, width, channel)
            img = Image.fromarray(observation)
            img = img.resize(INPUT_SHAPE).convert('L')
            processed_observation = np.array(img)
            assert processed_observation.shape == INPUT_SHAPE
            return processed_observation.astype('uint8')

        def process_state_batch(self, batch):
            processed_batch = batch.astype('float32') / 255.
            return processed_batch

        def process_reward(self, reward):
            return np.clip(reward, -1., 1.)

```

1. Implementación de la red neuronal

```

In [ ]: from tensorflow.keras.layers import Layer, Input
        from tensorflow.keras.models import Model

        if IN_COLAB:

```

```

# Solo necesario en Colab con TF 2.18 / Keras 3, para permitir
# La deserialización del Lambda de combinación del Dueling DQN
from tf_keras.src.saving import serialization_lib
serialization_lib.in_safe_mode = lambda: False

class DuelingCombine(Layer):
    def call(self, inputs):
        value, advantage = inputs
        return value + (advantage - K.mean(advantage, axis=1, keepdims=True))

def build_model(window_length, input_shape, nb_actions, dueling=False):
    input_shape_full = (window_length,) + input_shape
    frame = Input(shape=input_shape_full)
    x = Permute((2, 3, 1))(frame)

    x = Convolution2D(32, (8, 8), strides=(4, 4), activation='relu')(x)
    x = Convolution2D(64, (4, 4), strides=(2, 2), activation='relu')(x)
    x = Convolution2D(64, (3, 3), strides=(1, 1), activation='relu')(x)
    x = Flatten()(x)
    x = Dense(512, activation='relu')(x)

    if dueling:
        value = Dense(1, activation='linear')(x)
        advantage = Dense(nb_actions, activation='linear')(x)
        outputs = DuelingCombine()([value, advantage])
    else:
        outputs = Dense(nb_actions, activation='linear')(x)

    model = Model(inputs=frame, outputs=outputs)
    print(model.summary())
    return model

```

La arquitectura convolucional utilizada es muy similar a la propuesta originalmente por DeepMind para Atari.

Las capas convolucionales permiten extraer automáticamente características espaciales relevantes de las imágenes del juego, mientras que la capa totalmente conectada aprende una representación de alto nivel antes de estimar los valores Q asociados a cada acción.

2. Implementación de la solución DQN

```
In [ ]: def build_agent(nb_actions, window_length, input_shape,
                        double_dqn=False, dueling=False):
    model = build_model(window_length, input_shape, nb_actions, dueling=dueling)

    memory = SequentialMemory(limit=1000000, window_length=window_length)
    processor = AtariProcessor()

    policy = LinearAnnealedPolicy(
        EpsGreedyQPolicy(), attr='eps',
        value_max=1.0, value_min=0.1, value_test=0.05,
        nb_steps=250000
    )

    dqn = DQNAgent(
        model=model, nb_actions=nb_actions, policy=policy, memory=memory,
        processor=processor, nb_steps_warmup=10000, gamma=0.99,
        target_model_update=5000, train_interval=4, delta_clip=1.0,
        enable_double_dqn=double_dqn, enable_dueling_network=dueling,
        dueling_type='avg' if dueling else None
    )
    dqn.compile(Adam(learning_rate=0.00025), metrics=['mae'])
    return dqn

histories = {}
```

Entrenamiento

```
In [ ]: from datetime import datetime
        from os import name

        def train_agent(name):

            # Timestamp único para este arranque de fit()
            run_timestamp = datetime.now().strftime('%Y%m%d_%H%M%S')

            cfg = experiments[name]

            print(f"\n==== Entrenando: {name} a las {run_timestamp} =====")
            env = gym.make(env_name)
```



```

env.seed(123)

dqn = build_agent(nb_actions, WINDOW_LENGTH, INPUT_SHAPE, **cfg)
dqn.step = 0

weights_filename = f'dqn_{env_name}_{name}_weights_{run_timestamp}.h5f'
checkpoint_filename = f'dqn_{env_name}_{name}_checkpoint_{run_timestamp}.h5f'
log_filename = f'dqn_{env_name}_{name}_log_{run_timestamp}.json'

callbacks = [
    ModelIntervalCheckpoint(checkpoint_filename, interval=100000),
    FileLogger(log_filename, interval=10000)
]

history = dqn.fit(env, callbacks=callbacks, nb_steps=NB_STEPS_TRAIN,
                  log_interval=10000, visualize=False)

dqn.save_weights(weights_filename, overwrite=True)
histories[name] = log_filename
env.close()

```

Selección de las variantes

Con el objetivo de comparar distintas estrategias de mejora sobre el algoritmo DQN original, se han seleccionado las siguientes variantes:

Baseline DQN, que actúa como modelo de referencia.

Double DQN, que incorpora una estrategia para reducir la sobreestimación de los valores Q durante el aprendizaje.

Dueling DQN, que modifica la arquitectura de la red neuronal para estimar de forma independiente el valor del estado y la ventaja asociada a cada acción.

Estas tres variantes representan algunas de las mejoras más relevantes propuestas sobre DQN y permiten analizar distintos aspectos del proceso de aprendizaje. La versión Baseline sirve como punto de comparación, mientras que Double DQN busca obtener estimaciones más precisas de los valores Q. Por su parte, Dueling DQN resulta especialmente interesante en un entorno como Enduro, donde durante gran parte de la partida muchas acciones producen efectos similares. En estas situaciones, separar el valor del estado de la ventaja de cada acción puede facilitar el aprendizaje y mejorar el rendimiento del agente.

Baseline DQN

```
In [ ]: train_agent('baseline_dqn')
```

```
===== Entrenando: baseline_dqn a las 20260725_164222 =====
```

```
Model: "model"
```

Layer (type)	Output Shape	Param #
=====		
input_1 (InputLayer)	[(None, 4, 84, 84)]	0
permute (Permute)	(None, 84, 84, 4)	0
conv2d (Conv2D)	(None, 20, 20, 32)	8224
conv2d_1 (Conv2D)	(None, 9, 9, 64)	32832
conv2d_2 (Conv2D)	(None, 7, 7, 64)	36928
flatten (Flatten)	(None, 3136)	0
dense (Dense)	(None, 512)	1606144
dense_1 (Dense)	(None, 9)	4617
=====		
Total params: 1,688,745		
Trainable params: 1,688,745		
Non-trainable params: 0		

```
None
```

```
Training for 1750000 steps ...
```

```
Interval 1 (0 steps performed)
```

```
14/10000 [.....] - ETA: 46s - reward: 0.0000e+00
```

```
c:\Users\rosai\OneDrive\Documentos\Learning\VIU - Master IA\08MIAR - Aprendizaje por Refuerzo\Proyecto\gym_env\lib\site-package
s\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state_updates` will be removed in a future version. This
property should not be used in TensorFlow 2.0, as `updates` are applied automatically.
```

```
warnings.warn(`Model.state_updates` will be removed in a future version. '
```

10000/10000 [=====] - 59s 6ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - ale.lives: 0.000

Interval 2 (10000 steps performed)

10000/10000 [=====] - 402s 40ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.099 - mean_q: 0.117 - mean_eps: 0.946 - ale.lives: 0.000

Interval 3 (20000 steps performed)

10000/10000 [=====] - 455s 45ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.111 - mean_q: 0.128 - mean_eps: 0.910 - ale.lives: 0.000

Interval 4 (30000 steps performed)

10000/10000 [=====] - 252s 25ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.112 - mean_q: 0.129 - mean_eps: 0.874 - ale.lives: 0.000

Interval 5 (40000 steps performed)

10000/10000 [=====] - 223s 22ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.114 - mean_q: 0.131 - mean_eps: 0.838 - ale.lives: 0.000

Interval 6 (50000 steps performed)

10000/10000 [=====] - 222s 22ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.113 - mean_q: 0.130 - mean_eps: 0.802 - ale.lives: 0.000

Interval 7 (60000 steps performed)

10000/10000 [=====] - 223s 22ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.113 - mean_q: 0.130 - mean_eps: 0.766 - ale.lives: 0.000

Interval 8 (70000 steps performed)

10000/10000 [=====] - 222s 22ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.111 - mean_q: 0.127 - mean_eps: 0.730 - ale.lives: 0.000

Interval 9 (80000 steps performed)

10000/10000 [=====] - 229s 23ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.110 - mean_q: 0.125 - mean_eps: 0.694 - ale.lives: 0.000

00

Interval 10 (90000 steps performed)

10000/10000 [=====] - 299s 30ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.110 - mean_q: 0.125 - mean_eps: 0.658 - ale.lives: 0.000

Interval 11 (100000 steps performed)

10000/10000 [=====] - 376s 38ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.111 - mean_q: 0.127 - mean_eps: 0.622 - ale.lives: 0.000

Interval 12 (110000 steps performed)

10000/10000 [=====] - 326s 33ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.111 - mean_q: 0.126 - mean_eps: 0.586 - ale.lives: 0.000

Interval 13 (120000 steps performed)

10000/10000 [=====] - 330s 33ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.110 - mean_q: 0.125 - mean_eps: 0.550 - ale.lives: 0.000

Interval 14 (130000 steps performed)

10000/10000 [=====] - 358s 36ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.109 - mean_q: 0.124 - mean_eps: 0.514 - ale.lives: 0.000

Interval 15 (140000 steps performed)

10000/10000 [=====] - 418s 42ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.108 - mean_q: 0.123 - mean_eps: 0.478 - ale.lives: 0.000

Interval 16 (150000 steps performed)

10000/10000 [=====] - 381s 38ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.107 - mean_q: 0.122 - mean_eps: 0.442 - ale.lives: 0.000

Interval 17 (160000 steps performed)

10000/10000 [=====] - 419s 42ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.107 - mean_q: 0.123 - mean_eps: 0.406 - ale.lives: 0.000

Interval 18 (170000 steps performed)

10000/10000 [=====] - 374s 37ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.111 - mean_q: 0.128 - mean_eps: 0.370 - ale.lives: 0.000

Interval 19 (180000 steps performed)

10000/10000 [=====] - 419s 42ms/step - reward: 1.0000e-04

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.112 - mean_q: 0.130 - mean_eps: 0.334 - ale.lives: 0.000

Interval 20 (190000 steps performed)

10000/10000 [=====] - 381s 38ms/step - reward: -1.0000e-04

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.001 - mae: 0.117 - mean_q: 0.135 - mean_eps: 0.298 - ale.lives: 0.000

Interval 21 (200000 steps performed)

10000/10000 [=====] - 412s 41ms/step - reward: 1.0000e-04

2 episodes - episode_reward: 0.500 [0.000, 1.000] - loss: 0.001 - mae: 0.124 - mean_q: 0.144 - mean_eps: 0.262 - ale.lives: 0.000

Interval 22 (210000 steps performed)

10000/10000 [=====] - 309s 31ms/step - reward: 1.0000e-04

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.001 - mae: 0.138 - mean_q: 0.161 - mean_eps: 0.226 - ale.lives: 0.000

Interval 23 (220000 steps performed)

10000/10000 [=====] - 309s 31ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.500 [0.000, 1.000] - loss: 0.001 - mae: 0.144 - mean_q: 0.170 - mean_eps: 0.190 - ale.lives: 0.000

Interval 24 (230000 steps performed)

10000/10000 [=====] - 339s 34ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.002 - mae: 0.159 - mean_q: 0.185 - mean_eps: 0.154 - ale.lives: 0.000

Interval 25 (240000 steps performed)

10000/10000 [=====] - 320s 32ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.002 - mae: 0.175 - mean_q: 0.201 - mean_eps: 0.118 - ale.lives: 0.000

Interval 26 (250000 steps performed)
10000/10000 [=====] - 334s 33ms/step - reward: 0.0000e+00
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.003 - mae: 0.196 - mean_q: 0.222 - mean_eps: 0.100 - ale.lives: 0.000

Interval 27 (260000 steps performed)
10000/10000 [=====] - 338s 34ms/step - reward: 1.0000e-03
2 episodes - episode_reward: 1.500 [0.000, 3.000] - loss: 0.003 - mae: 0.229 - mean_q: 0.257 - mean_eps: 0.100 - ale.lives: 0.000

Interval 28 (270000 steps performed)
10000/10000 [=====] - 339s 34ms/step - reward: -7.0000e-04
3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.004 - mae: 0.249 - mean_q: 0.276 - mean_eps: 0.100 - ale.lives: 0.000

Interval 29 (280000 steps performed)
10000/10000 [=====] - 342s 34ms/step - reward: 0.0018
2 episodes - episode_reward: 3.000 [2.000, 4.000] - loss: 0.005 - mae: 0.284 - mean_q: 0.309 - mean_eps: 0.100 - ale.lives: 0.000

Interval 30 (290000 steps performed)
10000/10000 [=====] - 335s 34ms/step - reward: -0.0011
2 episodes - episode_reward: 0.500 [0.000, 1.000] - loss: 0.006 - mae: 0.330 - mean_q: 0.361 - mean_eps: 0.100 - ale.lives: 0.000

Interval 31 (300000 steps performed)
10000/10000 [=====] - 353s 35ms/step - reward: 8.0000e-04
2 episodes - episode_reward: 4.000 [1.000, 7.000] - loss: 0.007 - mae: 0.377 - mean_q: 0.408 - mean_eps: 0.100 - ale.lives: 0.000

Interval 32 (310000 steps performed)
10000/10000 [=====] - 333s 33ms/step - reward: 0.0024
3 episodes - episode_reward: 8.000 [0.000, 17.000] - loss: 0.009 - mae: 0.412 - mean_q: 0.435 - mean_eps: 0.100 - ale.lives: 0.000

Interval 33 (320000 steps performed)
10000/10000 [=====] - 338s 34ms/step - reward: 7.0000e-04
2 episodes - episode_reward: 3.000 [3.000, 3.000] - loss: 0.010 - mae: 0.479 - mean_q: 0.499 - mean_eps: 0.100 - ale.lives: 0.000

Interval 34 (330000 steps performed)

10000/10000 [=====] - 350s 35ms/step - reward: 0.0049
2 episodes - episode_reward: 19.500 [15.000, 24.000] - loss: 0.012 - mae: 0.542 - mean_q: 0.571 - mean_eps: 0.100 - ale.lives: 0.000

Interval 35 (340000 steps performed)
10000/10000 [=====] - 335s 34ms/step - reward: 6.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.013 - mae: 0.617 - mean_q: 0.660 - mean_eps: 0.100 - ale.lives: 0.000

Interval 36 (350000 steps performed)
10000/10000 [=====] - 376s 38ms/step - reward: 0.0041
3 episodes - episode_reward: 17.333 [13.000, 26.000] - loss: 0.015 - mae: 0.671 - mean_q: 0.712 - mean_eps: 0.100 - ale.lives: 0.000

Interval 37 (360000 steps performed)
10000/10000 [=====] - 340s 34ms/step - reward: 0.0113
2 episodes - episode_reward: 59.500 [56.000, 63.000] - loss: 0.017 - mae: 0.763 - mean_q: 0.805 - mean_eps: 0.100 - ale.lives: 0.000

Interval 38 (370000 steps performed)
10000/10000 [=====] - 295s 29ms/step - reward: 0.0059
2 episodes - episode_reward: 15.500 [15.000, 16.000] - loss: 0.018 - mae: 0.822 - mean_q: 0.876 - mean_eps: 0.100 - ale.lives: 0.000

Interval 39 (380000 steps performed)
10000/10000 [=====] - 273s 27ms/step - reward: 0.0070
2 episodes - episode_reward: 40.500 [13.000, 68.000] - loss: 0.018 - mae: 0.882 - mean_q: 0.953 - mean_eps: 0.100 - ale.lives: 0.000

Interval 40 (390000 steps performed)
10000/10000 [=====] - 273s 27ms/step - reward: 0.0169
3 episodes - episode_reward: 60.667 [46.000, 83.000] - loss: 0.021 - mae: 0.954 - mean_q: 1.043 - mean_eps: 0.100 - ale.lives: 0.000

Interval 41 (400000 steps performed)
10000/10000 [=====] - 394s 39ms/step - reward: 0.0152
2 episodes - episode_reward: 75.500 [52.000, 99.000] - loss: 0.022 - mae: 1.007 - mean_q: 1.105 - mean_eps: 0.100 - ale.lives: 0.000

Interval 42 (410000 steps performed)
10000/10000 [=====] - 349s 35ms/step - reward: 0.0132

2 episodes - episode_reward: 49.500 [49.000, 50.000] - loss: 0.022 - mae: 1.042 - mean_q: 1.148 - mean_eps: 0.100 - ale.lives: 0.000

Interval 43 (420000 steps performed)

10000/10000 [=====] - 393s 39ms/step - reward: 0.0169

2 episodes - episode_reward: 77.500 [68.000, 87.000] - loss: 0.024 - mae: 1.123 - mean_q: 1.240 - mean_eps: 0.100 - ale.lives: 0.000

Interval 44 (430000 steps performed)

10000/10000 [=====] - 382s 38ms/step - reward: 0.0137

3 episodes - episode_reward: 61.667 [29.000, 92.000] - loss: 0.024 - mae: 1.213 - mean_q: 1.339 - mean_eps: 0.100 - ale.lives: 0.000

Interval 45 (440000 steps performed)

10000/10000 [=====] - 444s 44ms/step - reward: 0.0154

2 episodes - episode_reward: 68.500 [56.000, 81.000] - loss: 0.024 - mae: 1.269 - mean_q: 1.388 - mean_eps: 0.100 - ale.lives: 0.000

Interval 46 (450000 steps performed)

10000/10000 [=====] - 312s 31ms/step - reward: 0.0186

2 episodes - episode_reward: 80.500 [48.000, 113.000] - loss: 0.027 - mae: 1.367 - mean_q: 1.495 - mean_eps: 0.100 - ale.lives: 0.000

Interval 47 (460000 steps performed)

10000/10000 [=====] - 260s 26ms/step - reward: 0.0206

2 episodes - episode_reward: 99.000 [98.000, 100.000] - loss: 0.029 - mae: 1.480 - mean_q: 1.619 - mean_eps: 0.100 - ale.lives: 0.000

Interval 48 (470000 steps performed)

10000/10000 [=====] - 264s 26ms/step - reward: 0.0186

3 episodes - episode_reward: 68.667 [56.000, 90.000] - loss: 0.031 - mae: 1.565 - mean_q: 1.728 - mean_eps: 0.100 - ale.lives: 0.000

Interval 49 (480000 steps performed)

10000/10000 [=====] - 263s 26ms/step - reward: 0.0124

2 episodes - episode_reward: 63.000 [51.000, 75.000] - loss: 0.032 - mae: 1.637 - mean_q: 1.823 - mean_eps: 0.100 - ale.lives: 0.000

Interval 50 (490000 steps performed)

10000/10000 [=====] - 263s 26ms/step - reward: 0.0279

2 episodes - episode_reward: 108.500 [97.000, 120.000] - loss: 0.032 - mae: 1.732 - mean_q: 1.940 - mean_eps: 0.100 - ale.live

s: 0.000

Interval 51 (500000 steps performed)

10000/10000 [=====] - 422s 42ms/step - reward: 0.0272

2 episodes - episode_reward: 137.500 [131.000, 144.000] - loss: 0.034 - mae: 1.836 - mean_q: 2.065 - mean_eps: 0.100 - ale.lives: 0.000

Interval 52 (510000 steps performed)

10000/10000 [=====] - 380s 38ms/step - reward: 0.0202

3 episodes - episode_reward: 95.667 [84.000, 111.000] - loss: 0.035 - mae: 1.954 - mean_q: 2.210 - mean_eps: 0.100 - ale.lives: 0.000

Interval 53 (520000 steps performed)

10000/10000 [=====] - 443s 44ms/step - reward: 0.0206

2 episodes - episode_reward: 93.000 [65.000, 121.000] - loss: 0.035 - mae: 2.018 - mean_q: 2.294 - mean_eps: 0.100 - ale.lives: 0.000

Interval 54 (530000 steps performed)

10000/10000 [=====] - 385s 39ms/step - reward: 0.0276

2 episodes - episode_reward: 122.000 [109.000, 135.000] - loss: 0.036 - mae: 2.085 - mean_q: 2.371 - mean_eps: 0.100 - ale.lives: 0.000

Interval 55 (540000 steps performed)

10000/10000 [=====] - 317s 32ms/step - reward: 0.0253

2 episodes - episode_reward: 95.500 [85.000, 106.000] - loss: 0.036 - mae: 2.171 - mean_q: 2.482 - mean_eps: 0.100 - ale.lives: 0.000

Interval 56 (550000 steps performed)

10000/10000 [=====] - 279s 28ms/step - reward: 0.0265

3 episodes - episode_reward: 119.000 [110.000, 131.000] - loss: 0.039 - mae: 2.369 - mean_q: 2.721 - mean_eps: 0.100 - ale.lives: 0.000

Interval 57 (560000 steps performed)

10000/10000 [=====] - 282s 28ms/step - reward: 0.0142

2 episodes - episode_reward: 66.500 [25.000, 108.000] - loss: 0.040 - mae: 2.536 - mean_q: 2.915 - mean_eps: 0.100 - ale.lives: 0.000

Interval 58 (570000 steps performed)

10000/10000 [=====] - 340s 34ms/step - reward: 0.0253

2 episodes - episode_reward: 113.000 [107.000, 119.000] - loss: 0.041 - mae: 2.660 - mean_q: 3.059 - mean_eps: 0.100 - ale.lives: 0.000

Interval 59 (580000 steps performed)

10000/10000 [=====] - 337s 34ms/step - reward: 0.0256

2 episodes - episode_reward: 93.500 [62.000, 125.000] - loss: 0.041 - mae: 2.740 - mean_q: 3.147 - mean_eps: 0.100 - ale.lives: 0.000

Interval 60 (590000 steps performed)

10000/10000 [=====] - 328s 33ms/step - reward: 0.0238

3 episodes - episode_reward: 120.333 [98.000, 150.000] - loss: 0.041 - mae: 2.828 - mean_q: 3.253 - mean_eps: 0.100 - ale.lives: 0.000

Interval 61 (600000 steps performed)

10000/10000 [=====] - 579s 58ms/step - reward: 0.0249

2 episodes - episode_reward: 118.000 [97.000, 139.000] - loss: 0.043 - mae: 2.954 - mean_q: 3.401 - mean_eps: 0.100 - ale.lives: 0.000

Interval 62 (610000 steps performed)

10000/10000 [=====] - 571s 57ms/step - reward: 0.0284

2 episodes - episode_reward: 117.500 [109.000, 126.000] - loss: 0.045 - mae: 3.029 - mean_q: 3.483 - mean_eps: 0.100 - ale.lives: 0.000

Interval 63 (620000 steps performed)

10000/10000 [=====] - 542s 54ms/step - reward: 0.0234

2 episodes - episode_reward: 93.000 [84.000, 102.000] - loss: 0.047 - mae: 3.191 - mean_q: 3.678 - mean_eps: 0.100 - ale.lives: 0.000

Interval 64 (630000 steps performed)

10000/10000 [=====] - 446s 45ms/step - reward: 0.0305

3 episodes - episode_reward: 130.333 [129.000, 132.000] - loss: 0.050 - mae: 3.225 - mean_q: 3.716 - mean_eps: 0.100 - ale.lives: 0.000

Interval 65 (640000 steps performed)

10000/10000 [=====] - 360s 36ms/step - reward: 0.0279

2 episodes - episode_reward: 148.000 [142.000, 154.000] - loss: 0.052 - mae: 3.343 - mean_q: 3.851 - mean_eps: 0.100 - ale.lives: 0.000

Interval 66 (650000 steps performed)

10000/10000 [=====] - 360s 36ms/step - reward: 0.0176

2 episodes - episode_reward: 77.500 [63.000, 92.000] - loss: 0.050 - mae: 3.441 - mean_q: 3.962 - mean_eps: 0.100 - ale.lives: 0.000

Interval 67 (660000 steps performed)
10000/10000 [=====] - 357s 36ms/step - reward: 0.0134
2 episodes - episode_reward: 60.500 [60.000, 61.000] - loss: 0.050 - mae: 3.498 - mean_q: 4.031 - mean_eps: 0.100 - ale.lives: 0.000

Interval 68 (670000 steps performed)
10000/10000 [=====] - 340s 34ms/step - reward: 0.0194
3 episodes - episode_reward: 80.000 [52.000, 96.000] - loss: 0.049 - mae: 3.591 - mean_q: 4.135 - mean_eps: 0.100 - ale.lives: 0.000

Interval 69 (680000 steps performed)
10000/10000 [=====] - 328s 33ms/step - reward: 0.0257
2 episodes - episode_reward: 99.500 [85.000, 114.000] - loss: 0.050 - mae: 3.705 - mean_q: 4.264 - mean_eps: 0.100 - ale.lives: 0.000

Interval 70 (690000 steps performed)
10000/10000 [=====] - 343s 34ms/step - reward: 0.0350
2 episodes - episode_reward: 156.000 [142.000, 170.000] - loss: 0.052 - mae: 3.831 - mean_q: 4.405 - mean_eps: 0.100 - ale.lives: 0.000

Interval 71 (700000 steps performed)
10000/10000 [=====] - 606s 61ms/step - reward: 0.0329
3 episodes - episode_reward: 143.333 [128.000, 167.000] - loss: 0.052 - mae: 3.889 - mean_q: 4.475 - mean_eps: 0.100 - ale.lives: 0.000

Interval 72 (710000 steps performed)
10000/10000 [=====] - 607s 61ms/step - reward: 0.0266
2 episodes - episode_reward: 125.000 [121.000, 129.000] - loss: 0.052 - mae: 3.998 - mean_q: 4.602 - mean_eps: 0.100 - ale.lives: 0.000

Interval 73 (720000 steps performed)
10000/10000 [=====] - 555s 55ms/step - reward: 0.0280
2 episodes - episode_reward: 123.000 [119.000, 127.000] - loss: 0.051 - mae: 4.071 - mean_q: 4.682 - mean_eps: 0.100 - ale.lives: 0.000

Interval 74 (730000 steps performed)
10000/10000 [=====] - 416s 42ms/step - reward: 0.0424
2 episodes - episode_reward: 168.500 [168.000, 169.000] - loss: 0.050 - mae: 4.107 - mean_q: 4.721 - mean_eps: 0.100 - ale.lives: 0.000

Interval 75 (740000 steps performed)

10000/10000 [=====] - 380s 38ms/step - reward: 0.0346
3 episodes - episode_reward: 160.333 [135.000, 192.000] - loss: 0.051 - mae: 4.224 - mean_q: 4.852 - mean_eps: 0.100 - ale.lives: 0.000

Interval 76 (750000 steps performed)

10000/10000 [=====] - 386s 39ms/step - reward: 0.0340
2 episodes - episode_reward: 147.500 [147.000, 148.000] - loss: 0.056 - mae: 4.306 - mean_q: 4.945 - mean_eps: 0.100 - ale.lives: 0.000

Interval 77 (760000 steps performed)

10000/10000 [=====] - 373s 37ms/step - reward: 0.0361
2 episodes - episode_reward: 172.500 [160.000, 185.000] - loss: 0.052 - mae: 4.353 - mean_q: 5.001 - mean_eps: 0.100 - ale.lives: 0.000

Interval 78 (770000 steps performed)

10000/10000 [=====] - 359s 36ms/step - reward: 0.0315
2 episodes - episode_reward: 131.000 [120.000, 142.000] - loss: 0.055 - mae: 4.454 - mean_q: 5.111 - mean_eps: 0.100 - ale.lives: 0.000

Interval 79 (780000 steps performed)

10000/10000 [=====] - 355s 35ms/step - reward: 0.0318
3 episodes - episode_reward: 144.667 [118.000, 163.000] - loss: 0.054 - mae: 4.608 - mean_q: 5.289 - mean_eps: 0.100 - ale.lives: 0.000

Interval 80 (790000 steps performed)

10000/10000 [=====] - 350s 35ms/step - reward: 0.0276
2 episodes - episode_reward: 125.000 [124.000, 126.000] - loss: 0.056 - mae: 4.675 - mean_q: 5.362 - mean_eps: 0.100 - ale.lives: 0.000

Interval 81 (800000 steps performed)

10000/10000 [=====] - 648s 65ms/step - reward: 0.0323
2 episodes - episode_reward: 152.000 [133.000, 171.000] - loss: 0.057 - mae: 4.726 - mean_q: 5.418 - mean_eps: 0.100 - ale.lives: 0.000

Interval 82 (810000 steps performed)

10000/10000 [=====] - 608s 61ms/step - reward: 0.0318
2 episodes - episode_reward: 129.000 [113.000, 145.000] - loss: 0.058 - mae: 4.778 - mean_q: 5.482 - mean_eps: 0.100 - ale.lives: 0.000

Interval 83 (820000 steps performed)

10000/10000 [=====] - 428s 43ms/step - reward: 0.0312

3 episodes - episode_reward: 136.333 [125.000, 144.000] - loss: 0.055 - mae: 4.862 - mean_q: 5.574 - mean_eps: 0.100 - ale.lives: 0.000

Interval 84 (830000 steps performed)

10000/10000 [=====] - 532s 53ms/step - reward: 0.0319

2 episodes - episode_reward: 144.000 [129.000, 159.000] - loss: 0.056 - mae: 4.945 - mean_q: 5.669 - mean_eps: 0.100 - ale.lives: 0.000

Interval 85 (840000 steps performed)

10000/10000 [=====] - 459s 46ms/step - reward: 0.0339

2 episodes - episode_reward: 156.500 [145.000, 168.000] - loss: 0.055 - mae: 5.010 - mean_q: 5.741 - mean_eps: 0.100 - ale.lives: 0.000

Interval 86 (850000 steps performed)

10000/10000 [=====] - 368s 37ms/step - reward: 0.0316

2 episodes - episode_reward: 141.000 [134.000, 148.000] - loss: 0.055 - mae: 5.071 - mean_q: 5.805 - mean_eps: 0.100 - ale.lives: 0.000

Interval 87 (860000 steps performed)

10000/10000 [=====] - 337s 34ms/step - reward: 0.0395

3 episodes - episode_reward: 162.000 [150.000, 179.000] - loss: 0.056 - mae: 5.120 - mean_q: 5.863 - mean_eps: 0.100 - ale.lives: 0.000

Interval 88 (870000 steps performed)

10000/10000 [=====] - 337s 34ms/step - reward: 0.0298

2 episodes - episode_reward: 129.000 [126.000, 132.000] - loss: 0.056 - mae: 5.170 - mean_q: 5.919 - mean_eps: 0.100 - ale.lives: 0.000

Interval 89 (880000 steps performed)

10000/10000 [=====] - 370s 37ms/step - reward: 0.0353

2 episodes - episode_reward: 174.000 [166.000, 182.000] - loss: 0.059 - mae: 5.193 - mean_q: 5.949 - mean_eps: 0.100 - ale.lives: 0.000

Interval 90 (890000 steps performed)

10000/10000 [=====] - 366s 37ms/step - reward: 0.0354

2 episodes - episode_reward: 147.000 [146.000, 148.000] - loss: 0.058 - mae: 5.331 - mean_q: 6.107 - mean_eps: 0.100 - ale.lives: 0.000

Interval 91 (900000 steps performed)

10000/10000 [=====] - 602s 60ms/step - reward: 0.0341

3 episodes - episode_reward: 148.333 [126.000, 168.000] - loss: 0.059 - mae: 5.459 - mean_q: 6.248 - mean_eps: 0.100 - ale.lives: 0.000

s: 0.000

Interval 92 (910000 steps performed)

10000/10000 [=====] - 617s 62ms/step - reward: 0.0365

2 episodes - episode_reward: 148.000 [127.000, 169.000] - loss: 0.059 - mae: 5.504 - mean_q: 6.294 - mean_eps: 0.100 - ale.lives: 0.000

Interval 93 (920000 steps performed)

10000/10000 [=====] - 551s 55ms/step - reward: 0.0385

1 episodes - episode_reward: 377.000 [377.000, 377.000] - loss: 0.057 - mae: 5.474 - mean_q: 6.257 - mean_eps: 0.100 - ale.lives: 0.000

Interval 94 (930000 steps performed)

10000/10000 [=====] - 418s 42ms/step - reward: 0.0439

1 episodes - episode_reward: 402.000 [402.000, 402.000] - loss: 0.059 - mae: 5.549 - mean_q: 6.349 - mean_eps: 0.100 - ale.lives: 0.000

Interval 95 (940000 steps performed)

10000/10000 [=====] - 405s 40ms/step - reward: 0.0426

3 episodes - episode_reward: 177.333 [156.000, 191.000] - loss: 0.062 - mae: 5.572 - mean_q: 6.371 - mean_eps: 0.100 - ale.lives: 0.000

Interval 96 (950000 steps performed)

10000/10000 [=====] - 407s 41ms/step - reward: 0.0317

2 episodes - episode_reward: 158.000 [138.000, 178.000] - loss: 0.062 - mae: 5.609 - mean_q: 6.419 - mean_eps: 0.100 - ale.lives: 0.000

Interval 97 (960000 steps performed)

10000/10000 [=====] - 389s 39ms/step - reward: 0.0398

1 episodes - episode_reward: 133.000 [133.000, 133.000] - loss: 0.060 - mae: 5.647 - mean_q: 6.455 - mean_eps: 0.100 - ale.lives: 0.000

Interval 98 (970000 steps performed)

10000/10000 [=====] - 379s 38ms/step - reward: 0.0377

1 episodes - episode_reward: 367.000 [367.000, 367.000] - loss: 0.058 - mae: 5.658 - mean_q: 6.473 - mean_eps: 0.100 - ale.lives: 0.000

Interval 99 (980000 steps performed)

10000/10000 [=====] - 382s 38ms/step - reward: 0.0362

2 episodes - episode_reward: 222.500 [117.000, 328.000] - loss: 0.059 - mae: 5.775 - mean_q: 6.601 - mean_eps: 0.100 - ale.lives: 0.000

Interval 100 (990000 steps performed)

10000/10000 [=====] - 394s 39ms/step - reward: 0.0275

2 episodes - episode_reward: 210.500 [172.000, 249.000] - loss: 0.061 - mae: 5.931 - mean_q: 6.784 - mean_eps: 0.100 - ale.lives: 0.000

Interval 101 (1000000 steps performed)

10000/10000 [=====] - 641s 64ms/step - reward: 0.0348

2 episodes - episode_reward: 164.000 [141.000, 187.000] - loss: 0.060 - mae: 5.953 - mean_q: 6.811 - mean_eps: 0.100 - ale.lives: 0.000

Interval 102 (1010000 steps performed)

10000/10000 [=====] - 647s 65ms/step - reward: 0.0425

1 episodes - episode_reward: 177.000 [177.000, 177.000] - loss: 0.064 - mae: 6.026 - mean_q: 6.893 - mean_eps: 0.100 - ale.lives: 0.000

Interval 103 (1020000 steps performed)

10000/10000 [=====] - 641s 64ms/step - reward: 0.0393

3 episodes - episode_reward: 237.000 [173.000, 363.000] - loss: 0.067 - mae: 6.132 - mean_q: 7.018 - mean_eps: 0.100 - ale.lives: 0.000

Interval 104 (1030000 steps performed)

10000/10000 [=====] - 590s 59ms/step - reward: 0.0393

2 episodes - episode_reward: 187.000 [181.000, 193.000] - loss: 0.067 - mae: 6.173 - mean_q: 7.062 - mean_eps: 0.100 - ale.lives: 0.000

Interval 105 (1040000 steps performed)

10000/10000 [=====] - 429s 43ms/step - reward: 0.0384

2 episodes - episode_reward: 146.000 [123.000, 169.000] - loss: 0.063 - mae: 6.244 - mean_q: 7.138 - mean_eps: 0.100 - ale.lives: 0.000

Interval 106 (1050000 steps performed)

10000/10000 [=====] - 427s 43ms/step - reward: 0.0417

1 episodes - episode_reward: 366.000 [366.000, 366.000] - loss: 0.067 - mae: 6.406 - mean_q: 7.327 - mean_eps: 0.100 - ale.lives: 0.000

Interval 107 (1060000 steps performed)

10000/10000 [=====] - 426s 43ms/step - reward: 0.0347

1 episodes - episode_reward: 307.000 [307.000, 307.000] - loss: 0.067 - mae: 6.470 - mean_q: 7.400 - mean_eps: 0.100 - ale.lives: 0.000

Interval 108 (1070000 steps performed)
10000/10000 [=====] - 396s 40ms/step - reward: 0.0280
2 episodes - episode_reward: 234.000 [174.000, 294.000] - loss: 0.069 - mae: 6.629 - mean_q: 7.578 - mean_eps: 0.100 - ale.lives: 0.000

Interval 109 (1080000 steps performed)
10000/10000 [=====] - 392s 39ms/step - reward: 0.0386
2 episodes - episode_reward: 161.500 [141.000, 182.000] - loss: 0.072 - mae: 6.764 - mean_q: 7.742 - mean_eps: 0.100 - ale.lives: 0.000

Interval 110 (1090000 steps performed)
10000/10000 [=====] - 393s 39ms/step - reward: 0.0379
2 episodes - episode_reward: 157.500 [131.000, 184.000] - loss: 0.069 - mae: 6.891 - mean_q: 7.879 - mean_eps: 0.100 - ale.lives: 0.000

Interval 111 (1100000 steps performed)
10000/10000 [=====] - 674s 67ms/step - reward: 0.0408
2 episodes - episode_reward: 266.500 [167.000, 366.000] - loss: 0.071 - mae: 6.944 - mean_q: 7.942 - mean_eps: 0.100 - ale.lives: 0.000

Interval 112 (1110000 steps performed)
10000/10000 [=====] - 659s 66ms/step - reward: 0.0377
2 episodes - episode_reward: 185.500 [183.000, 188.000] - loss: 0.075 - mae: 7.056 - mean_q: 8.067 - mean_eps: 0.100 - ale.lives: 0.000

Interval 113 (1120000 steps performed)
10000/10000 [=====] - 545s 54ms/step - reward: 0.0400
2 episodes - episode_reward: 155.000 [146.000, 164.000] - loss: 0.072 - mae: 7.224 - mean_q: 8.258 - mean_eps: 0.100 - ale.lives: 0.000

Interval 114 (1130000 steps performed)
10000/10000 [=====] - 430s 43ms/step - reward: 0.0380
2 episodes - episode_reward: 186.000 [175.000, 197.000] - loss: 0.076 - mae: 7.417 - mean_q: 8.477 - mean_eps: 0.100 - ale.lives: 0.000

Interval 115 (1140000 steps performed)
10000/10000 [=====] - 419s 42ms/step - reward: 0.0404
3 episodes - episode_reward: 173.000 [155.000, 194.000] - loss: 0.081 - mae: 7.483 - mean_q: 8.559 - mean_eps: 0.100 - ale.lives: 0.000

Interval 116 (1150000 steps performed)

10000/10000 [=====] - 413s 41ms/step - reward: 0.0367
2 episodes - episode_reward: 164.000 [155.000, 173.000] - loss: 0.079 - mae: 7.527 - mean_q: 8.613 - mean_eps: 0.100 - ale.lives: 0.000

Interval 117 (1160000 steps performed)

10000/10000 [=====] - 386s 39ms/step - reward: 0.0437
2 episodes - episode_reward: 183.000 [179.000, 187.000] - loss: 0.084 - mae: 7.623 - mean_q: 8.721 - mean_eps: 0.100 - ale.lives: 0.000

Interval 118 (1170000 steps performed)

10000/10000 [=====] - 389s 39ms/step - reward: 0.0349
1 episodes - episode_reward: 278.000 [278.000, 278.000] - loss: 0.082 - mae: 7.854 - mean_q: 8.979 - mean_eps: 0.100 - ale.lives: 0.000

Interval 119 (1180000 steps performed)

10000/10000 [=====] - 388s 39ms/step - reward: 0.0321
2 episodes - episode_reward: 240.500 [149.000, 332.000] - loss: 0.087 - mae: 7.993 - mean_q: 9.136 - mean_eps: 0.100 - ale.lives: 0.000

Interval 120 (1190000 steps performed)

10000/10000 [=====] - 391s 39ms/step - reward: 0.0391
1 episodes - episode_reward: 190.000 [190.000, 190.000] - loss: 0.084 - mae: 8.061 - mean_q: 9.209 - mean_eps: 0.100 - ale.lives: 0.000

Interval 121 (1200000 steps performed)

10000/10000 [=====] - 640s 64ms/step - reward: 0.0386
2 episodes - episode_reward: 279.000 [171.000, 387.000] - loss: 0.085 - mae: 8.126 - mean_q: 9.279 - mean_eps: 0.100 - ale.lives: 0.000

Interval 122 (1210000 steps performed)

10000/10000 [=====] - 646s 65ms/step - reward: 0.0344
2 episodes - episode_reward: 135.000 [119.000, 151.000] - loss: 0.084 - mae: 8.266 - mean_q: 9.427 - mean_eps: 0.100 - ale.lives: 0.000

Interval 123 (1220000 steps performed)

10000/10000 [=====] - 562s 56ms/step - reward: 0.0379
2 episodes - episode_reward: 153.000 [153.000, 153.000] - loss: 0.085 - mae: 8.340 - mean_q: 9.516 - mean_eps: 0.100 - ale.lives: 0.000

Interval 124 (1230000 steps performed)

10000/10000 [=====] - 428s 43ms/step - reward: 0.0340

1 episodes - episode_reward: 284.000 [284.000, 284.000] - loss: 0.084 - mae: 8.361 - mean_q: 9.539 - mean_eps: 0.100 - ale.lives: 0.000

Interval 125 (1240000 steps performed)

10000/10000 [=====] - 430s 43ms/step - reward: 0.0330

2 episodes - episode_reward: 246.000 [133.000, 359.000] - loss: 0.086 - mae: 8.436 - mean_q: 9.632 - mean_eps: 0.100 - ale.lives: 0.000

Interval 126 (1250000 steps performed)

10000/10000 [=====] - 425s 42ms/step - reward: 0.0349

2 episodes - episode_reward: 143.000 [138.000, 148.000] - loss: 0.085 - mae: 8.562 - mean_q: 9.777 - mean_eps: 0.100 - ale.lives: 0.000

Interval 127 (1260000 steps performed)

10000/10000 [=====] - 395s 39ms/step - reward: 0.0445

3 episodes - episode_reward: 179.333 [165.000, 188.000] - loss: 0.085 - mae: 8.607 - mean_q: 9.828 - mean_eps: 0.100 - ale.lives: 0.000

Interval 128 (1270000 steps performed)

10000/10000 [=====] - 663s 66ms/step - reward: 0.0303

1 episodes - episode_reward: 331.000 [331.000, 331.000] - loss: 0.084 - mae: 8.684 - mean_q: 9.910 - mean_eps: 0.100 - ale.lives: 0.000

Interval 129 (1280000 steps performed)

10000/10000 [=====] - 512s 51ms/step - reward: 0.0338

1 episodes - episode_reward: 169.000 [169.000, 169.000] - loss: 0.082 - mae: 8.793 - mean_q: 10.029 - mean_eps: 0.100 - ale.lives: 0.000

Interval 130 (1290000 steps performed)

10000/10000 [=====] - 432s 43ms/step - reward: 0.0296

2 episodes - episode_reward: 196.500 [147.000, 246.000] - loss: 0.089 - mae: 8.813 - mean_q: 10.055 - mean_eps: 0.100 - ale.lives: 0.000

Interval 131 (1300000 steps performed)

10000/10000 [=====] - 664s 66ms/step - reward: 0.0365

3 episodes - episode_reward: 153.000 [131.000, 168.000] - loss: 0.088 - mae: 8.918 - mean_q: 10.172 - mean_eps: 0.100 - ale.lives: 0.000

Interval 132 (1310000 steps performed)

10000/10000 [=====] - 663s 66ms/step - reward: 0.0413

1 episodes - episode_reward: 358.000 [358.000, 358.000] - loss: 0.088 - mae: 9.038 - mean_q: 10.310 - mean_eps: 0.100 - ale.lives: 0.000

es: 0.000

Interval 133 (1320000 steps performed)

10000/10000 [=====] - 532s 53ms/step - reward: 0.0437

1 episodes - episode_reward: 387.000 [387.000, 387.000] - loss: 0.082 - mae: 8.961 - mean_q: 10.219 - mean_eps: 0.100 - ale.lives: 0.000

Interval 134 (1330000 steps performed)

10000/10000 [=====] - 429s 43ms/step - reward: 0.0410

2 episodes - episode_reward: 185.500 [185.000, 186.000] - loss: 0.081 - mae: 8.878 - mean_q: 10.121 - mean_eps: 0.100 - ale.lives: 0.000

Interval 135 (1340000 steps performed)

10000/10000 [=====] - 431s 43ms/step - reward: 0.0470

3 episodes - episode_reward: 192.000 [191.000, 193.000] - loss: 0.085 - mae: 8.938 - mean_q: 10.193 - mean_eps: 0.100 - ale.lives: 0.000

Interval 136 (1350000 steps performed)

10000/10000 [=====] - 428s 43ms/step - reward: 0.0398

1 episodes - episode_reward: 386.000 [386.000, 386.000] - loss: 0.081 - mae: 8.952 - mean_q: 10.210 - mean_eps: 0.100 - ale.lives: 0.000

Interval 137 (1360000 steps performed)

10000/10000 [=====] - 436s 44ms/step - reward: 0.0406

1 episodes - episode_reward: 359.000 [359.000, 359.000] - loss: 0.082 - mae: 8.957 - mean_q: 10.210 - mean_eps: 0.100 - ale.lives: 0.000

Interval 138 (1370000 steps performed)

10000/10000 [=====] - 401s 40ms/step - reward: 0.0422

2 episodes - episode_reward: 174.500 [173.000, 176.000] - loss: 0.087 - mae: 9.076 - mean_q: 10.342 - mean_eps: 0.100 - ale.lives: 0.000

Interval 139 (1380000 steps performed)

10000/10000 [=====] - 502s 50ms/step - reward: 0.0295

2 episodes - episode_reward: 236.000 [145.000, 327.000] - loss: 0.085 - mae: 9.125 - mean_q: 10.401 - mean_eps: 0.100 - ale.lives: 0.000

Interval 140 (1390000 steps performed)

10000/10000 [=====] - 615s 61ms/step - reward: 0.0380

2 episodes - episode_reward: 162.000 [154.000, 170.000] - loss: 0.087 - mae: 9.031 - mean_q: 10.291 - mean_eps: 0.100 - ale.lives: 0.000

Interval 141 (1400000 steps performed)

10000/10000 [=====] - 647s 65ms/step - reward: 0.0447

1 episodes - episode_reward: 192.000 [192.000, 192.000] - loss: 0.086 - mae: 8.910 - mean_q: 10.160 - mean_eps: 0.100 - ale.lives: 0.000

Interval 142 (1410000 steps performed)

10000/10000 [=====] - 520s 52ms/step - reward: 0.0411

2 episodes - episode_reward: 371.500 [360.000, 383.000] - loss: 0.084 - mae: 9.050 - mean_q: 10.321 - mean_eps: 0.100 - ale.lives: 0.000

Interval 143 (1420000 steps performed)

10000/10000 [=====] - 430s 43ms/step - reward: 0.0378

2 episodes - episode_reward: 167.500 [155.000, 180.000] - loss: 0.084 - mae: 9.067 - mean_q: 10.335 - mean_eps: 0.100 - ale.lives: 0.000

Interval 144 (1430000 steps performed)

10000/10000 [=====] - 431s 43ms/step - reward: 0.0421

2 episodes - episode_reward: 191.000 [186.000, 196.000] - loss: 0.089 - mae: 9.028 - mean_q: 10.295 - mean_eps: 0.100 - ale.lives: 0.000

Interval 145 (1440000 steps performed)

10000/10000 [=====] - 421s 42ms/step - reward: 0.0349

2 episodes - episode_reward: 155.000 [126.000, 184.000] - loss: 0.088 - mae: 9.042 - mean_q: 10.315 - mean_eps: 0.100 - ale.lives: 0.000

Interval 146 (1450000 steps performed)

10000/10000 [=====] - 398s 40ms/step - reward: 0.0406

3 episodes - episode_reward: 172.333 [155.000, 184.000] - loss: 0.085 - mae: 9.048 - mean_q: 10.315 - mean_eps: 0.100 - ale.lives: 0.000

Interval 147 (1460000 steps performed)

10000/10000 [=====] - 389s 39ms/step - reward: 0.0336

2 episodes - episode_reward: 150.500 [131.000, 170.000] - loss: 0.085 - mae: 9.114 - mean_q: 10.383 - mean_eps: 0.100 - ale.lives: 0.000

Interval 148 (1470000 steps performed)

10000/10000 [=====] - 397s 40ms/step - reward: 0.0431

1 episodes - episode_reward: 161.000 [161.000, 161.000] - loss: 0.084 - mae: 9.101 - mean_q: 10.380 - mean_eps: 0.100 - ale.lives: 0.000

Interval 149 (1480000 steps performed)

10000/10000 [=====] - 392s 39ms/step - reward: 0.0361

2 episodes - episode_reward: 275.500 [171.000, 380.000] - loss: 0.085 - mae: 9.120 - mean_q: 10.393 - mean_eps: 0.100 - ale.lives: 0.000

Interval 150 (1490000 steps performed)

10000/10000 [=====] - 386s 39ms/step - reward: 0.0391

3 episodes - episode_reward: 171.333 [158.000, 188.000] - loss: 0.083 - mae: 9.222 - mean_q: 10.499 - mean_eps: 0.100 - ale.lives: 0.000

Interval 151 (1500000 steps performed)

10000/10000 [=====] - 665s 66ms/step - reward: 0.0395

1 episodes - episode_reward: 359.000 [359.000, 359.000] - loss: 0.086 - mae: 9.301 - mean_q: 10.593 - mean_eps: 0.100 - ale.lives: 0.000

Interval 152 (1510000 steps performed)

10000/10000 [=====] - 674s 67ms/step - reward: 0.0356

1 episodes - episode_reward: 317.000 [317.000, 317.000] - loss: 0.088 - mae: 9.286 - mean_q: 10.582 - mean_eps: 0.100 - ale.lives: 0.000

Interval 153 (1520000 steps performed)

10000/10000 [=====] - 533s 53ms/step - reward: 0.0407

2 episodes - episode_reward: 156.500 [143.000, 170.000] - loss: 0.088 - mae: 9.252 - mean_q: 10.544 - mean_eps: 0.100 - ale.lives: 0.000

Interval 154 (1530000 steps performed)

10000/10000 [=====] - 438s 44ms/step - reward: 0.0394

3 episodes - episode_reward: 188.333 [175.000, 198.000] - loss: 0.084 - mae: 9.317 - mean_q: 10.617 - mean_eps: 0.100 - ale.lives: 0.000

Interval 155 (1540000 steps performed)

10000/10000 [=====] - 435s 43ms/step - reward: 0.0356

2 episodes - episode_reward: 151.000 [122.000, 180.000] - loss: 0.087 - mae: 9.254 - mean_q: 10.544 - mean_eps: 0.100 - ale.lives: 0.000

Interval 156 (1550000 steps performed)

10000/10000 [=====] - 418s 42ms/step - reward: 0.0402

2 episodes - episode_reward: 177.500 [177.000, 178.000] - loss: 0.092 - mae: 9.369 - mean_q: 10.678 - mean_eps: 0.100 - ale.lives: 0.000

Interval 157 (1560000 steps performed)

10000/10000 [=====] - 403s 40ms/step - reward: 0.0363
2 episodes - episode_reward: 169.500 [159.000, 180.000] - loss: 0.088 - mae: 9.355 - mean_q: 10.661 - mean_eps: 0.100 - ale.lives: 0.000

Interval 158 (1570000 steps performed)

10000/10000 [=====] - 402s 40ms/step - reward: 0.0336
3 episodes - episode_reward: 153.333 [112.000, 188.000] - loss: 0.085 - mae: 9.487 - mean_q: 10.808 - mean_eps: 0.100 - ale.lives: 0.000

Interval 159 (1580000 steps performed)

10000/10000 [=====] - 361s 36ms/step - reward: 0.0337
1 episodes - episode_reward: 286.000 [286.000, 286.000] - loss: 0.089 - mae: 9.514 - mean_q: 10.844 - mean_eps: 0.100 - ale.lives: 0.000

Interval 160 (1590000 steps performed)

10000/10000 [=====] - 345s 35ms/step - reward: 0.0328
2 episodes - episode_reward: 167.500 [142.000, 193.000] - loss: 0.091 - mae: 9.529 - mean_q: 10.866 - mean_eps: 0.100 - ale.lives: 0.000

Interval 161 (1600000 steps performed)

10000/10000 [=====] - 559s 56ms/step - reward: 0.0378
2 episodes - episode_reward: 138.000 [123.000, 153.000] - loss: 0.090 - mae: 9.586 - mean_q: 10.931 - mean_eps: 0.100 - ale.lives: 0.000

Interval 162 (1610000 steps performed)

10000/10000 [=====] - 506s 51ms/step - reward: 0.0356
2 episodes - episode_reward: 248.000 [105.000, 391.000] - loss: 0.096 - mae: 9.670 - mean_q: 11.033 - mean_eps: 0.100 - ale.lives: 0.000

Interval 163 (1620000 steps performed)

10000/10000 [=====] - 477s 48ms/step - reward: 0.0427
1 episodes - episode_reward: 183.000 [183.000, 183.000] - loss: 0.094 - mae: 9.714 - mean_q: 11.080 - mean_eps: 0.100 - ale.lives: 0.000

Interval 164 (1630000 steps performed)

10000/10000 [=====] - 532s 53ms/step - reward: 0.0471
1 episodes - episode_reward: 400.000 [400.000, 400.000] - loss: 0.097 - mae: 9.756 - mean_q: 11.128 - mean_eps: 0.100 - ale.lives: 0.000

Interval 165 (1640000 steps performed)

10000/10000 [=====] - 544s 54ms/step - reward: 0.0384

1 episodes - episode_reward: 416.000 [416.000, 416.000] - loss: 0.095 - mae: 9.764 - mean_q: 11.137 - mean_eps: 0.100 - ale.lives: 0.000

Interval 166 (1650000 steps performed)

10000/10000 [=====] - 454s 45ms/step - reward: 0.0442

2 episodes - episode_reward: 264.000 [166.000, 362.000] - loss: 0.095 - mae: 9.812 - mean_q: 11.193 - mean_eps: 0.100 - ale.lives: 0.000

Interval 167 (1660000 steps performed)

10000/10000 [=====] - 428s 43ms/step - reward: 0.0376

1 episodes - episode_reward: 340.000 [340.000, 340.000] - loss: 0.093 - mae: 9.886 - mean_q: 11.261 - mean_eps: 0.100 - ale.lives: 0.000

Interval 168 (1670000 steps performed)

10000/10000 [=====] - 432s 43ms/step - reward: 0.0355

2 episodes - episode_reward: 249.000 [175.000, 323.000] - loss: 0.093 - mae: 10.016 - mean_q: 11.414 - mean_eps: 0.100 - ale.lives: 0.000

Interval 169 (1680000 steps performed)

10000/10000 [=====] - 397s 40ms/step - reward: 0.0353

2 episodes - episode_reward: 152.000 [147.000, 157.000] - loss: 0.098 - mae: 10.006 - mean_q: 11.402 - mean_eps: 0.100 - ale.lives: 0.000

Interval 170 (1690000 steps performed)

10000/10000 [=====] - 389s 39ms/step - reward: 0.0395

2 episodes - episode_reward: 168.500 [146.000, 191.000] - loss: 0.096 - mae: 10.036 - mean_q: 11.436 - mean_eps: 0.100 - ale.lives: 0.000

Interval 171 (1700000 steps performed)

10000/10000 [=====] - 687s 69ms/step - reward: 0.0368

1 episodes - episode_reward: 325.000 [325.000, 325.000] - loss: 0.089 - mae: 10.058 - mean_q: 11.457 - mean_eps: 0.100 - ale.lives: 0.000

Interval 172 (1710000 steps performed)

10000/10000 [=====] - 573s 57ms/step - reward: 0.0409

2 episodes - episode_reward: 267.000 [163.000, 371.000] - loss: 0.093 - mae: 10.028 - mean_q: 11.426 - mean_eps: 0.100 - ale.lives: 0.000

Interval 173 (1720000 steps performed)

10000/10000 [=====] - 466s 47ms/step - reward: 0.0284

1 episodes - episode_reward: 268.000 [268.000, 268.000] - loss: 0.090 - mae: 10.032 - mean_q: 11.431 - mean_eps: 0.100 - ale.lives: 0.000

ves: 0.000

Interval 174 (1730000 steps performed)

10000/10000 [=====] - 342s 34ms/step - reward: 0.0416

1 episodes - episode_reward: 345.000 [345.000, 345.000] - loss: 0.095 - mae: 10.059 - mean_q: 11.461 - mean_eps: 0.100 - ale.lives: 0.000

Interval 175 (1740000 steps performed)

10000/10000 [=====] - 348s 35ms/step - reward: 0.0439

done, took 73590.542 seconds

Double DQN

```
In [ ]: train_agent('double_dqn')
```


==== Entrenando: double_dqn a las 20260731_234157 ====

Model: "model"

Layer (type)	Output Shape	Param #
=====		
input_1 (InputLayer)	[(None, 4, 84, 84)]	0
permute (Permute)	(None, 84, 84, 4)	0
conv2d (Conv2D)	(None, 20, 20, 32)	8224
conv2d_1 (Conv2D)	(None, 9, 9, 64)	32832
conv2d_2 (Conv2D)	(None, 7, 7, 64)	36928
flatten (Flatten)	(None, 3136)	0
dense (Dense)	(None, 512)	1606144
dense_1 (Dense)	(None, 9)	4617
=====		
Total params: 1,688,745		
Trainable params: 1,688,745		
Non-trainable params: 0		

None

Training for 1750000 steps ...

Interval 1 (0 steps performed)

1/10000 [.....] - ETA: 32:06 - reward: 0.0000e+00

c:\Users\rosai\OneDrive\Documentos\Learning\VIU - Master IA\08MIAR - Aprendizaje por Refuerzo\Proyecto\gym_env\lib\site-packages\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.
warnings.warn("`Model.state\_updates` will be removed in a future version. ')

10000/10000 [=====] - 53s 5ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - ale.lives: 0.000

Interval 2 (10000 steps performed)

10000/10000 [=====] - 528s 53ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.001 - mae: 0.009 - mean_q: 0.007 - mean_eps: 0.946 - ale.lives: 0.000

Interval 3 (20000 steps performed)

10000/10000 [=====] - 565s 57ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: -0.000 - mean_eps: 0.910 - ale.lives: 0.000

Interval 4 (30000 steps performed)

10000/10000 [=====] - 338s 34ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: -0.000 - mean_eps: 0.874 - ale.lives: 0.000

Interval 5 (40000 steps performed)

10000/10000 [=====] - 380s 38ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.838 - ale.lives: 0.000

Interval 6 (50000 steps performed)

10000/10000 [=====] - 306s 31ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.802 - ale.lives: 0.000

Interval 7 (60000 steps performed)

10000/10000 [=====] - 481s 48ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.766 - ale.lives: 0.000

Interval 8 (70000 steps performed)

10000/10000 [=====] - 579s 58ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.730 - ale.lives: 0.000

Interval 9 (80000 steps performed)

10000/10000 [=====] - 365s 37ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.694 - ale.lives: 0.000

00

Interval 10 (90000 steps performed)

10000/10000 [=====] - 552s 55ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.658 - ale.lives: 0.000

Interval 11 (100000 steps performed)

10000/10000 [=====] - 346s 35ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.622 - ale.lives: 0.000

Interval 12 (110000 steps performed)

10000/10000 [=====] - 417s 42ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.586 - ale.lives: 0.000

Interval 13 (120000 steps performed)

10000/10000 [=====] - 415s 42ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.550 - ale.lives: 0.000

Interval 14 (130000 steps performed)

10000/10000 [=====] - 476s 48ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.514 - ale.lives: 0.000

Interval 15 (140000 steps performed)

10000/10000 [=====] - 595s 59ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: -0.000 - mean_eps: 0.478 - ale.lives: 0.000

Interval 16 (150000 steps performed)

10000/10000 [=====] - 581s 58ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: -0.000 - mean_eps: 0.442 - ale.lives: 0.000

Interval 17 (160000 steps performed)

10000/10000 [=====] - 498s 50ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.000 - mean_eps: 0.406 - ale.lives: 0.000

Interval 18 (170000 steps performed)

10000/10000 [=====] - 448s 45ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.000 - mean_q: 0.000 - mean_eps: 0.370 - ale.lives: 0.000

Interval 19 (180000 steps performed)

10000/10000 [=====] - 449s 45ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.000 - mean_q: 0.000 - mean_eps: 0.334 - ale.lives: 0.000

Interval 20 (190000 steps performed)

10000/10000 [=====] - 457s 46ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.000 - mean_q: 0.000 - mean_eps: 0.298 - ale.lives: 0.000

Interval 21 (200000 steps performed)

10000/10000 [=====] - 514s 51ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.000 - mean_q: 0.000 - mean_eps: 0.262 - ale.lives: 0.000

Interval 22 (210000 steps performed)

10000/10000 [=====] - 567s 57ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.000 - mean_q: 0.000 - mean_eps: 0.226 - ale.lives: 0.000

Interval 23 (220000 steps performed)

10000/10000 [=====] - 530s 53ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.000 - mean_q: 0.001 - mean_eps: 0.190 - ale.lives: 0.000

Interval 24 (230000 steps performed)

10000/10000 [=====] - 440s 44ms/step - reward: 1.0000e-04

3 episodes - episode_reward: 0.333 [0.000, 1.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.001 - mean_eps: 0.154 - ale.lives: 0.000

Interval 25 (240000 steps performed)

10000/10000 [=====] - 450s 45ms/step - reward: 7.0000e-04

2 episodes - episode_reward: 3.500 [3.000, 4.000] - loss: 0.000 - mae: 0.001 - mean_q: 0.002 - mean_eps: 0.118 - ale.lives: 0.000

Interval 26 (250000 steps performed)
10000/10000 [=====] - 465s 47ms/step - reward: 0.0015
2 episodes - episode_reward: 3.000 [0.000, 6.000] - loss: 0.001 - mae: 0.004 - mean_q: 0.006 - mean_eps: 0.100 - ale.lives: 0.000

Interval 27 (260000 steps performed)
10000/10000 [=====] - 467s 47ms/step - reward: 3.0000e-04
2 episodes - episode_reward: 1.500 [1.000, 2.000] - loss: 0.001 - mae: 0.003 - mean_q: 0.007 - mean_eps: 0.100 - ale.lives: 0.000

Interval 28 (270000 steps performed)
10000/10000 [=====] - 456s 46ms/step - reward: 8.0000e-04
3 episodes - episode_reward: 2.333 [0.000, 5.000] - loss: 0.002 - mae: 0.006 - mean_q: 0.009 - mean_eps: 0.100 - ale.lives: 0.000

Interval 29 (280000 steps performed)
10000/10000 [=====] - 459s 46ms/step - reward: -8.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.002 - mae: 0.008 - mean_q: 0.012 - mean_eps: 0.100 - ale.lives: 0.000

Interval 30 (290000 steps performed)
10000/10000 [=====] - 450s 45ms/step - reward: 0.0021
2 episodes - episode_reward: 11.000 [0.000, 22.000] - loss: 0.002 - mae: 0.008 - mean_q: 0.013 - mean_eps: 0.100 - ale.lives: 0.000

Interval 31 (300000 steps performed)
10000/10000 [=====] - 539s 54ms/step - reward: 8.0000e-04
2 episodes - episode_reward: 4.500 [0.000, 9.000] - loss: 0.002 - mae: 0.010 - mean_q: 0.014 - mean_eps: 0.100 - ale.lives: 0.000

Interval 32 (310000 steps performed)
10000/10000 [=====] - 592s 59ms/step - reward: 3.0000e-04
3 episodes - episode_reward: 0.333 [0.000, 1.000] - loss: 0.003 - mae: 0.009 - mean_q: 0.013 - mean_eps: 0.100 - ale.lives: 0.000

Interval 33 (320000 steps performed)
10000/10000 [=====] - 550s 55ms/step - reward: 9.0000e-04
2 episodes - episode_reward: 2.500 [0.000, 5.000] - loss: 0.003 - mae: 0.012 - mean_q: 0.016 - mean_eps: 0.100 - ale.lives: 0.000

Interval 34 (330000 steps performed)

10000/10000 [=====] - 459s 46ms/step - reward: 0.0013
2 episodes - episode_reward: 2.000 [0.000, 4.000] - loss: 0.004 - mae: 0.018 - mean_q: 0.023 - mean_eps: 0.100 - ale.lives: 0.000

Interval 35 (340000 steps performed)

10000/10000 [=====] - 459s 46ms/step - reward: 0.0028
2 episodes - episode_reward: 12.500 [8.000, 17.000] - loss: 0.004 - mae: 0.026 - mean_q: 0.033 - mean_eps: 0.100 - ale.lives: 0.000

Interval 36 (350000 steps performed)

10000/10000 [=====] - 467s 47ms/step - reward: 0.0023
3 episodes - episode_reward: 12.667 [0.000, 27.000] - loss: 0.005 - mae: 0.030 - mean_q: 0.038 - mean_eps: 0.100 - ale.lives: 0.000

Interval 37 (360000 steps performed)

10000/10000 [=====] - 482s 48ms/step - reward: 0.0012
2 episodes - episode_reward: 0.500 [0.000, 1.000] - loss: 0.004 - mae: 0.038 - mean_q: 0.046 - mean_eps: 0.100 - ale.lives: 0.000

Interval 38 (370000 steps performed)

10000/10000 [=====] - 476s 48ms/step - reward: 6.0000e-04
2 episodes - episode_reward: 9.500 [1.000, 18.000] - loss: 0.005 - mae: 0.049 - mean_q: 0.059 - mean_eps: 0.100 - ale.lives: 0.000

Interval 39 (380000 steps performed)

10000/10000 [=====] - 474s 47ms/step - reward: 8.0000e-04
2 episodes - episode_reward: 3.000 [2.000, 4.000] - loss: 0.005 - mae: 0.052 - mean_q: 0.060 - mean_eps: 0.100 - ale.lives: 0.000

Interval 40 (390000 steps performed)

10000/10000 [=====] - 470s 47ms/step - reward: 9.0000e-04
3 episodes - episode_reward: 2.000 [0.000, 6.000] - loss: 0.005 - mae: 0.047 - mean_q: 0.058 - mean_eps: 0.100 - ale.lives: 0.000

Interval 41 (400000 steps performed)

10000/10000 [=====] - 561s 56ms/step - reward: 0.0011
2 episodes - episode_reward: 3.500 [2.000, 5.000] - loss: 0.006 - mae: 0.048 - mean_q: 0.058 - mean_eps: 0.100 - ale.lives: 0.000

Interval 42 (410000 steps performed)

10000/10000 [=====] - 609s 61ms/step - reward: -7.0000e-04

2 episodes - episode_reward: 0.500 [0.000, 1.000] - loss: 0.006 - mae: 0.058 - mean_q: 0.070 - mean_eps: 0.100 - ale.lives: 0.000

Interval 43 (420000 steps performed)

10000/10000 [=====] - 568s 57ms/step - reward: 0.0022

2 episodes - episode_reward: 9.000 [4.000, 14.000] - loss: 0.006 - mae: 0.066 - mean_q: 0.078 - mean_eps: 0.100 - ale.lives: 0.000

Interval 44 (430000 steps performed)

10000/10000 [=====] - 505s 50ms/step - reward: 0.0011

3 episodes - episode_reward: 4.000 [2.000, 5.000] - loss: 0.006 - mae: 0.065 - mean_q: 0.076 - mean_eps: 0.100 - ale.lives: 0.000

Interval 45 (440000 steps performed)

10000/10000 [=====] - 502s 50ms/step - reward: 0.0026

2 episodes - episode_reward: 6.500 [4.000, 9.000] - loss: 0.007 - mae: 0.069 - mean_q: 0.082 - mean_eps: 0.100 - ale.lives: 0.000

Interval 46 (450000 steps performed)

10000/10000 [=====] - 480s 48ms/step - reward: 0.0011

2 episodes - episode_reward: 13.000 [2.000, 24.000] - loss: 0.007 - mae: 0.071 - mean_q: 0.084 - mean_eps: 0.100 - ale.lives: 0.000

Interval 47 (460000 steps performed)

10000/10000 [=====] - 489s 49ms/step - reward: 0.0028

2 episodes - episode_reward: 11.500 [1.000, 22.000] - loss: 0.007 - mae: 0.083 - mean_q: 0.100 - mean_eps: 0.100 - ale.lives: 0.000

Interval 48 (470000 steps performed)

10000/10000 [=====] - 486s 49ms/step - reward: 0.0024

3 episodes - episode_reward: 8.667 [0.000, 17.000] - loss: 0.007 - mae: 0.086 - mean_q: 0.100 - mean_eps: 0.100 - ale.lives: 0.000

Interval 49 (480000 steps performed)

10000/10000 [=====] - 500s 50ms/step - reward: 0.0052

2 episodes - episode_reward: 23.000 [15.000, 31.000] - loss: 0.007 - mae: 0.086 - mean_q: 0.101 - mean_eps: 0.100 - ale.lives: 0.000

Interval 50 (490000 steps performed)

10000/10000 [=====] - 490s 49ms/step - reward: 5.0000e-04

2 episodes - episode_reward: 5.000 [0.000, 10.000] - loss: 0.007 - mae: 0.090 - mean_q: 0.105 - mean_eps: 0.100 - ale.lives: 0.

000

Interval 51 (500000 steps performed)

10000/10000 [=====] - 627s 63ms/step - reward: 6.0000e-04

2 episodes - episode_reward: 3.000 [0.000, 6.000] - loss: 0.008 - mae: 0.088 - mean_q: 0.105 - mean_eps: 0.100 - ale.lives: 0.000

Interval 52 (510000 steps performed)

10000/10000 [=====] - 642s 64ms/step - reward: 0.0013

3 episodes - episode_reward: 6.667 [3.000, 11.000] - loss: 0.008 - mae: 0.089 - mean_q: 0.105 - mean_eps: 0.100 - ale.lives: 0.000

Interval 53 (520000 steps performed)

10000/10000 [=====] - 593s 59ms/step - reward: 0.0064

2 episodes - episode_reward: 15.500 [0.000, 31.000] - loss: 0.008 - mae: 0.090 - mean_q: 0.106 - mean_eps: 0.100 - ale.lives: 0.000

Interval 54 (530000 steps performed)

10000/10000 [=====] - 521s 52ms/step - reward: 0.0027

2 episodes - episode_reward: 24.000 [14.000, 34.000] - loss: 0.008 - mae: 0.101 - mean_q: 0.118 - mean_eps: 0.100 - ale.lives: 0.000

Interval 55 (540000 steps performed)

10000/10000 [=====] - 502s 50ms/step - reward: 0.0041

2 episodes - episode_reward: 24.000 [22.000, 26.000] - loss: 0.008 - mae: 0.115 - mean_q: 0.136 - mean_eps: 0.100 - ale.lives: 0.000

Interval 56 (550000 steps performed)

10000/10000 [=====] - 500s 50ms/step - reward: 0.0024

3 episodes - episode_reward: 5.667 [2.000, 10.000] - loss: 0.009 - mae: 0.118 - mean_q: 0.138 - mean_eps: 0.100 - ale.lives: 0.000

Interval 57 (560000 steps performed)

10000/10000 [=====] - 525s 52ms/step - reward: -6.0000e-04

2 episodes - episode_reward: 0.500 [0.000, 1.000] - loss: 0.009 - mae: 0.125 - mean_q: 0.145 - mean_eps: 0.100 - ale.lives: 0.000

Interval 58 (570000 steps performed)

10000/10000 [=====] - 548s 55ms/step - reward: 0.0013

2 episodes - episode_reward: 5.000 [0.000, 10.000] - loss: 0.009 - mae: 0.139 - mean_q: 0.161 - mean_eps: 0.100 - ale.lives: 0.000

Interval 59 (580000 steps performed)
10000/10000 [=====] - 541s 54ms/step - reward: 0.0020
2 episodes - episode_reward: 8.000 [2.000, 14.000] - loss: 0.009 - mae: 0.136 - mean_q: 0.157 - mean_eps: 0.100 - ale.lives: 0.000

Interval 60 (590000 steps performed)
10000/10000 [=====] - 551s 55ms/step - reward: 5.0000e-04
3 episodes - episode_reward: 5.333 [0.000, 11.000] - loss: 0.009 - mae: 0.139 - mean_q: 0.161 - mean_eps: 0.100 - ale.lives: 0.000

Interval 61 (600000 steps performed)
10000/10000 [=====] - 610s 61ms/step - reward: 0.0024
2 episodes - episode_reward: 9.500 [5.000, 14.000] - loss: 0.009 - mae: 0.147 - mean_q: 0.173 - mean_eps: 0.100 - ale.lives: 0.000

Interval 62 (610000 steps performed)
10000/10000 [=====] - 658s 66ms/step - reward: 0.0000e+00
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.009 - mae: 0.146 - mean_q: 0.170 - mean_eps: 0.100 - ale.lives: 0.000

Interval 63 (620000 steps performed)
10000/10000 [=====] - 668s 67ms/step - reward: 1.0000e-03
2 episodes - episode_reward: 7.000 [1.000, 13.000] - loss: 0.009 - mae: 0.153 - mean_q: 0.177 - mean_eps: 0.100 - ale.lives: 0.000

Interval 64 (630000 steps performed)
10000/10000 [=====] - 639s 64ms/step - reward: 9.0000e-04
3 episodes - episode_reward: 2.667 [0.000, 6.000] - loss: 0.010 - mae: 0.157 - mean_q: 0.182 - mean_eps: 0.100 - ale.lives: 0.000

Interval 65 (640000 steps performed)
10000/10000 [=====] - 561s 56ms/step - reward: 0.0011
2 episodes - episode_reward: 1.500 [0.000, 3.000] - loss: 0.009 - mae: 0.159 - mean_q: 0.184 - mean_eps: 0.100 - ale.lives: 0.000

Interval 66 (650000 steps performed)
10000/10000 [=====] - 545s 55ms/step - reward: 0.0012
2 episodes - episode_reward: 11.500 [2.000, 21.000] - loss: 0.010 - mae: 0.167 - mean_q: 0.192 - mean_eps: 0.100 - ale.lives: 0.000

Interval 67 (660000 steps performed)

10000/10000 [=====] - 542s 54ms/step - reward: 0.0020

2 episodes - episode_reward: 10.000 [8.000, 12.000] - loss: 0.010 - mae: 0.158 - mean_q: 0.183 - mean_eps: 0.100 - ale.lives: 0.000

Interval 68 (670000 steps performed)

10000/10000 [=====] - 537s 54ms/step - reward: 6.0000e-04

3 episodes - episode_reward: 1.333 [0.000, 3.000] - loss: 0.010 - mae: 0.157 - mean_q: 0.181 - mean_eps: 0.100 - ale.lives: 0.000

Interval 69 (680000 steps performed)

10000/10000 [=====] - 539s 54ms/step - reward: 0.0045

2 episodes - episode_reward: 18.500 [14.000, 23.000] - loss: 0.010 - mae: 0.157 - mean_q: 0.181 - mean_eps: 0.100 - ale.lives: 0.000

Interval 70 (690000 steps performed)

10000/10000 [=====] - 544s 54ms/step - reward: -3.0000e-04

2 episodes - episode_reward: 3.500 [3.000, 4.000] - loss: 0.010 - mae: 0.161 - mean_q: 0.187 - mean_eps: 0.100 - ale.lives: 0.000

Interval 71 (700000 steps performed)

10000/10000 [=====] - 666s 67ms/step - reward: 2.0000e-04

3 episodes - episode_reward: 0.667 [0.000, 1.000] - loss: 0.010 - mae: 0.159 - mean_q: 0.183 - mean_eps: 0.100 - ale.lives: 0.000

Interval 72 (710000 steps performed)

10000/10000 [=====] - 688s 69ms/step - reward: 9.0000e-04

2 episodes - episode_reward: 4.500 [3.000, 6.000] - loss: 0.011 - mae: 0.168 - mean_q: 0.194 - mean_eps: 0.100 - ale.lives: 0.000

Interval 73 (720000 steps performed)

10000/10000 [=====] - 639s 64ms/step - reward: 0.0011

2 episodes - episode_reward: 2.000 [0.000, 4.000] - loss: 0.011 - mae: 0.162 - mean_q: 0.187 - mean_eps: 0.100 - ale.lives: 0.000

Interval 74 (730000 steps performed)

10000/10000 [=====] - 568s 57ms/step - reward: 7.0000e-04

2 episodes - episode_reward: 2.000 [0.000, 4.000] - loss: 0.010 - mae: 0.162 - mean_q: 0.185 - mean_eps: 0.100 - ale.lives: 0.000

Interval 75 (740000 steps performed)

10000/10000 [=====] - 686s 69ms/step - reward: 8.0000e-04
3 episodes - episode_reward: 6.000 [1.000, 12.000] - loss: 0.011 - mae: 0.165 - mean_q: 0.192 - mean_eps: 0.100 - ale.lives: 0.000

Interval 76 (750000 steps performed)
10000/10000 [=====] - 614s 61ms/step - reward: 0.0019
2 episodes - episode_reward: 6.500 [5.000, 8.000] - loss: 0.011 - mae: 0.171 - mean_q: 0.197 - mean_eps: 0.100 - ale.lives: 0.000

Interval 77 (760000 steps performed)
10000/10000 [=====] - 595s 60ms/step - reward: 0.0011
2 episodes - episode_reward: 3.500 [0.000, 7.000] - loss: 0.011 - mae: 0.178 - mean_q: 0.207 - mean_eps: 0.100 - ale.lives: 0.000

Interval 78 (770000 steps performed)
10000/10000 [=====] - 611s 61ms/step - reward: 1.0000e-04
2 episodes - episode_reward: 1.500 [0.000, 3.000] - loss: 0.011 - mae: 0.181 - mean_q: 0.206 - mean_eps: 0.100 - ale.lives: 0.000

Interval 79 (780000 steps performed)
10000/10000 [=====] - 705s 70ms/step - reward: 0.0025
3 episodes - episode_reward: 10.000 [5.000, 15.000] - loss: 0.011 - mae: 0.177 - mean_q: 0.205 - mean_eps: 0.100 - ale.lives: 0.000

Interval 80 (790000 steps performed)
10000/10000 [=====] - 631s 63ms/step - reward: 0.0021
2 episodes - episode_reward: 12.000 [9.000, 15.000] - loss: 0.011 - mae: 0.178 - mean_q: 0.206 - mean_eps: 0.100 - ale.lives: 0.000

Interval 81 (800000 steps performed)
10000/10000 [=====] - 678s 68ms/step - reward: 0.0015
2 episodes - episode_reward: 2.500 [1.000, 4.000] - loss: 0.011 - mae: 0.185 - mean_q: 0.213 - mean_eps: 0.100 - ale.lives: 0.000

Interval 82 (810000 steps performed)
10000/10000 [=====] - 366s 37ms/step - reward: 0.0013
2 episodes - episode_reward: 4.000 [0.000, 8.000] - loss: 0.011 - mae: 0.186 - mean_q: 0.213 - mean_eps: 0.100 - ale.lives: 0.000

Interval 83 (820000 steps performed)
10000/10000 [=====] - 381s 38ms/step - reward: 0.0038

3 episodes - episode_reward: 17.000 [9.000, 25.000] - loss: 0.011 - mae: 0.188 - mean_q: 0.217 - mean_eps: 0.100 - ale.lives: 0.000

Interval 84 (830000 steps performed)

10000/10000 [=====] - 397s 40ms/step - reward: 0.0025

2 episodes - episode_reward: 10.500 [0.000, 21.000] - loss: 0.011 - mae: 0.193 - mean_q: 0.223 - mean_eps: 0.100 - ale.lives: 0.000

Interval 85 (840000 steps performed)

10000/10000 [=====] - 354s 35ms/step - reward: 0.0041

2 episodes - episode_reward: 13.000 [6.000, 20.000] - loss: 0.011 - mae: 0.189 - mean_q: 0.217 - mean_eps: 0.100 - ale.lives: 0.000

Interval 86 (850000 steps performed)

10000/10000 [=====] - 352s 35ms/step - reward: 0.0018

2 episodes - episode_reward: 12.500 [7.000, 18.000] - loss: 0.011 - mae: 0.190 - mean_q: 0.219 - mean_eps: 0.100 - ale.lives: 0.000

Interval 87 (860000 steps performed)

10000/10000 [=====] - 373s 37ms/step - reward: 6.0000e-04

3 episodes - episode_reward: 6.333 [2.000, 13.000] - loss: 0.011 - mae: 0.194 - mean_q: 0.224 - mean_eps: 0.100 - ale.lives: 0.000

Interval 88 (870000 steps performed)

10000/10000 [=====] - 360s 36ms/step - reward: 3.0000e-04

2 episodes - episode_reward: 2.000 [2.000, 2.000] - loss: 0.011 - mae: 0.205 - mean_q: 0.236 - mean_eps: 0.100 - ale.lives: 0.000

Interval 89 (880000 steps performed)

10000/10000 [=====] - 325s 33ms/step - reward: 0.0049

2 episodes - episode_reward: 22.000 [3.000, 41.000] - loss: 0.012 - mae: 0.211 - mean_q: 0.242 - mean_eps: 0.100 - ale.lives: 0.000

Interval 90 (890000 steps performed)

10000/10000 [=====] - 349s 35ms/step - reward: 7.0000e-04

2 episodes - episode_reward: 5.500 [1.000, 10.000] - loss: 0.012 - mae: 0.220 - mean_q: 0.254 - mean_eps: 0.100 - ale.lives: 0.000

Interval 91 (900000 steps performed)

10000/10000 [=====] - 475s 47ms/step - reward: 0.0016

3 episodes - episode_reward: 4.333 [0.000, 12.000] - loss: 0.012 - mae: 0.222 - mean_q: 0.255 - mean_eps: 0.100 - ale.lives: 0.000

000

Interval 92 (910000 steps performed)

10000/10000 [=====] - 507s 51ms/step - reward: 8.0000e-04

2 episodes - episode_reward: 4.000 [0.000, 8.000] - loss: 0.012 - mae: 0.227 - mean_q: 0.262 - mean_eps: 0.100 - ale.lives: 0.000

Interval 93 (920000 steps performed)

10000/10000 [=====] - 462s 46ms/step - reward: 0.0015

2 episodes - episode_reward: 7.500 [6.000, 9.000] - loss: 0.012 - mae: 0.232 - mean_q: 0.266 - mean_eps: 0.100 - ale.lives: 0.000

Interval 94 (930000 steps performed)

10000/10000 [=====] - 367s 37ms/step - reward: 1.0000e-04

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.012 - mae: 0.232 - mean_q: 0.266 - mean_eps: 0.100 - ale.lives: 0.000

Interval 95 (940000 steps performed)

10000/10000 [=====] - 375s 37ms/step - reward: 0.0060

3 episodes - episode_reward: 21.333 [9.000, 42.000] - loss: 0.012 - mae: 0.234 - mean_q: 0.269 - mean_eps: 0.100 - ale.lives: 0.000

Interval 96 (950000 steps performed)

10000/10000 [=====] - 378s 38ms/step - reward: 0.0015

2 episodes - episode_reward: 4.500 [3.000, 6.000] - loss: 0.012 - mae: 0.240 - mean_q: 0.276 - mean_eps: 0.100 - ale.lives: 0.000

Interval 97 (960000 steps performed)

10000/10000 [=====] - 376s 38ms/step - reward: 4.0000e-04

2 episodes - episode_reward: 1.000 [0.000, 2.000] - loss: 0.012 - mae: 0.247 - mean_q: 0.284 - mean_eps: 0.100 - ale.lives: 0.000

Interval 98 (970000 steps performed)

10000/10000 [=====] - 476s 48ms/step - reward: 0.0019

2 episodes - episode_reward: 13.500 [7.000, 20.000] - loss: 0.013 - mae: 0.261 - mean_q: 0.300 - mean_eps: 0.100 - ale.lives: 0.000

Interval 99 (980000 steps performed)

10000/10000 [=====] - 627s 63ms/step - reward: 0.0030

3 episodes - episode_reward: 10.333 [2.000, 24.000] - loss: 0.012 - mae: 0.265 - mean_q: 0.304 - mean_eps: 0.100 - ale.lives: 0.000

Interval 100 (990000 steps performed)
10000/10000 [=====] - 527s 53ms/step - reward: 0.0016
2 episodes - episode_reward: 7.500 [5.000, 10.000] - loss: 0.012 - mae: 0.260 - mean_q: 0.299 - mean_eps: 0.100 - ale.lives: 0.000

Interval 101 (1000000 steps performed)
10000/10000 [=====] - 608s 61ms/step - reward: 0.0032
2 episodes - episode_reward: 4.000 [0.000, 8.000] - loss: 0.013 - mae: 0.258 - mean_q: 0.296 - mean_eps: 0.100 - ale.lives: 0.000

Interval 102 (1010000 steps performed)
10000/10000 [=====] - 665s 66ms/step - reward: -0.0015
2 episodes - episode_reward: 3.000 [2.000, 4.000] - loss: 0.012 - mae: 0.258 - mean_q: 0.296 - mean_eps: 0.100 - ale.lives: 0.000

Interval 103 (1020000 steps performed)
10000/10000 [=====] - 680s 68ms/step - reward: 0.0018
3 episodes - episode_reward: 5.333 [0.000, 15.000] - loss: 0.013 - mae: 0.257 - mean_q: 0.295 - mean_eps: 0.100 - ale.lives: 0.000

Interval 104 (1030000 steps performed)
10000/10000 [=====] - 618s 62ms/step - reward: 0.0011
2 episodes - episode_reward: 2.500 [0.000, 5.000] - loss: 0.013 - mae: 0.264 - mean_q: 0.303 - mean_eps: 0.100 - ale.lives: 0.000

Interval 105 (1040000 steps performed)
10000/10000 [=====] - 501s 50ms/step - reward: -1.0000e-03
2 episodes - episode_reward: 1.000 [0.000, 2.000] - loss: 0.012 - mae: 0.264 - mean_q: 0.302 - mean_eps: 0.100 - ale.lives: 0.000

Interval 106 (1050000 steps performed)
10000/10000 [=====] - 550s 55ms/step - reward: 0.0016
2 episodes - episode_reward: 6.000 [0.000, 12.000] - loss: 0.012 - mae: 0.252 - mean_q: 0.289 - mean_eps: 0.100 - ale.lives: 0.000

Interval 107 (1060000 steps performed)
10000/10000 [=====] - 668s 67ms/step - reward: 0.0070
3 episodes - episode_reward: 22.667 [10.000, 29.000] - loss: 0.013 - mae: 0.250 - mean_q: 0.289 - mean_eps: 0.100 - ale.lives: 0.000

Interval 108 (1070000 steps performed)

10000/10000 [=====] - 517s 52ms/step - reward: 0.0052

2 episodes - episode_reward: 21.000 [2.000, 40.000] - loss: 0.013 - mae: 0.248 - mean_q: 0.285 - mean_eps: 0.100 - ale.lives: 0.000

Interval 109 (1080000 steps performed)

10000/10000 [=====] - 504s 50ms/step - reward: 0.0021

2 episodes - episode_reward: 16.500 [4.000, 29.000] - loss: 0.013 - mae: 0.250 - mean_q: 0.287 - mean_eps: 0.100 - ale.lives: 0.000

Interval 110 (1090000 steps performed)

10000/10000 [=====] - 601s 60ms/step - reward: 0.0050

2 episodes - episode_reward: 17.000 [2.000, 32.000] - loss: 0.013 - mae: 0.247 - mean_q: 0.284 - mean_eps: 0.100 - ale.lives: 0.000

Interval 111 (1100000 steps performed)

10000/10000 [=====] - 709s 71ms/step - reward: 6.0000e-04

3 episodes - episode_reward: 7.333 [1.000, 16.000] - loss: 0.014 - mae: 0.240 - mean_q: 0.276 - mean_eps: 0.100 - ale.lives: 0.000

Interval 112 (1110000 steps performed)

10000/10000 [=====] - 645s 65ms/step - reward: 1.0000e-03

2 episodes - episode_reward: 7.000 [2.000, 12.000] - loss: 0.014 - mae: 0.238 - mean_q: 0.276 - mean_eps: 0.100 - ale.lives: 0.000

Interval 113 (1120000 steps performed)

10000/10000 [=====] - 531s 53ms/step - reward: 0.0052

2 episodes - episode_reward: 11.500 [6.000, 17.000] - loss: 0.015 - mae: 0.228 - mean_q: 0.264 - mean_eps: 0.100 - ale.lives: 0.000

Interval 114 (1130000 steps performed)

10000/10000 [=====] - 457s 46ms/step - reward: -9.0000e-04

2 episodes - episode_reward: 8.000 [1.000, 15.000] - loss: 0.014 - mae: 0.232 - mean_q: 0.268 - mean_eps: 0.100 - ale.lives: 0.000

Interval 115 (1140000 steps performed)

10000/10000 [=====] - 407s 41ms/step - reward: 0.0011

3 episodes - episode_reward: 4.667 [0.000, 12.000] - loss: 0.014 - mae: 0.238 - mean_q: 0.274 - mean_eps: 0.100 - ale.lives: 0.000

Interval 116 (1150000 steps performed)

10000/10000 [=====] - 406s 41ms/step - reward: 0.0014
2 episodes - episode_reward: 7.500 [1.000, 14.000] - loss: 0.015 - mae: 0.252 - mean_q: 0.292 - mean_eps: 0.100 - ale.lives: 0.000

Interval 117 (1160000 steps performed)

10000/10000 [=====] - 386s 39ms/step - reward: 0.0026
2 episodes - episode_reward: 5.500 [5.000, 6.000] - loss: 0.015 - mae: 0.250 - mean_q: 0.288 - mean_eps: 0.100 - ale.lives: 0.000

Interval 118 (1170000 steps performed)

10000/10000 [=====] - 348s 35ms/step - reward: 0.0035
2 episodes - episode_reward: 21.500 [20.000, 23.000] - loss: 0.015 - mae: 0.251 - mean_q: 0.291 - mean_eps: 0.100 - ale.lives: 0.000

Interval 119 (1180000 steps performed)

10000/10000 [=====] - 394s 39ms/step - reward: 0.0014
3 episodes - episode_reward: 6.000 [3.000, 9.000] - loss: 0.016 - mae: 0.254 - mean_q: 0.293 - mean_eps: 0.100 - ale.lives: 0.000

Interval 120 (1190000 steps performed)

10000/10000 [=====] - 354s 35ms/step - reward: 0.0014
2 episodes - episode_reward: 6.000 [6.000, 6.000] - loss: 0.016 - mae: 0.258 - mean_q: 0.298 - mean_eps: 0.100 - ale.lives: 0.000

Interval 121 (1200000 steps performed)

10000/10000 [=====] - 356s 36ms/step - reward: -3.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.016 - mae: 0.262 - mean_q: 0.302 - mean_eps: 0.100 - ale.lives: 0.000

Interval 122 (1210000 steps performed)

10000/10000 [=====] - 347s 35ms/step - reward: 0.0011
2 episodes - episode_reward: 6.500 [1.000, 12.000] - loss: 0.015 - mae: 0.266 - mean_q: 0.307 - mean_eps: 0.100 - ale.lives: 0.000

Interval 123 (1220000 steps performed)

10000/10000 [=====] - 350s 35ms/step - reward: 0.0037
3 episodes - episode_reward: 10.000 [2.000, 24.000] - loss: 0.016 - mae: 0.259 - mean_q: 0.299 - mean_eps: 0.100 - ale.lives: 0.000

Interval 124 (1230000 steps performed)

10000/10000 [=====] - 340s 34ms/step - reward: 1.0000e-04

2 episodes - episode_reward: 1.000 [0.000, 2.000] - loss: 0.016 - mae: 0.260 - mean_q: 0.300 - mean_eps: 0.100 - ale.lives: 0.000

Interval 125 (1240000 steps performed)

10000/10000 [=====] - 425s 42ms/step - reward: 0.0021

2 episodes - episode_reward: 13.500 [9.000, 18.000] - loss: 0.017 - mae: 0.247 - mean_q: 0.287 - mean_eps: 0.100 - ale.lives: 0.000

Interval 126 (1250000 steps performed)

10000/10000 [=====] - 417s 42ms/step - reward: 0.0044

2 episodes - episode_reward: 20.000 [14.000, 26.000] - loss: 0.016 - mae: 0.241 - mean_q: 0.279 - mean_eps: 0.100 - ale.lives: 0.000

Interval 127 (1260000 steps performed)

10000/10000 [=====] - 415s 41ms/step - reward: 0.0011

3 episodes - episode_reward: 3.667 [2.000, 6.000] - loss: 0.016 - mae: 0.248 - mean_q: 0.288 - mean_eps: 0.100 - ale.lives: 0.000

Interval 128 (1270000 steps performed)

10000/10000 [=====] - 413s 41ms/step - reward: 0.0041

2 episodes - episode_reward: 15.000 [3.000, 27.000] - loss: 0.017 - mae: 0.260 - mean_q: 0.302 - mean_eps: 0.100 - ale.lives: 0.000

Interval 129 (1280000 steps performed)

10000/10000 [=====] - 415s 41ms/step - reward: 0.0023

2 episodes - episode_reward: 8.000 [1.000, 15.000] - loss: 0.017 - mae: 0.265 - mean_q: 0.308 - mean_eps: 0.100 - ale.lives: 0.000

Interval 130 (1290000 steps performed)

10000/10000 [=====] - 423s 42ms/step - reward: -7.0000e-04

2 episodes - episode_reward: 5.000 [1.000, 9.000] - loss: 0.017 - mae: 0.271 - mean_q: 0.313 - mean_eps: 0.100 - ale.lives: 0.000

Interval 131 (1300000 steps performed)

10000/10000 [=====] - 419s 42ms/step - reward: 0.0027

3 episodes - episode_reward: 6.000 [0.000, 15.000] - loss: 0.017 - mae: 0.263 - mean_q: 0.307 - mean_eps: 0.100 - ale.lives: 0.000

Interval 132 (1310000 steps performed)

10000/10000 [=====] - 422s 42ms/step - reward: 0.0040

2 episodes - episode_reward: 23.500 [19.000, 28.000] - loss: 0.017 - mae: 0.266 - mean_q: 0.310 - mean_eps: 0.100 - ale.lives:

0.000

Interval 133 (1320000 steps performed)

10000/10000 [=====] - 424s 42ms/step - reward: 0.0039

2 episodes - episode_reward: 13.000 [2.000, 24.000] - loss: 0.017 - mae: 0.266 - mean_q: 0.309 - mean_eps: 0.100 - ale.lives: 0.000

Interval 134 (1330000 steps performed)

10000/10000 [=====] - 426s 43ms/step - reward: -8.0000e-04

2 episodes - episode_reward: 6.000 [3.000, 9.000] - loss: 0.017 - mae: 0.266 - mean_q: 0.308 - mean_eps: 0.100 - ale.lives: 0.000

Interval 135 (1340000 steps performed)

10000/10000 [=====] - 411s 41ms/step - reward: 0.0037

3 episodes - episode_reward: 2.000 [0.000, 4.000] - loss: 0.017 - mae: 0.256 - mean_q: 0.298 - mean_eps: 0.100 - ale.lives: 0.000

Interval 136 (1350000 steps performed)

10000/10000 [=====] - 416s 42ms/step - reward: 0.0025

2 episodes - episode_reward: 28.000 [24.000, 32.000] - loss: 0.017 - mae: 0.261 - mean_q: 0.305 - mean_eps: 0.100 - ale.lives: 0.000

Interval 137 (1360000 steps performed)

10000/10000 [=====] - 433s 43ms/step - reward: 0.0024

2 episodes - episode_reward: 12.000 [6.000, 18.000] - loss: 0.016 - mae: 0.247 - mean_q: 0.286 - mean_eps: 0.100 - ale.lives: 0.000

Interval 138 (1370000 steps performed)

10000/10000 [=====] - 422s 42ms/step - reward: 4.0000e-04

2 episodes - episode_reward: 2.000 [1.000, 3.000] - loss: 0.017 - mae: 0.244 - mean_q: 0.281 - mean_eps: 0.100 - ale.lives: 0.000

Interval 139 (1380000 steps performed)

10000/10000 [=====] - 408s 41ms/step - reward: 0.0016

3 episodes - episode_reward: 5.333 [0.000, 11.000] - loss: 0.017 - mae: 0.240 - mean_q: 0.278 - mean_eps: 0.100 - ale.lives: 0.000

Interval 140 (1390000 steps performed)

10000/10000 [=====] - 357s 36ms/step - reward: 0.0020

2 episodes - episode_reward: 4.500 [4.000, 5.000] - loss: 0.017 - mae: 0.233 - mean_q: 0.269 - mean_eps: 0.100 - ale.lives: 0.000

Interval 141 (1400000 steps performed)

10000/10000 [=====] - 354s 35ms/step - reward: 0.0030

2 episodes - episode_reward: 15.500 [0.000, 31.000] - loss: 0.017 - mae: 0.233 - mean_q: 0.271 - mean_eps: 0.100 - ale.lives: 0.000

Interval 142 (1410000 steps performed)

10000/10000 [=====] - 360s 36ms/step - reward: 0.0019

3 episodes - episode_reward: 9.333 [4.000, 14.000] - loss: 0.017 - mae: 0.234 - mean_q: 0.272 - mean_eps: 0.100 - ale.lives: 0.000

Interval 143 (1420000 steps performed)

10000/10000 [=====] - 345s 35ms/step - reward: 0.0020

2 episodes - episode_reward: 8.500 [5.000, 12.000] - loss: 0.017 - mae: 0.241 - mean_q: 0.280 - mean_eps: 0.100 - ale.lives: 0.000

Interval 144 (1430000 steps performed)

10000/10000 [=====] - 345s 34ms/step - reward: 0.0011

2 episodes - episode_reward: 2.000 [0.000, 4.000] - loss: 0.016 - mae: 0.243 - mean_q: 0.284 - mean_eps: 0.100 - ale.lives: 0.000

Interval 145 (1440000 steps performed)

10000/10000 [=====] - 354s 35ms/step - reward: 0.0043

2 episodes - episode_reward: 20.500 [0.000, 41.000] - loss: 0.017 - mae: 0.248 - mean_q: 0.286 - mean_eps: 0.100 - ale.lives: 0.000

Interval 146 (1450000 steps performed)

10000/10000 [=====] - 342s 34ms/step - reward: 0.0013

3 episodes - episode_reward: 8.667 [0.000, 23.000] - loss: 0.016 - mae: 0.243 - mean_q: 0.280 - mean_eps: 0.100 - ale.lives: 0.000

Interval 147 (1460000 steps performed)

10000/10000 [=====] - 350s 35ms/step - reward: 0.0016

2 episodes - episode_reward: 5.500 [0.000, 11.000] - loss: 0.017 - mae: 0.239 - mean_q: 0.279 - mean_eps: 0.100 - ale.lives: 0.000

Interval 148 (1470000 steps performed)

10000/10000 [=====] - 342s 34ms/step - reward: 6.0000e-04

2 episodes - episode_reward: 5.500 [2.000, 9.000] - loss: 0.017 - mae: 0.224 - mean_q: 0.259 - mean_eps: 0.100 - ale.lives: 0.000

Interval 149 (1480000 steps performed)

10000/10000 [=====] - 357s 36ms/step - reward: 0.0047

2 episodes - episode_reward: 14.500 [8.000, 21.000] - loss: 0.017 - mae: 0.217 - mean_q: 0.250 - mean_eps: 0.100 - ale.lives: 0.000

Interval 150 (1490000 steps performed)

10000/10000 [=====] - 406s 41ms/step - reward: 0.0015

3 episodes - episode_reward: 10.000 [2.000, 17.000] - loss: 0.017 - mae: 0.225 - mean_q: 0.261 - mean_eps: 0.100 - ale.lives: 0.000

Interval 151 (1500000 steps performed)

10000/10000 [=====] - 415s 42ms/step - reward: 0.0018

2 episodes - episode_reward: 8.000 [0.000, 16.000] - loss: 0.017 - mae: 0.224 - mean_q: 0.261 - mean_eps: 0.100 - ale.lives: 0.000

Interval 152 (1510000 steps performed)

10000/10000 [=====] - 423s 42ms/step - reward: 0.0022

2 episodes - episode_reward: 11.500 [0.000, 23.000] - loss: 0.017 - mae: 0.222 - mean_q: 0.259 - mean_eps: 0.100 - ale.lives: 0.000

Interval 153 (1520000 steps performed)

10000/10000 [=====] - 408s 41ms/step - reward: 0.0013

2 episodes - episode_reward: 8.500 [7.000, 10.000] - loss: 0.017 - mae: 0.231 - mean_q: 0.266 - mean_eps: 0.100 - ale.lives: 0.000

Interval 154 (1530000 steps performed)

10000/10000 [=====] - 413s 41ms/step - reward: 9.0000e-04

3 episodes - episode_reward: 2.000 [0.000, 6.000] - loss: 0.017 - mae: 0.232 - mean_q: 0.269 - mean_eps: 0.100 - ale.lives: 0.000

Interval 155 (1540000 steps performed)

10000/10000 [=====] - 422s 42ms/step - reward: 0.0030

2 episodes - episode_reward: 9.000 [1.000, 17.000] - loss: 0.017 - mae: 0.232 - mean_q: 0.267 - mean_eps: 0.100 - ale.lives: 0.000

Interval 156 (1550000 steps performed)

10000/10000 [=====] - 419s 42ms/step - reward: 7.0000e-04

2 episodes - episode_reward: 2.500 [1.000, 4.000] - loss: 0.017 - mae: 0.222 - mean_q: 0.256 - mean_eps: 0.100 - ale.lives: 0.000

Interval 157 (1560000 steps performed)

10000/10000 [=====] - 425s 42ms/step - reward: 0.0017
2 episodes - episode_reward: 14.000 [7.000, 21.000] - loss: 0.017 - mae: 0.224 - mean_q: 0.260 - mean_eps: 0.100 - ale.lives:
0.000

Interval 158 (1570000 steps performed)
10000/10000 [=====] - 429s 43ms/step - reward: 0.0035
3 episodes - episode_reward: 12.333 [8.000, 16.000] - loss: 0.017 - mae: 0.231 - mean_q: 0.268 - mean_eps: 0.100 - ale.lives:
0.000

Interval 159 (1580000 steps performed)
10000/10000 [=====] - 425s 42ms/step - reward: 0.0048
2 episodes - episode_reward: 22.000 [16.000, 28.000] - loss: 0.017 - mae: 0.241 - mean_q: 0.280 - mean_eps: 0.100 - ale.lives:
0.000

Interval 160 (1590000 steps performed)
10000/10000 [=====] - 420s 42ms/step - reward: 0.0041
2 episodes - episode_reward: 19.500 [1.000, 38.000] - loss: 0.017 - mae: 0.241 - mean_q: 0.280 - mean_eps: 0.100 - ale.lives:
0.000

Interval 161 (1600000 steps performed)
10000/10000 [=====] - 421s 42ms/step - reward: 0.0072
2 episodes - episode_reward: 16.500 [0.000, 33.000] - loss: 0.016 - mae: 0.239 - mean_q: 0.277 - mean_eps: 0.100 - ale.lives:
0.000

Interval 162 (1610000 steps performed)
10000/10000 [=====] - 418s 42ms/step - reward: 1.0000e-04
3 episodes - episode_reward: 15.333 [0.000, 45.000] - loss: 0.017 - mae: 0.240 - mean_q: 0.276 - mean_eps: 0.100 - ale.lives:
0.000

Interval 163 (1620000 steps performed)
10000/10000 [=====] - 421s 42ms/step - reward: 0.0063
2 episodes - episode_reward: 29.000 [26.000, 32.000] - loss: 0.017 - mae: 0.234 - mean_q: 0.272 - mean_eps: 0.100 - ale.lives:
0.000

Interval 164 (1630000 steps performed)
10000/10000 [=====] - 363s 36ms/step - reward: 0.0037
2 episodes - episode_reward: 23.000 [14.000, 32.000] - loss: 0.017 - mae: 0.237 - mean_q: 0.273 - mean_eps: 0.100 - ale.lives:
0.000

Interval 165 (1640000 steps performed)
10000/10000 [=====] - 360s 36ms/step - reward: 0.0037

2 episodes - episode_reward: 18.000 [7.000, 29.000] - loss: 0.017 - mae: 0.247 - mean_q: 0.286 - mean_eps: 0.100 - ale.lives: 0.000

Interval 166 (1650000 steps performed)

10000/10000 [=====] - 343s 34ms/step - reward: 0.0012

3 episodes - episode_reward: 3.667 [0.000, 8.000] - loss: 0.017 - mae: 0.252 - mean_q: 0.291 - mean_eps: 0.100 - ale.lives: 0.000

Interval 167 (1660000 steps performed)

10000/10000 [=====] - 420s 42ms/step - reward: 0.0055

2 episodes - episode_reward: 20.500 [8.000, 33.000] - loss: 0.017 - mae: 0.258 - mean_q: 0.297 - mean_eps: 0.100 - ale.lives: 0.000

Interval 168 (1670000 steps performed)

10000/10000 [=====] - 1057s 106ms/step - reward: -0.0011

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.018 - mae: 0.258 - mean_q: 0.298 - mean_eps: 0.100 - ale.lives: 0.000

Interval 169 (1680000 steps performed)

10000/10000 [=====] - 1201s 120ms/step - reward: 0.0013

2 episodes - episode_reward: 8.500 [8.000, 9.000] - loss: 0.018 - mae: 0.259 - mean_q: 0.301 - mean_eps: 0.100 - ale.lives: 0.000

Interval 170 (1690000 steps performed)

10000/10000 [=====] - 382s 38ms/step - reward: 0.0027

3 episodes - episode_reward: 9.000 [1.000, 22.000] - loss: 0.017 - mae: 0.263 - mean_q: 0.305 - mean_eps: 0.100 - ale.lives: 0.000

Interval 171 (1700000 steps performed)

10000/10000 [=====] - 426s 43ms/step - reward: 0.0034

2 episodes - episode_reward: 15.000 [14.000, 16.000] - loss: 0.017 - mae: 0.272 - mean_q: 0.315 - mean_eps: 0.100 - ale.lives: 0.000

Interval 172 (1710000 steps performed)

10000/10000 [=====] - 368s 37ms/step - reward: -1.0000e-04

2 episodes - episode_reward: 0.500 [0.000, 1.000] - loss: 0.017 - mae: 0.278 - mean_q: 0.321 - mean_eps: 0.100 - ale.lives: 0.000

Interval 173 (1720000 steps performed)

10000/10000 [=====] - 372s 37ms/step - reward: 0.0024

2 episodes - episode_reward: 13.000 [10.000, 16.000] - loss: 0.017 - mae: 0.283 - mean_q: 0.326 - mean_eps: 0.100 - ale.lives:

0.000

Interval 174 (1730000 steps performed)

10000/10000 [=====] - 383s 38ms/step - reward: 0.0017

3 episodes - episode_reward: 4.333 [0.000, 11.000] - loss: 0.017 - mae: 0.285 - mean_q: 0.328 - mean_eps: 0.100 - ale.lives: 0.000

Interval 175 (1740000 steps performed)

10000/10000 [=====] - 412s 41ms/step - reward: 1.0000e-03

done, took 83787.395 seconds

Dueling DQN

```
In [ ]: train_agent('dueling_dqn')
```

==== Entrenando: dueling_dqn a las 20260728_142544 ====

Model: "model"

Layer (type)	Output Shape	Param #	Connected to
=====			
input_1 (InputLayer)	[(None, 4, 84, 84)]	0	
permute (Permute)	(None, 84, 84, 4)	0	input_1[0][0]
conv2d (Conv2D)	(None, 20, 20, 32)	8224	permute[0][0]
conv2d_1 (Conv2D)	(None, 9, 9, 64)	32832	conv2d[0][0]
conv2d_2 (Conv2D)	(None, 7, 7, 64)	36928	conv2d_1[0][0]
flatten (Flatten)	(None, 3136)	0	conv2d_2[0][0]
dense (Dense)	(None, 512)	1606144	flatten[0][0]
dense_1 (Dense)	(None, 1)	513	dense[0][0]
dense_2 (Dense)	(None, 9)	4617	dense[0][0]
dueling_combine (DuelingCombine)	(None, 9)	0	dense_1[0][0] dense_2[0][0]
=====			
Total params: 1,689,258			
Trainable params: 1,689,258			
Non-trainable params: 0			

None

Training for 1750000 steps ...

Interval 1 (0 steps performed)

12/10000 [.....] - ETA: 48s - reward: 0.0000e+00

c:\Users\rosai\OneDrive\Documentos\Learning\VIU - Master IA\08MIAR - Aprendizaje por Refuerzo\Proyecto\gym_env\lib\site-packages\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.

warnings.warn("`Model.state\_updates` will be removed in a future version. ")

10000/10000 [=====] - 43s 4ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - ale.lives: 0.000

Interval 2 (10000 steps performed)

10000/10000 [=====] - 253s 25ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.030 - mean_q: -0.022 - mean_eps: 0.946 - ale.lives: 0.000

Interval 3 (20000 steps performed)

10000/10000 [=====] - 257s 26ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.028 - mean_q: -0.019 - mean_eps: 0.910 - ale.lives: 0.000

Interval 4 (30000 steps performed)

10000/10000 [=====] - 303s 30ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.026 - mean_q: -0.016 - mean_eps: 0.874 - ale.lives: 0.000

Interval 5 (40000 steps performed)

10000/10000 [=====] - 266s 27ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.024 - mean_q: -0.014 - mean_eps: 0.838 - ale.lives: 0.000

Interval 6 (50000 steps performed)

10000/10000 [=====] - 262s 26ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.024 - mean_q: -0.013 - mean_eps: 0.802 - ale.lives: 0.000

Interval 7 (60000 steps performed)

10000/10000 [=====] - 258s 26ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.023 - mean_q: -0.012 - mean_eps: 0.766 - ale.lives: 0.000

Interval 8 (70000 steps performed)

10000/10000 [=====] - 262s 26ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.022 - mean_q: -0.010 - mean_eps: 0.730 - ale.lives: 0.000

Interval 9 (80000 steps performed)

10000/10000 [=====] - 267s 27ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.020 - mean_q: -0.009 - mean_eps: 0.694 - ale.lives: 0.

000

Interval 10 (90000 steps performed)

10000/10000 [=====] - 262s 26ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.020 - mean_q: -0.008 - mean_eps: 0.658 - ale.lives: 0.000

Interval 11 (100000 steps performed)

10000/10000 [=====] - 317s 32ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.019 - mean_q: -0.004 - mean_eps: 0.622 - ale.lives: 0.000

Interval 12 (110000 steps performed)

10000/10000 [=====] - 323s 32ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.019 - mean_q: -0.002 - mean_eps: 0.586 - ale.lives: 0.000

Interval 13 (120000 steps performed)

10000/10000 [=====] - 494s 49ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.018 - mean_q: 0.000 - mean_eps: 0.550 - ale.lives: 0.000

Interval 14 (130000 steps performed)

10000/10000 [=====] - 537s 54ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.018 - mean_q: 0.001 - mean_eps: 0.514 - ale.lives: 0.000

Interval 15 (140000 steps performed)

10000/10000 [=====] - 581s 58ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.018 - mean_q: 0.003 - mean_eps: 0.478 - ale.lives: 0.000

Interval 16 (150000 steps performed)

10000/10000 [=====] - 581s 58ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.018 - mean_q: 0.005 - mean_eps: 0.442 - ale.lives: 0.000

Interval 17 (160000 steps performed)

10000/10000 [=====] - 697s 70ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.018 - mean_q: 0.007 - mean_eps: 0.406 - ale.lives: 0.000

Interval 18 (170000 steps performed)

10000/10000 [=====] - 700s 70ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.019 - mean_q: 0.008 - mean_eps: 0.370 - ale.lives: 0.000

Interval 19 (180000 steps performed)

10000/10000 [=====] - 666s 67ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.019 - mean_q: 0.010 - mean_eps: 0.334 - ale.lives: 0.000

Interval 20 (190000 steps performed)

10000/10000 [=====] - 722s 72ms/step - reward: 0.0000e+00

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.019 - mean_q: 0.010 - mean_eps: 0.298 - ale.lives: 0.000

Interval 21 (200000 steps performed)

10000/10000 [=====] - 440s 44ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.018 - mean_q: 0.012 - mean_eps: 0.262 - ale.lives: 0.000

Interval 22 (210000 steps performed)

10000/10000 [=====] - 303s 30ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.018 - mean_q: 0.012 - mean_eps: 0.226 - ale.lives: 0.000

Interval 23 (220000 steps performed)

10000/10000 [=====] - 362s 36ms/step - reward: 0.0000e+00

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.020 - mean_q: 0.020 - mean_eps: 0.190 - ale.lives: 0.000

Interval 24 (230000 steps performed)

10000/10000 [=====] - 292s 29ms/step - reward: 1.0000e-04

3 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.000 - mae: 0.021 - mean_q: 0.024 - mean_eps: 0.154 - ale.lives: 0.000

Interval 25 (240000 steps performed)

10000/10000 [=====] - 312s 31ms/step - reward: 6.0000e-04

2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.001 - mae: 0.024 - mean_q: 0.029 - mean_eps: 0.118 - ale.lives: 0.000

Interval 26 (250000 steps performed)
10000/10000 [=====] - 303s 30ms/step - reward: -7.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.001 - mae: 0.029 - mean_q: 0.036 - mean_eps: 0.100 - ale.lives: 0.000

Interval 27 (260000 steps performed)
10000/10000 [=====] - 266s 27ms/step - reward: 5.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.001 - mae: 0.038 - mean_q: 0.045 - mean_eps: 0.100 - ale.lives: 0.000

Interval 28 (270000 steps performed)
10000/10000 [=====] - 257s 26ms/step - reward: -3.0000e-04
3 episodes - episode_reward: 0.667 [0.000, 2.000] - loss: 0.002 - mae: 0.044 - mean_q: 0.049 - mean_eps: 0.100 - ale.lives: 0.000

Interval 29 (280000 steps performed)
10000/10000 [=====] - 251s 25ms/step - reward: 7.0000e-04
2 episodes - episode_reward: 1.500 [0.000, 3.000] - loss: 0.002 - mae: 0.055 - mean_q: 0.056 - mean_eps: 0.100 - ale.lives: 0.000

Interval 30 (290000 steps performed)
10000/10000 [=====] - 253s 25ms/step - reward: -4.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.002 - mae: 0.075 - mean_q: 0.074 - mean_eps: 0.100 - ale.lives: 0.000

Interval 31 (300000 steps performed)
10000/10000 [=====] - 333s 33ms/step - reward: 5.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.003 - mae: 0.092 - mean_q: 0.087 - mean_eps: 0.100 - ale.lives: 0.000

Interval 32 (310000 steps performed)
10000/10000 [=====] - 392s 39ms/step - reward: -2.0000e-04
3 episodes - episode_reward: 1.000 [0.000, 2.000] - loss: 0.003 - mae: 0.104 - mean_q: 0.096 - mean_eps: 0.100 - ale.lives: 0.000

Interval 33 (320000 steps performed)
10000/10000 [=====] - 506s 51ms/step - reward: 0.0012
2 episodes - episode_reward: 6.000 [0.000, 12.000] - loss: 0.004 - mae: 0.121 - mean_q: 0.115 - mean_eps: 0.100 - ale.lives: 0.000

Interval 34 (330000 steps performed)

10000/10000 [=====] - 385s 39ms/step - reward: 0.0012
2 episodes - episode_reward: 2.000 [1.000, 3.000] - loss: 0.004 - mae: 0.129 - mean_q: 0.121 - mean_eps: 0.100 - ale.lives: 0.000

Interval 35 (340000 steps performed)

10000/10000 [=====] - 439s 44ms/step - reward: -4.0000e-04
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.004 - mae: 0.158 - mean_q: 0.144 - mean_eps: 0.100 - ale.lives: 0.000

Interval 36 (350000 steps performed)

10000/10000 [=====] - 339s 34ms/step - reward: -1.0000e-04
3 episodes - episode_reward: 1.000 [0.000, 3.000] - loss: 0.005 - mae: 0.191 - mean_q: 0.170 - mean_eps: 0.100 - ale.lives: 0.000

Interval 37 (360000 steps performed)

10000/10000 [=====] - 310s 31ms/step - reward: 1.0000e-03
2 episodes - episode_reward: 0.000 [0.000, 0.000] - loss: 0.005 - mae: 0.208 - mean_q: 0.189 - mean_eps: 0.100 - ale.lives: 0.000

Interval 38 (370000 steps performed)

10000/10000 [=====] - 315s 31ms/step - reward: 0.0000e+00
2 episodes - episode_reward: 5.000 [0.000, 10.000] - loss: 0.006 - mae: 0.231 - mean_q: 0.208 - mean_eps: 0.100 - ale.lives: 0.000

Interval 39 (380000 steps performed)

10000/10000 [=====] - 316s 32ms/step - reward: 0.0012
2 episodes - episode_reward: 6.000 [0.000, 12.000] - loss: 0.007 - mae: 0.244 - mean_q: 0.217 - mean_eps: 0.100 - ale.lives: 0.000

Interval 40 (390000 steps performed)

10000/10000 [=====] - 276s 28ms/step - reward: 0.0041
3 episodes - episode_reward: 8.667 [3.000, 15.000] - loss: 0.007 - mae: 0.277 - mean_q: 0.243 - mean_eps: 0.100 - ale.lives: 0.000

Interval 41 (400000 steps performed)

10000/10000 [=====] - 431s 43ms/step - reward: 0.0047
2 episodes - episode_reward: 31.000 [17.000, 45.000] - loss: 0.008 - mae: 0.301 - mean_q: 0.265 - mean_eps: 0.100 - ale.lives: 0.000

Interval 42 (410000 steps performed)

10000/10000 [=====] - 612s 61ms/step - reward: 0.0044

2 episodes - episode_reward: 14.500 [4.000, 25.000] - loss: 0.008 - mae: 0.336 - mean_q: 0.306 - mean_eps: 0.100 - ale.lives: 0.000

Interval 43 (420000 steps performed)

10000/10000 [=====] - 526s 53ms/step - reward: 0.0036

2 episodes - episode_reward: 22.000 [17.000, 27.000] - loss: 0.010 - mae: 0.388 - mean_q: 0.363 - mean_eps: 0.100 - ale.lives: 0.000

Interval 44 (430000 steps performed)

10000/10000 [=====] - 3219s 322ms/step - reward: 0.0046

3 episodes - episode_reward: 15.667 [5.000, 24.000] - loss: 0.011 - mae: 0.426 - mean_q: 0.394 - mean_eps: 0.100 - ale.lives: 0.000

Interval 45 (440000 steps performed)

10000/10000 [=====] - 301s 30ms/step - reward: 0.0143

2 episodes - episode_reward: 70.500 [33.000, 108.000] - loss: 0.012 - mae: 0.467 - mean_q: 0.438 - mean_eps: 0.100 - ale.lives: 0.000

Interval 46 (450000 steps performed)

10000/10000 [=====] - 336s 34ms/step - reward: 0.0190

2 episodes - episode_reward: 68.000 [47.000, 89.000] - loss: 0.013 - mae: 0.506 - mean_q: 0.470 - mean_eps: 0.100 - ale.lives: 0.000

Interval 47 (460000 steps performed)

10000/10000 [=====] - 308s 31ms/step - reward: 0.0104

2 episodes - episode_reward: 52.500 [11.000, 94.000] - loss: 0.015 - mae: 0.564 - mean_q: 0.538 - mean_eps: 0.100 - ale.lives: 0.000

Interval 48 (470000 steps performed)

10000/10000 [=====] - 317s 32ms/step - reward: 0.0109

3 episodes - episode_reward: 56.667 [29.000, 94.000] - loss: 0.015 - mae: 0.624 - mean_q: 0.593 - mean_eps: 0.100 - ale.lives: 0.000

Interval 49 (480000 steps performed)

10000/10000 [=====] - 302s 30ms/step - reward: 0.0137

2 episodes - episode_reward: 58.000 [43.000, 73.000] - loss: 0.016 - mae: 0.677 - mean_q: 0.658 - mean_eps: 0.100 - ale.lives: 0.000

Interval 50 (490000 steps performed)

10000/10000 [=====] - 326s 33ms/step - reward: 0.0174

2 episodes - episode_reward: 71.000 [39.000, 103.000] - loss: 0.018 - mae: 0.724 - mean_q: 0.702 - mean_eps: 0.100 - ale.lives: 0.000

0.000

Interval 51 (500000 steps performed)

10000/10000 [=====] - 408s 41ms/step - reward: 0.0218

2 episodes - episode_reward: 99.000 [87.000, 111.000] - loss: 0.017 - mae: 0.746 - mean_q: 0.723 - mean_eps: 0.100 - ale.lives: 0.000

Interval 52 (510000 steps performed)

10000/10000 [=====] - 413s 41ms/step - reward: 0.0221

3 episodes - episode_reward: 91.667 [65.000, 114.000] - loss: 0.018 - mae: 0.796 - mean_q: 0.780 - mean_eps: 0.100 - ale.lives: 0.000

Interval 53 (520000 steps performed)

10000/10000 [=====] - 447s 45ms/step - reward: 0.0198

2 episodes - episode_reward: 104.500 [90.000, 119.000] - loss: 0.020 - mae: 0.861 - mean_q: 0.863 - mean_eps: 0.100 - ale.lives: 0.000

Interval 54 (530000 steps performed)

10000/10000 [=====] - 422s 42ms/step - reward: 0.0201

2 episodes - episode_reward: 86.000 [46.000, 126.000] - loss: 0.020 - mae: 0.901 - mean_q: 0.905 - mean_eps: 0.100 - ale.lives: 0.000

Interval 55 (540000 steps performed)

10000/10000 [=====] - 387s 39ms/step - reward: 0.0247

2 episodes - episode_reward: 111.000 [110.000, 112.000] - loss: 0.021 - mae: 0.940 - mean_q: 0.954 - mean_eps: 0.100 - ale.lives: 0.000

Interval 56 (550000 steps performed)

10000/10000 [=====] - 347s 35ms/step - reward: 0.0228

3 episodes - episode_reward: 90.667 [71.000, 116.000] - loss: 0.020 - mae: 0.981 - mean_q: 0.996 - mean_eps: 0.100 - ale.lives: 0.000

Interval 57 (560000 steps performed)

10000/10000 [=====] - 341s 34ms/step - reward: 0.0210

2 episodes - episode_reward: 109.500 [104.000, 115.000] - loss: 0.021 - mae: 1.019 - mean_q: 1.050 - mean_eps: 0.100 - ale.lives: 0.000

Interval 58 (570000 steps performed)

10000/10000 [=====] - 338s 34ms/step - reward: 0.0237

2 episodes - episode_reward: 99.500 [93.000, 106.000] - loss: 0.021 - mae: 1.043 - mean_q: 1.081 - mean_eps: 0.100 - ale.lives: 0.000

Interval 59 (580000 steps performed)

10000/10000 [=====] - 346s 35ms/step - reward: 0.0293

2 episodes - episode_reward: 110.500 [91.000, 130.000] - loss: 0.023 - mae: 1.076 - mean_q: 1.122 - mean_eps: 0.100 - ale.lives: 0.000

Interval 60 (590000 steps performed)

10000/10000 [=====] - 338s 34ms/step - reward: 0.0265

3 episodes - episode_reward: 122.333 [110.000, 132.000] - loss: 0.023 - mae: 1.128 - mean_q: 1.189 - mean_eps: 0.100 - ale.lives: 0.000

Interval 61 (600000 steps performed)

10000/10000 [=====] - 433s 43ms/step - reward: 0.0251

2 episodes - episode_reward: 124.500 [118.000, 131.000] - loss: 0.024 - mae: 1.155 - mean_q: 1.206 - mean_eps: 0.100 - ale.lives: 0.000

Interval 62 (610000 steps performed)

10000/10000 [=====] - 452s 45ms/step - reward: 0.0333

2 episodes - episode_reward: 134.500 [129.000, 140.000] - loss: 0.024 - mae: 1.217 - mean_q: 1.283 - mean_eps: 0.100 - ale.lives: 0.000

Interval 63 (620000 steps performed)

10000/10000 [=====] - 437s 44ms/step - reward: 0.0225

2 episodes - episode_reward: 121.500 [107.000, 136.000] - loss: 0.026 - mae: 1.275 - mean_q: 1.362 - mean_eps: 0.100 - ale.lives: 0.000

Interval 64 (630000 steps performed)

10000/10000 [=====] - 455s 45ms/step - reward: 0.0324

3 episodes - episode_reward: 126.333 [84.000, 153.000] - loss: 0.028 - mae: 1.359 - mean_q: 1.470 - mean_eps: 0.100 - ale.lives: 0.000

Interval 65 (640000 steps performed)

10000/10000 [=====] - 365s 37ms/step - reward: 0.0265

2 episodes - episode_reward: 120.500 [116.000, 125.000] - loss: 0.028 - mae: 1.426 - mean_q: 1.560 - mean_eps: 0.100 - ale.lives: 0.000

Interval 66 (650000 steps performed)

10000/10000 [=====] - 365s 37ms/step - reward: 0.0326

2 episodes - episode_reward: 130.500 [121.000, 140.000] - loss: 0.028 - mae: 1.487 - mean_q: 1.632 - mean_eps: 0.100 - ale.lives: 0.000

Interval 67 (660000 steps performed)
10000/10000 [=====] - 365s 36ms/step - reward: 0.0313
2 episodes - episode_reward: 133.000 [132.000, 134.000] - loss: 0.030 - mae: 1.560 - mean_q: 1.714 - mean_eps: 0.100 - ale.lives: 0.000

Interval 68 (670000 steps performed)
10000/10000 [=====] - 363s 36ms/step - reward: 0.0297
3 episodes - episode_reward: 138.667 [123.000, 147.000] - loss: 0.031 - mae: 1.602 - mean_q: 1.774 - mean_eps: 0.100 - ale.lives: 0.000

Interval 69 (680000 steps performed)
10000/10000 [=====] - 358s 36ms/step - reward: 0.0379
2 episodes - episode_reward: 157.000 [146.000, 168.000] - loss: 0.031 - mae: 1.646 - mean_q: 1.815 - mean_eps: 0.100 - ale.lives: 0.000

Interval 70 (690000 steps performed)
10000/10000 [=====] - 352s 35ms/step - reward: 0.0208
2 episodes - episode_reward: 134.000 [109.000, 159.000] - loss: 0.033 - mae: 1.757 - mean_q: 1.949 - mean_eps: 0.100 - ale.lives: 0.000

Interval 71 (700000 steps performed)
10000/10000 [=====] - 455s 45ms/step - reward: 0.0338
2 episodes - episode_reward: 85.000 [39.000, 131.000] - loss: 0.033 - mae: 1.870 - mean_q: 2.094 - mean_eps: 0.100 - ale.lives: 0.000

Interval 72 (710000 steps performed)
10000/10000 [=====] - 509s 51ms/step - reward: 0.0367
2 episodes - episode_reward: 251.000 [184.000, 318.000] - loss: 0.035 - mae: 1.925 - mean_q: 2.164 - mean_eps: 0.100 - ale.lives: 0.000

Interval 73 (720000 steps performed)
10000/10000 [=====] - 460s 46ms/step - reward: 0.0429
1 episodes - episode_reward: 181.000 [181.000, 181.000] - loss: 0.038 - mae: 2.039 - mean_q: 2.299 - mean_eps: 0.100 - ale.lives: 0.000

Interval 74 (730000 steps performed)
10000/10000 [=====] - 469s 47ms/step - reward: 0.0234
2 episodes - episode_reward: 227.500 [111.000, 344.000] - loss: 0.038 - mae: 2.084 - mean_q: 2.343 - mean_eps: 0.100 - ale.lives: 0.000

Interval 75 (740000 steps performed)

10000/10000 [=====] - 391s 39ms/step - reward: 0.0317
3 episodes - episode_reward: 134.667 [115.000, 154.000] - loss: 0.038 - mae: 2.114 - mean_q: 2.379 - mean_eps: 0.100 - ale.lives: 0.000

Interval 76 (750000 steps performed)

10000/10000 [=====] - 386s 39ms/step - reward: 0.0194
2 episodes - episode_reward: 89.000 [80.000, 98.000] - loss: 0.038 - mae: 2.187 - mean_q: 2.472 - mean_eps: 0.100 - ale.lives: 0.000

Interval 77 (760000 steps performed)

10000/10000 [=====] - 393s 39ms/step - reward: 0.0321
2 episodes - episode_reward: 134.000 [123.000, 145.000] - loss: 0.040 - mae: 2.284 - mean_q: 2.581 - mean_eps: 0.100 - ale.lives: 0.000

Interval 78 (770000 steps performed)

10000/10000 [=====] - 412s 41ms/step - reward: 0.0380
2 episodes - episode_reward: 162.500 [148.000, 177.000] - loss: 0.045 - mae: 2.373 - mean_q: 2.669 - mean_eps: 0.100 - ale.lives: 0.000

Interval 79 (780000 steps performed)

10000/10000 [=====] - 386s 39ms/step - reward: 0.0319
3 episodes - episode_reward: 148.000 [125.000, 186.000] - loss: 0.045 - mae: 2.462 - mean_q: 2.784 - mean_eps: 0.100 - ale.lives: 0.000

Interval 80 (790000 steps performed)

10000/10000 [=====] - 364s 36ms/step - reward: 0.0376
1 episodes - episode_reward: 146.000 [146.000, 146.000] - loss: 0.044 - mae: 2.506 - mean_q: 2.834 - mean_eps: 0.100 - ale.lives: 0.000

Interval 81 (800000 steps performed)

10000/10000 [=====] - 508s 51ms/step - reward: 0.0442
1 episodes - episode_reward: 375.000 [375.000, 375.000] - loss: 0.048 - mae: 2.588 - mean_q: 2.924 - mean_eps: 0.100 - ale.lives: 0.000

Interval 82 (810000 steps performed)

10000/10000 [=====] - 585s 59ms/step - reward: 0.0331
1 episodes - episode_reward: 390.000 [390.000, 390.000] - loss: 0.046 - mae: 2.644 - mean_q: 2.993 - mean_eps: 0.100 - ale.lives: 0.000

Interval 83 (820000 steps performed)

10000/10000 [=====] - 533s 53ms/step - reward: 0.0349

3 episodes - episode_reward: 196.333 [155.000, 269.000] - loss: 0.048 - mae: 2.747 - mean_q: 3.112 - mean_eps: 0.100 - ale.lives: 0.000

Interval 84 (830000 steps performed)

10000/10000 [=====] - 475s 47ms/step - reward: 0.0370

2 episodes - episode_reward: 169.500 [169.000, 170.000] - loss: 0.052 - mae: 2.913 - mean_q: 3.308 - mean_eps: 0.100 - ale.lives: 0.000

Interval 85 (840000 steps performed)

10000/10000 [=====] - 422s 42ms/step - reward: 0.0430

2 episodes - episode_reward: 186.000 [178.000, 194.000] - loss: 0.054 - mae: 3.053 - mean_q: 3.463 - mean_eps: 0.100 - ale.lives: 0.000

Interval 86 (850000 steps performed)

10000/10000 [=====] - 344s 34ms/step - reward: 0.0366

2 episodes - episode_reward: 141.500 [126.000, 157.000] - loss: 0.055 - mae: 3.166 - mean_q: 3.600 - mean_eps: 0.100 - ale.lives: 0.000

Interval 87 (860000 steps performed)

10000/10000 [=====] - 327s 33ms/step - reward: 0.0352

1 episodes - episode_reward: 323.000 [323.000, 323.000] - loss: 0.055 - mae: 3.264 - mean_q: 3.720 - mean_eps: 0.100 - ale.lives: 0.000

Interval 88 (870000 steps performed)

10000/10000 [=====] - 325s 33ms/step - reward: 0.0244

2 episodes - episode_reward: 218.000 [157.000, 279.000] - loss: 0.055 - mae: 3.339 - mean_q: 3.816 - mean_eps: 0.100 - ale.lives: 0.000

Interval 89 (880000 steps performed)

10000/10000 [=====] - 327s 33ms/step - reward: 0.0458

2 episodes - episode_reward: 181.000 [171.000, 191.000] - loss: 0.059 - mae: 3.456 - mean_q: 3.953 - mean_eps: 0.100 - ale.lives: 0.000

Interval 90 (890000 steps performed)

10000/10000 [=====] - 372s 37ms/step - reward: 0.0357

1 episodes - episode_reward: 317.000 [317.000, 317.000] - loss: 0.058 - mae: 3.547 - mean_q: 4.059 - mean_eps: 0.100 - ale.lives: 0.000

Interval 91 (900000 steps performed)

10000/10000 [=====] - 411s 41ms/step - reward: 0.0319

3 episodes - episode_reward: 152.333 [115.000, 191.000] - loss: 0.062 - mae: 3.613 - mean_q: 4.133 - mean_eps: 0.100 - ale.lives: 0.000

s: 0.000

Interval 92 (910000 steps performed)

10000/10000 [=====] - 486s 49ms/step - reward: 0.0352

2 episodes - episode_reward: 154.500 [139.000, 170.000] - loss: 0.061 - mae: 3.650 - mean_q: 4.186 - mean_eps: 0.100 - ale.lives: 0.000

Interval 93 (920000 steps performed)

10000/10000 [=====] - 409s 41ms/step - reward: 0.0534

1 episodes - episode_reward: 465.000 [465.000, 465.000] - loss: 0.060 - mae: 3.685 - mean_q: 4.215 - mean_eps: 0.100 - ale.lives: 0.000

Interval 94 (930000 steps performed)

10000/10000 [=====] - 426s 43ms/step - reward: 0.0380

1 episodes - episode_reward: 374.000 [374.000, 374.000] - loss: 0.059 - mae: 3.693 - mean_q: 4.223 - mean_eps: 0.100 - ale.lives: 0.000

Interval 95 (940000 steps performed)

10000/10000 [=====] - 364s 36ms/step - reward: 0.0418

2 episodes - episode_reward: 261.000 [165.000, 357.000] - loss: 0.059 - mae: 3.737 - mean_q: 4.277 - mean_eps: 0.100 - ale.lives: 0.000

Interval 96 (950000 steps performed)

10000/10000 [=====] - 349s 35ms/step - reward: 0.0384

1 episodes - episode_reward: 366.000 [366.000, 366.000] - loss: 0.061 - mae: 3.863 - mean_q: 4.419 - mean_eps: 0.100 - ale.lives: 0.000

Interval 97 (960000 steps performed)

10000/10000 [=====] - 408s 41ms/step - reward: 0.0378

2 episodes - episode_reward: 169.500 [157.000, 182.000] - loss: 0.062 - mae: 4.007 - mean_q: 4.584 - mean_eps: 0.100 - ale.lives: 0.000

Interval 98 (970000 steps performed)

10000/10000 [=====] - 438s 44ms/step - reward: 0.0406

2 episodes - episode_reward: 170.500 [151.000, 190.000] - loss: 0.064 - mae: 4.073 - mean_q: 4.664 - mean_eps: 0.100 - ale.lives: 0.000

Interval 99 (980000 steps performed)

10000/10000 [=====] - 520s 52ms/step - reward: 0.0396

2 episodes - episode_reward: 160.000 [142.000, 178.000] - loss: 0.067 - mae: 4.201 - mean_q: 4.813 - mean_eps: 0.100 - ale.lives: 0.000

Interval 100 (990000 steps performed)
10000/10000 [=====] - 588s 59ms/step - reward: 0.0393
2 episodes - episode_reward: 280.000 [195.000, 365.000] - loss: 0.068 - mae: 4.263 - mean_q: 4.882 - mean_eps: 0.100 - ale.lives: 0.000

Interval 101 (1000000 steps performed)
10000/10000 [=====] - 563s 56ms/step - reward: 0.0384
2 episodes - episode_reward: 187.500 [185.000, 190.000] - loss: 0.067 - mae: 4.387 - mean_q: 5.031 - mean_eps: 0.100 - ale.lives: 0.000

Interval 102 (1010000 steps performed)
10000/10000 [=====] - 626s 63ms/step - reward: 0.0420
1 episodes - episode_reward: 150.000 [150.000, 150.000] - loss: 0.068 - mae: 4.415 - mean_q: 5.055 - mean_eps: 0.100 - ale.lives: 0.000

Interval 103 (1020000 steps performed)
10000/10000 [=====] - 629s 63ms/step - reward: 0.0402
2 episodes - episode_reward: 359.500 [355.000, 364.000] - loss: 0.067 - mae: 4.496 - mean_q: 5.145 - mean_eps: 0.100 - ale.lives: 0.000

Interval 104 (1030000 steps performed)
10000/10000 [=====] - 559s 56ms/step - reward: 0.0348
2 episodes - episode_reward: 153.500 [149.000, 158.000] - loss: 0.068 - mae: 4.610 - mean_q: 5.276 - mean_eps: 0.100 - ale.lives: 0.000

Interval 105 (1040000 steps performed)
10000/10000 [=====] - 480s 48ms/step - reward: 0.0434
1 episodes - episode_reward: 368.000 [368.000, 368.000] - loss: 0.071 - mae: 4.669 - mean_q: 5.347 - mean_eps: 0.100 - ale.lives: 0.000

Interval 106 (1050000 steps performed)
10000/10000 [=====] - 454s 45ms/step - reward: 0.0451
1 episodes - episode_reward: 417.000 [417.000, 417.000] - loss: 0.072 - mae: 4.771 - mean_q: 5.465 - mean_eps: 0.100 - ale.lives: 0.000

Interval 107 (1060000 steps performed)
10000/10000 [=====] - 457s 46ms/step - reward: 0.0472
1 episodes - episode_reward: 398.000 [398.000, 398.000] - loss: 0.073 - mae: 4.846 - mean_q: 5.548 - mean_eps: 0.100 - ale.lives: 0.000

Interval 108 (1070000 steps performed)

10000/10000 [=====] - 451s 45ms/step - reward: 0.0418

2 episodes - episode_reward: 288.000 [186.000, 390.000] - loss: 0.074 - mae: 4.962 - mean_q: 5.684 - mean_eps: 0.100 - ale.lives: 0.000

Interval 109 (1080000 steps performed)

10000/10000 [=====] - 418s 42ms/step - reward: 0.0481

1 episodes - episode_reward: 471.000 [471.000, 471.000] - loss: 0.074 - mae: 4.994 - mean_q: 5.721 - mean_eps: 0.100 - ale.lives: 0.000

Interval 110 (1090000 steps performed)

10000/10000 [=====] - 404s 40ms/step - reward: 0.0459

1 episodes - episode_reward: 174.000 [174.000, 174.000] - loss: 0.074 - mae: 5.069 - mean_q: 5.806 - mean_eps: 0.100 - ale.lives: 0.000

Interval 111 (1100000 steps performed)

10000/10000 [=====] - 542s 54ms/step - reward: 0.0444

2 episodes - episode_reward: 377.000 [376.000, 378.000] - loss: 0.072 - mae: 5.134 - mean_q: 5.875 - mean_eps: 0.100 - ale.lives: 0.000

Interval 112 (1110000 steps performed)

10000/10000 [=====] - 630s 63ms/step - reward: 0.0348

1 episodes - episode_reward: 358.000 [358.000, 358.000] - loss: 0.074 - mae: 5.227 - mean_q: 5.976 - mean_eps: 0.100 - ale.lives: 0.000

Interval 113 (1120000 steps performed)

10000/10000 [=====] - 580s 58ms/step - reward: 0.0409

2 episodes - episode_reward: 167.500 [139.000, 196.000] - loss: 0.076 - mae: 5.364 - mean_q: 6.136 - mean_eps: 0.100 - ale.lives: 0.000

Interval 114 (1130000 steps performed)

10000/10000 [=====] - 470s 47ms/step - reward: 0.0428

1 episodes - episode_reward: 387.000 [387.000, 387.000] - loss: 0.077 - mae: 5.488 - mean_q: 6.272 - mean_eps: 0.100 - ale.lives: 0.000

Interval 115 (1140000 steps performed)

10000/10000 [=====] - 449s 45ms/step - reward: 0.0415

2 episodes - episode_reward: 278.000 [189.000, 367.000] - loss: 0.075 - mae: 5.583 - mean_q: 6.383 - mean_eps: 0.100 - ale.lives: 0.000

Interval 116 (1150000 steps performed)

10000/10000 [=====] - 452s 45ms/step - reward: 0.0475
1 episodes - episode_reward: 393.000 [393.000, 393.000] - loss: 0.080 - mae: 5.702 - mean_q: 6.522 - mean_eps: 0.100 - ale.lives: 0.000

Interval 117 (1160000 steps performed)

10000/10000 [=====] - 454s 45ms/step - reward: 0.0356
1 episodes - episode_reward: 342.000 [342.000, 342.000] - loss: 0.083 - mae: 5.815 - mean_q: 6.648 - mean_eps: 0.100 - ale.lives: 0.000

Interval 118 (1170000 steps performed)

10000/10000 [=====] - 428s 43ms/step - reward: 0.0440
1 episodes - episode_reward: 358.000 [358.000, 358.000] - loss: 0.083 - mae: 5.873 - mean_q: 6.719 - mean_eps: 0.100 - ale.lives: 0.000

Interval 119 (1180000 steps performed)

10000/10000 [=====] - 531s 53ms/step - reward: 0.0383
2 episodes - episode_reward: 271.000 [195.000, 347.000] - loss: 0.083 - mae: 5.955 - mean_q: 6.805 - mean_eps: 0.100 - ale.lives: 0.000

Interval 120 (1190000 steps performed)

10000/10000 [=====] - 579s 58ms/step - reward: 0.0421
1 episodes - episode_reward: 392.000 [392.000, 392.000] - loss: 0.089 - mae: 6.116 - mean_q: 6.989 - mean_eps: 0.100 - ale.lives: 0.000

Interval 121 (1200000 steps performed)

10000/10000 [=====] - 554s 55ms/step - reward: 0.0381
2 episodes - episode_reward: 159.000 [124.000, 194.000] - loss: 0.083 - mae: 6.138 - mean_q: 7.006 - mean_eps: 0.100 - ale.lives: 0.000

Interval 122 (1210000 steps performed)

10000/10000 [=====] - 628s 63ms/step - reward: 0.0418
1 episodes - episode_reward: 365.000 [365.000, 365.000] - loss: 0.088 - mae: 6.199 - mean_q: 7.082 - mean_eps: 0.100 - ale.lives: 0.000

Interval 123 (1220000 steps performed)

10000/10000 [=====] - 641s 64ms/step - reward: 0.0415
2 episodes - episode_reward: 276.000 [177.000, 375.000] - loss: 0.089 - mae: 6.300 - mean_q: 7.201 - mean_eps: 0.100 - ale.lives: 0.000

Interval 124 (1230000 steps performed)

10000/10000 [=====] - 576s 58ms/step - reward: 0.0333

1 episodes - episode_reward: 321.000 [321.000, 321.000] - loss: 0.088 - mae: 6.385 - mean_q: 7.300 - mean_eps: 0.100 - ale.lives: 0.000

Interval 125 (1240000 steps performed)

10000/10000 [=====] - 459s 46ms/step - reward: 0.0408

1 episodes - episode_reward: 365.000 [365.000, 365.000] - loss: 0.085 - mae: 6.430 - mean_q: 7.343 - mean_eps: 0.100 - ale.lives: 0.000

Interval 126 (1250000 steps performed)

10000/10000 [=====] - 459s 46ms/step - reward: 0.0451

1 episodes - episode_reward: 148.000 [148.000, 148.000] - loss: 0.088 - mae: 6.418 - mean_q: 7.327 - mean_eps: 0.100 - ale.lives: 0.000

Interval 127 (1260000 steps performed)

10000/10000 [=====] - 449s 45ms/step - reward: 0.0397

2 episodes - episode_reward: 286.000 [162.000, 410.000] - loss: 0.086 - mae: 6.431 - mean_q: 7.343 - mean_eps: 0.100 - ale.lives: 0.000

Interval 128 (1270000 steps performed)

10000/10000 [=====] - 462s 46ms/step - reward: 0.0383

2 episodes - episode_reward: 253.000 [140.000, 366.000] - loss: 0.087 - mae: 6.520 - mean_q: 7.446 - mean_eps: 0.100 - ale.lives: 0.000

Interval 129 (1280000 steps performed)

10000/10000 [=====] - 433s 43ms/step - reward: 0.0433

1 episodes - episode_reward: 383.000 [383.000, 383.000] - loss: 0.091 - mae: 6.688 - mean_q: 7.634 - mean_eps: 0.100 - ale.lives: 0.000

Interval 130 (1290000 steps performed)

10000/10000 [=====] - 417s 42ms/step - reward: 0.0394

1 episodes - episode_reward: 387.000 [387.000, 387.000] - loss: 0.089 - mae: 6.808 - mean_q: 7.776 - mean_eps: 0.100 - ale.lives: 0.000

Interval 131 (1300000 steps performed)

10000/10000 [=====] - 544s 54ms/step - reward: 0.0431

2 episodes - episode_reward: 273.500 [182.000, 365.000] - loss: 0.088 - mae: 6.965 - mean_q: 7.949 - mean_eps: 0.100 - ale.lives: 0.000

Interval 132 (1310000 steps performed)

10000/10000 [=====] - 621s 62ms/step - reward: 0.0354

1 episodes - episode_reward: 329.000 [329.000, 329.000] - loss: 0.092 - mae: 7.026 - mean_q: 8.016 - mean_eps: 0.100 - ale.lives: 0.000

s: 0.000

Interval 133 (1320000 steps performed)

10000/10000 [=====] - 604s 60ms/step - reward: 0.0449

1 episodes - episode_reward: 388.000 [388.000, 388.000] - loss: 0.095 - mae: 7.048 - mean_q: 8.047 - mean_eps: 0.100 - ale.lives: 0.000

Interval 134 (1330000 steps performed)

10000/10000 [=====] - 561s 56ms/step - reward: 0.0513

1 episodes - episode_reward: 453.000 [453.000, 453.000] - loss: 0.095 - mae: 7.090 - mean_q: 8.092 - mean_eps: 0.100 - ale.lives: 0.000

Interval 135 (1340000 steps performed)

10000/10000 [=====] - 574s 57ms/step - reward: 0.0414

1 episodes - episode_reward: 353.000 [353.000, 353.000] - loss: 0.094 - mae: 7.088 - mean_q: 8.092 - mean_eps: 0.100 - ale.lives: 0.000

Interval 136 (1350000 steps performed)

10000/10000 [=====] - 588s 59ms/step - reward: 0.0424

1 episodes - episode_reward: 395.000 [395.000, 395.000] - loss: 0.094 - mae: 7.144 - mean_q: 8.148 - mean_eps: 0.100 - ale.lives: 0.000

Interval 137 (1360000 steps performed)

10000/10000 [=====] - 473s 47ms/step - reward: 0.0531

1 episodes - episode_reward: 440.000 [440.000, 440.000] - loss: 0.093 - mae: 7.164 - mean_q: 8.171 - mean_eps: 0.100 - ale.lives: 0.000

Interval 138 (1370000 steps performed)

10000/10000 [=====] - 458s 46ms/step - reward: 0.0398

2 episodes - episode_reward: 387.500 [356.000, 419.000] - loss: 0.090 - mae: 7.200 - mean_q: 8.209 - mean_eps: 0.100 - ale.lives: 0.000

Interval 139 (1380000 steps performed)

10000/10000 [=====] - 456s 46ms/step - reward: 0.0422

1 episodes - episode_reward: 375.000 [375.000, 375.000] - loss: 0.089 - mae: 7.247 - mean_q: 8.258 - mean_eps: 0.100 - ale.lives: 0.000

Interval 140 (1390000 steps performed)

10000/10000 [=====] - 455s 45ms/step - reward: 0.0429

1 episodes - episode_reward: 187.000 [187.000, 187.000] - loss: 0.094 - mae: 7.308 - mean_q: 8.335 - mean_eps: 0.100 - ale.lives: 0.000

Interval 141 (1400000 steps performed)

10000/10000 [=====] - 561s 56ms/step - reward: 0.0441

1 episodes - episode_reward: 396.000 [396.000, 396.000] - loss: 0.092 - mae: 7.278 - mean_q: 8.299 - mean_eps: 0.100 - ale.lives: 0.000

Interval 142 (1410000 steps performed)

10000/10000 [=====] - 624s 62ms/step - reward: 0.0473

2 episodes - episode_reward: 403.000 [401.000, 405.000] - loss: 0.091 - mae: 7.346 - mean_q: 8.384 - mean_eps: 0.100 - ale.lives: 0.000

Interval 143 (1420000 steps performed)

10000/10000 [=====] - 632s 63ms/step - reward: 0.0356

1 episodes - episode_reward: 313.000 [313.000, 313.000] - loss: 0.090 - mae: 7.371 - mean_q: 8.401 - mean_eps: 0.100 - ale.lives: 0.000

Interval 144 (1430000 steps performed)

10000/10000 [=====] - 573s 57ms/step - reward: 0.0424

1 episodes - episode_reward: 159.000 [159.000, 159.000] - loss: 0.089 - mae: 7.424 - mean_q: 8.460 - mean_eps: 0.100 - ale.lives: 0.000

Interval 145 (1440000 steps performed)

10000/10000 [=====] - 472s 47ms/step - reward: 0.0436

1 episodes - episode_reward: 421.000 [421.000, 421.000] - loss: 0.086 - mae: 7.488 - mean_q: 8.531 - mean_eps: 0.100 - ale.lives: 0.000

Interval 146 (1450000 steps performed)

10000/10000 [=====] - 454s 45ms/step - reward: 0.0525

2 episodes - episode_reward: 416.500 [402.000, 431.000] - loss: 0.087 - mae: 7.536 - mean_q: 8.586 - mean_eps: 0.100 - ale.lives: 0.000

Interval 147 (1460000 steps performed)

10000/10000 [=====] - 466s 47ms/step - reward: 0.0408

1 episodes - episode_reward: 146.000 [146.000, 146.000] - loss: 0.087 - mae: 7.560 - mean_q: 8.617 - mean_eps: 0.100 - ale.lives: 0.000

Interval 148 (1470000 steps performed)

10000/10000 [=====] - 448s 45ms/step - reward: 0.0500

2 episodes - episode_reward: 337.500 [190.000, 485.000] - loss: 0.086 - mae: 7.562 - mean_q: 8.612 - mean_eps: 0.100 - ale.lives: 0.000

Interval 149 (1480000 steps performed)

10000/10000 [=====] - 429s 43ms/step - reward: 0.0324

1 episodes - episode_reward: 297.000 [297.000, 297.000] - loss: 0.086 - mae: 7.630 - mean_q: 8.688 - mean_eps: 0.100 - ale.lives: 0.000

Interval 150 (1490000 steps performed)

10000/10000 [=====] - 416s 42ms/step - reward: 0.0458

2 episodes - episode_reward: 288.500 [192.000, 385.000] - loss: 0.091 - mae: 7.679 - mean_q: 8.755 - mean_eps: 0.100 - ale.lives: 0.000

Interval 151 (1500000 steps performed)

10000/10000 [=====] - 546s 55ms/step - reward: 0.0468

1 episodes - episode_reward: 441.000 [441.000, 441.000] - loss: 0.091 - mae: 7.789 - mean_q: 8.872 - mean_eps: 0.100 - ale.lives: 0.000

Interval 152 (1510000 steps performed)

10000/10000 [=====] - 640s 64ms/step - reward: 0.0397

2 episodes - episode_reward: 156.000 [128.000, 184.000] - loss: 0.093 - mae: 7.846 - mean_q: 8.935 - mean_eps: 0.100 - ale.lives: 0.000

Interval 153 (1520000 steps performed)

10000/10000 [=====] - 626s 63ms/step - reward: 0.0422

1 episodes - episode_reward: 176.000 [176.000, 176.000] - loss: 0.092 - mae: 7.839 - mean_q: 8.925 - mean_eps: 0.100 - ale.lives: 0.000

Interval 154 (1530000 steps performed)

10000/10000 [=====] - 575s 58ms/step - reward: 0.0531

2 episodes - episode_reward: 443.000 [416.000, 470.000] - loss: 0.095 - mae: 7.966 - mean_q: 9.075 - mean_eps: 0.100 - ale.lives: 0.000

Interval 155 (1540000 steps performed)

10000/10000 [=====] - 503s 50ms/step - reward: 0.0478

1 episodes - episode_reward: 459.000 [459.000, 459.000] - loss: 0.095 - mae: 7.967 - mean_q: 9.073 - mean_eps: 0.100 - ale.lives: 0.000

Interval 156 (1550000 steps performed)

10000/10000 [=====] - 457s 46ms/step - reward: 0.0375

1 episodes - episode_reward: 356.000 [356.000, 356.000] - loss: 0.091 - mae: 8.009 - mean_q: 9.119 - mean_eps: 0.100 - ale.lives: 0.000

Interval 157 (1560000 steps performed)

10000/10000 [=====] - 446s 45ms/step - reward: 0.0527
1 episodes - episode_reward: 162.000 [162.000, 162.000] - loss: 0.094 - mae: 8.064 - mean_q: 9.179 - mean_eps: 0.100 - ale.lives: 0.000

Interval 158 (1570000 steps performed)

10000/10000 [=====] - 451s 45ms/step - reward: 0.0502
2 episodes - episode_reward: 454.000 [410.000, 498.000] - loss: 0.094 - mae: 8.148 - mean_q: 9.276 - mean_eps: 0.100 - ale.lives: 0.000

Interval 159 (1580000 steps performed)

10000/10000 [=====] - 404s 40ms/step - reward: 0.0456
1 episodes - episode_reward: 421.000 [421.000, 421.000] - loss: 0.095 - mae: 8.166 - mean_q: 9.288 - mean_eps: 0.100 - ale.lives: 0.000

Interval 160 (1590000 steps performed)

10000/10000 [=====] - 361s 36ms/step - reward: 0.0427
1 episodes - episode_reward: 382.000 [382.000, 382.000] - loss: 0.093 - mae: 8.224 - mean_q: 9.354 - mean_eps: 0.100 - ale.lives: 0.000

Interval 161 (1600000 steps performed)

10000/10000 [=====] - 400s 40ms/step - reward: 0.0465
1 episodes - episode_reward: 389.000 [389.000, 389.000] - loss: 0.093 - mae: 8.226 - mean_q: 9.360 - mean_eps: 0.100 - ale.lives: 0.000

Interval 162 (1610000 steps performed)

10000/10000 [=====] - 445s 44ms/step - reward: 0.0453
1 episodes - episode_reward: 418.000 [418.000, 418.000] - loss: 0.094 - mae: 8.193 - mean_q: 9.325 - mean_eps: 0.100 - ale.lives: 0.000

Interval 163 (1620000 steps performed)

10000/10000 [=====] - 439s 44ms/step - reward: 0.0450
1 episodes - episode_reward: 390.000 [390.000, 390.000] - loss: 0.096 - mae: 8.215 - mean_q: 9.351 - mean_eps: 0.100 - ale.lives: 0.000

Interval 164 (1630000 steps performed)

10000/10000 [=====] - 624s 62ms/step - reward: 0.0395
2 episodes - episode_reward: 260.500 [161.000, 360.000] - loss: 0.093 - mae: 8.223 - mean_q: 9.354 - mean_eps: 0.100 - ale.lives: 0.000

Interval 165 (1640000 steps performed)

10000/10000 [=====] - 568s 57ms/step - reward: 0.0366

2 episodes - episode_reward: 186.500 [184.000, 189.000] - loss: 0.092 - mae: 8.306 - mean_q: 9.450 - mean_eps: 0.100 - ale.lives: 0.000

Interval 166 (1650000 steps performed)

10000/10000 [=====] - 470s 47ms/step - reward: 0.0436

3 episodes - episode_reward: 183.333 [163.000, 196.000] - loss: 0.088 - mae: 8.370 - mean_q: 9.519 - mean_eps: 0.100 - ale.lives: 0.000

Interval 167 (1660000 steps performed)

10000/10000 [=====] - 448s 45ms/step - reward: 0.0380

2 episodes - episode_reward: 171.500 [166.000, 177.000] - loss: 0.093 - mae: 8.463 - mean_q: 9.625 - mean_eps: 0.100 - ale.lives: 0.000

Interval 168 (1670000 steps performed)

10000/10000 [=====] - 363s 36ms/step - reward: 0.0427

1 episodes - episode_reward: 378.000 [378.000, 378.000] - loss: 0.089 - mae: 8.440 - mean_q: 9.600 - mean_eps: 0.100 - ale.lives: 0.000

Interval 169 (1680000 steps performed)

10000/10000 [=====] - 349s 35ms/step - reward: 0.0415

2 episodes - episode_reward: 171.000 [169.000, 173.000] - loss: 0.091 - mae: 8.457 - mean_q: 9.621 - mean_eps: 0.100 - ale.lives: 0.000

Interval 170 (1690000 steps performed)

10000/10000 [=====] - 350s 35ms/step - reward: 0.0418

2 episodes - episode_reward: 290.000 [195.000, 385.000] - loss: 0.088 - mae: 8.461 - mean_q: 9.626 - mean_eps: 0.100 - ale.lives: 0.000

Interval 171 (1700000 steps performed)

10000/10000 [=====] - 424s 42ms/step - reward: 0.0454

1 episodes - episode_reward: 164.000 [164.000, 164.000] - loss: 0.093 - mae: 8.500 - mean_q: 9.671 - mean_eps: 0.100 - ale.lives: 0.000

Interval 172 (1710000 steps performed)

10000/10000 [=====] - 499s 50ms/step - reward: 0.0469

1 episodes - episode_reward: 426.000 [426.000, 426.000] - loss: 0.100 - mae: 8.520 - mean_q: 9.699 - mean_eps: 0.100 - ale.lives: 0.000

Interval 173 (1720000 steps performed)

10000/10000 [=====] - 582s 58ms/step - reward: 0.0407

1 episodes - episode_reward: 349.000 [349.000, 349.000] - loss: 0.109 - mae: 8.550 - mean_q: 9.740 - mean_eps: 0.100 - ale.lives: 0.000

s: 0.000

Interval 174 (1730000 steps performed)

10000/10000 [=====] - 455s 45ms/step - reward: 0.0436

2 episodes - episode_reward: 406.500 [394.000, 419.000] - loss: 0.104 - mae: 8.644 - mean_q: 9.848 - mean_eps: 0.100 - ale.live
s: 0.000

Interval 175 (1740000 steps performed)

10000/10000 [=====] - 362s 36ms/step - reward: 0.0391

done, took 79400.905 seconds

Testing

```
In [ ]: import json

run_timestamps = {
    'baseline_dqn': "20260725_164222",
    'double_dqn':   "20260731_234157",
    'dueling_dqn':  "20260728_142544",
}

def test_agent(name):
    results = {}

    cfg = experiments[name]
    env = gym.make(env_name)
    dqn = build_agent(nb_actions, WINDOW_LENGTH, INPUT_SHAPE, **cfg)

    run_timestamp = run_timestamps.get(name)
    weights_filename = f'dqn_{env_name}_{name}_weights_{run_timestamp}.h5f'
    dqn.load_weights(weights_filename)

    test_history = dqn.test(env, nb_episodes=100, visualize=True)
    rewards = test_history.history['episode_reward']
    nb_steps = test_history.history['nb_steps']
    results[name] = rewards
    mean_reward = np.mean(rewards)
    pct_above_threshold = np.mean(np.array(rewards) >= REWARD_THRESHOLD) * 100

    print(f"{name}: media = {mean_reward:.2f} | "
```

```

        f"% episodios ≥ {REWARD_THRESHOLD} pts = {pct_above_threshold:.1f}%")

env.close()

# 'rewards' es el array que ya generaste con test_history.history['episode_reward']
test_results = {
    name: {
        'nb_steps': nb_steps.tolist() if hasattr(nb_steps, 'tolist') else list(nb_steps),
        'rewards': rewards.tolist() if hasattr(rewards, 'tolist') else list(rewards),
        'threshold': REWARD_THRESHOLD,
        'mean': mean_reward,
        'pct_above_threshold': pct_above_threshold,
    }
}

with open(f'test_results_{name}_{run_timestamp}.json', 'w') as f:
    json.dump(test_results, f, indent=2)

print(f"Guardado 'test_results_{name}_{run_timestamp}.json'.")

```

Baseline DQN

```
In [ ]: test_agent('baseline_dqn')
```

Model: "model_2"

Layer (type)	Output Shape	Param #
=====		
input_2 (InputLayer)	[(None, 4, 84, 84)]	0

permute_1 (Permute)	(None, 84, 84, 4)	0

conv2d_3 (Conv2D)	(None, 20, 20, 32)	8224

conv2d_4 (Conv2D)	(None, 9, 9, 64)	32832

conv2d_5 (Conv2D)	(None, 7, 7, 64)	36928

flatten_1 (Flatten)	(None, 3136)	0

dense_2 (Dense)	(None, 512)	1606144

dense_3 (Dense)	(None, 9)	4617
=====		
Total params: 1,688,745		
Trainable params: 1,688,745		
Non-trainable params: 0		

None

Testing for 100 episodes ...

c:\Users\rosai\OneDrive\Documentos\Learning\VIU - Master IA\08MIAR - Aprendizaje por Refuerzo\Proyecto\gym_env\lib\site-packages\tensorflow\python\keras\engine\training.py:2424: UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.
warnings.warn("`Model.state\_updates` will be removed in a future version. ')

Episode 1: reward: 419.000, steps: 8865
Episode 2: reward: 442.000, steps: 8924
Episode 3: reward: 484.000, steps: 8906
Episode 4: reward: 435.000, steps: 8860
Episode 5: reward: 495.000, steps: 8897
Episode 6: reward: 458.000, steps: 8903
Episode 7: reward: 578.000, steps: 10000
Episode 8: reward: 443.000, steps: 8831
Episode 9: reward: 589.000, steps: 10000
Episode 10: reward: 489.000, steps: 8872
Episode 11: reward: 440.000, steps: 8867
Episode 12: reward: 403.000, steps: 8881
Episode 13: reward: 469.000, steps: 8837
Episode 14: reward: 484.000, steps: 8932
Episode 15: reward: 472.000, steps: 8863
Episode 16: reward: 412.000, steps: 8905
Episode 17: reward: 465.000, steps: 8883
Episode 18: reward: 464.000, steps: 8849
Episode 19: reward: 402.000, steps: 8914
Episode 20: reward: 460.000, steps: 8859
Episode 21: reward: 368.000, steps: 8838
Episode 22: reward: 457.000, steps: 8887
Episode 23: reward: 470.000, steps: 8863
Episode 24: reward: 469.000, steps: 8879
Episode 25: reward: 411.000, steps: 8885
Episode 26: reward: 493.000, steps: 8863
Episode 27: reward: 436.000, steps: 8867
Episode 28: reward: 626.000, steps: 10000
Episode 29: reward: 444.000, steps: 8867
Episode 30: reward: 195.000, steps: 4419
Episode 31: reward: 562.000, steps: 10000
Episode 32: reward: 526.000, steps: 10000
Episode 33: reward: 487.000, steps: 8870
Episode 34: reward: 440.000, steps: 8889
Episode 35: reward: 449.000, steps: 8893
Episode 36: reward: 431.000, steps: 8872
Episode 37: reward: 528.000, steps: 10000
Episode 38: reward: 468.000, steps: 8834
Episode 39: reward: 467.000, steps: 8923
Episode 40: reward: 495.000, steps: 8874
Episode 41: reward: 432.000, steps: 8876

Episode 42: reward: 457.000, steps: 8870
Episode 43: reward: 493.000, steps: 8928
Episode 44: reward: 412.000, steps: 8884
Episode 45: reward: 450.000, steps: 8870
Episode 46: reward: 411.000, steps: 8849
Episode 47: reward: 407.000, steps: 8901
Episode 48: reward: 468.000, steps: 8902
Episode 49: reward: 394.000, steps: 8867
Episode 50: reward: 577.000, steps: 10000
Episode 51: reward: 452.000, steps: 8878
Episode 52: reward: 471.000, steps: 8897
Episode 53: reward: 575.000, steps: 10000
Episode 54: reward: 473.000, steps: 8891
Episode 55: reward: 393.000, steps: 8860
Episode 56: reward: 406.000, steps: 8882
Episode 57: reward: 411.000, steps: 8889
Episode 58: reward: 435.000, steps: 8827
Episode 59: reward: 463.000, steps: 8914
Episode 60: reward: 367.000, steps: 8912
Episode 61: reward: 446.000, steps: 8879
Episode 62: reward: 445.000, steps: 8887
Episode 63: reward: 386.000, steps: 8908
Episode 64: reward: 395.000, steps: 8825
Episode 65: reward: 482.000, steps: 8882
Episode 66: reward: 398.000, steps: 8911
Episode 67: reward: 437.000, steps: 8903
Episode 68: reward: 544.000, steps: 10000
Episode 69: reward: 409.000, steps: 8903
Episode 70: reward: 457.000, steps: 8908
Episode 71: reward: 494.000, steps: 8874
Episode 72: reward: 524.000, steps: 10000
Episode 73: reward: 492.000, steps: 8852
Episode 74: reward: 406.000, steps: 8855
Episode 75: reward: 482.000, steps: 8859
Episode 76: reward: 432.000, steps: 8875
Episode 77: reward: 393.000, steps: 8880
Episode 78: reward: 604.000, steps: 10000
Episode 79: reward: 455.000, steps: 8888
Episode 80: reward: 446.000, steps: 8899
Episode 81: reward: 565.000, steps: 10000
Episode 82: reward: 499.000, steps: 8853

```
Episode 83: reward: 164.000, steps: 4426
Episode 84: reward: 493.000, steps: 8881
Episode 85: reward: 483.000, steps: 8914
Episode 86: reward: 430.000, steps: 8890
Episode 87: reward: 438.000, steps: 8893
Episode 88: reward: 482.000, steps: 8832
Episode 89: reward: 441.000, steps: 8872
Episode 90: reward: 462.000, steps: 8860
Episode 91: reward: 461.000, steps: 8812
Episode 92: reward: 594.000, steps: 10000
Episode 93: reward: 585.000, steps: 10000
Episode 94: reward: 485.000, steps: 8876
Episode 95: reward: 466.000, steps: 8859
Episode 96: reward: 528.000, steps: 10000
Episode 97: reward: 566.000, steps: 10000
Episode 98: reward: 456.000, steps: 8887
Episode 99: reward: 552.000, steps: 10000
Episode 100: reward: 380.000, steps: 8861
baseline_dqn: media = 461.29 | % episodios ≥ 200 pts = 98.0%
Guardado 'test_results_baseline_dqn_20260725_164222.json'.
```

Double DQN

```
In [ ]: test_agent('double_dqn')
```

Model: "model_9"

Layer (type)	Output Shape	Param #
=====		
input_5 (InputLayer)	[(None, 4, 84, 84)]	0
permute_4 (Permute)	(None, 84, 84, 4)	0
conv2d_12 (Conv2D)	(None, 20, 20, 32)	8224
conv2d_13 (Conv2D)	(None, 9, 9, 64)	32832
conv2d_14 (Conv2D)	(None, 7, 7, 64)	36928
flatten_4 (Flatten)	(None, 3136)	0
dense_10 (Dense)	(None, 512)	1606144
dense_11 (Dense)	(None, 9)	4617

=====

Total params: 1,688,745

Trainable params: 1,688,745

Non-trainable params: 0

None

Testing for 100 episodes ...

Episode 1: reward: 8.000, steps: 4436

Episode 2: reward: 8.000, steps: 4454

Episode 3: reward: 8.000, steps: 4418

Episode 4: reward: 8.000, steps: 4407

Episode 5: reward: 8.000, steps: 4458

Episode 6: reward: 8.000, steps: 4456

Episode 7: reward: 8.000, steps: 4448

Episode 8: reward: 8.000, steps: 4441

Episode 9: reward: 8.000, steps: 4427

Episode 10: reward: 8.000, steps: 4451

Episode 11: reward: 8.000, steps: 4441

Episode 12: reward: 8.000, steps: 4439

Episode 13: reward: 8.000, steps: 4429

Episode 14: reward: 8.000, steps: 4420

Episode 15: reward: 8.000, steps: 4415

Episode 16: reward: 8.000, steps: 4440
Episode 17: reward: 8.000, steps: 4453
Episode 18: reward: 8.000, steps: 4463
Episode 19: reward: 8.000, steps: 4449
Episode 20: reward: 8.000, steps: 4446
Episode 21: reward: 8.000, steps: 4490
Episode 22: reward: 8.000, steps: 4441
Episode 23: reward: 8.000, steps: 4428
Episode 24: reward: 8.000, steps: 4441
Episode 25: reward: 8.000, steps: 4429
Episode 26: reward: 8.000, steps: 4440
Episode 27: reward: 8.000, steps: 4412
Episode 28: reward: 8.000, steps: 4460
Episode 29: reward: 8.000, steps: 4438
Episode 30: reward: 8.000, steps: 4438
Episode 31: reward: 8.000, steps: 4414
Episode 32: reward: 8.000, steps: 4473
Episode 33: reward: 8.000, steps: 4412
Episode 34: reward: 8.000, steps: 4434
Episode 35: reward: 8.000, steps: 4421
Episode 36: reward: 8.000, steps: 4459
Episode 37: reward: 8.000, steps: 4441
Episode 38: reward: 8.000, steps: 4462
Episode 39: reward: 8.000, steps: 4431
Episode 40: reward: 8.000, steps: 4420
Episode 41: reward: 8.000, steps: 4404
Episode 42: reward: 8.000, steps: 4421
Episode 43: reward: 8.000, steps: 4409
Episode 44: reward: 8.000, steps: 4455
Episode 45: reward: 8.000, steps: 4435
Episode 46: reward: 8.000, steps: 4448
Episode 47: reward: 8.000, steps: 4433
Episode 48: reward: 8.000, steps: 4425
Episode 49: reward: 8.000, steps: 4414
Episode 50: reward: 8.000, steps: 4421
Episode 51: reward: 8.000, steps: 4436
Episode 52: reward: 8.000, steps: 4435
Episode 53: reward: 8.000, steps: 4436
Episode 54: reward: 8.000, steps: 4414
Episode 55: reward: 8.000, steps: 4449
Episode 56: reward: 8.000, steps: 4435

Episode 57: reward: 8.000, steps: 4451
Episode 58: reward: 8.000, steps: 4433
Episode 59: reward: 8.000, steps: 4430
Episode 60: reward: 8.000, steps: 4447
Episode 61: reward: 8.000, steps: 4436
Episode 62: reward: 8.000, steps: 4464
Episode 63: reward: 8.000, steps: 4438
Episode 64: reward: 8.000, steps: 4431
Episode 65: reward: 8.000, steps: 4454
Episode 66: reward: 8.000, steps: 4469
Episode 67: reward: 8.000, steps: 4457
Episode 68: reward: 8.000, steps: 4447
Episode 69: reward: 8.000, steps: 4406
Episode 70: reward: 8.000, steps: 4421
Episode 71: reward: 8.000, steps: 4432
Episode 72: reward: 8.000, steps: 4435
Episode 73: reward: 8.000, steps: 4437
Episode 74: reward: 8.000, steps: 4410
Episode 75: reward: 8.000, steps: 4437
Episode 76: reward: 8.000, steps: 4436
Episode 77: reward: 8.000, steps: 4403
Episode 78: reward: 8.000, steps: 4447
Episode 79: reward: 8.000, steps: 4475
Episode 80: reward: 8.000, steps: 4465
Episode 81: reward: 8.000, steps: 4429
Episode 82: reward: 8.000, steps: 4443
Episode 83: reward: 8.000, steps: 4440
Episode 84: reward: 8.000, steps: 4437
Episode 85: reward: 8.000, steps: 4425
Episode 86: reward: 8.000, steps: 4434
Episode 87: reward: 8.000, steps: 4440
Episode 88: reward: 8.000, steps: 4437
Episode 89: reward: 8.000, steps: 4429
Episode 90: reward: 8.000, steps: 4430
Episode 91: reward: 8.000, steps: 4427
Episode 92: reward: 8.000, steps: 4447
Episode 93: reward: 8.000, steps: 4448
Episode 94: reward: 8.000, steps: 4431
Episode 95: reward: 8.000, steps: 4447
Episode 96: reward: 8.000, steps: 4455
Episode 97: reward: 8.000, steps: 4429

```
Episode 98: reward: 8.000, steps: 4425  
Episode 99: reward: 8.000, steps: 4434  
Episode 100: reward: 8.000, steps: 4430  
double_dqn: media = 8.00 | % episodios  $\geq$  200 pts = 0.0%  
Guardado 'test_results_double_dqn_20260731_234157.json'.
```

Dueling DQN

```
In [ ]: test_agent('dueling_dqn')
```

Model: "model_6"

Layer (type)	Output Shape	Param #	Connected to
=====			
input_4 (InputLayer)	[(None, 4, 84, 84)]	0	
permute_3 (Permute)	(None, 84, 84, 4)	0	input_4[0][0]
conv2d_9 (Conv2D)	(None, 20, 20, 32)	8224	permute_3[0][0]
conv2d_10 (Conv2D)	(None, 9, 9, 64)	32832	conv2d_9[0][0]
conv2d_11 (Conv2D)	(None, 7, 7, 64)	36928	conv2d_10[0][0]
flatten_3 (Flatten)	(None, 3136)	0	conv2d_11[0][0]
dense_6 (Dense)	(None, 512)	1606144	flatten_3[0][0]
dense_7 (Dense)	(None, 1)	513	dense_6[0][0]
dense_8 (Dense)	(None, 9)	4617	dense_6[0][0]
dueling_combine (DuelingCombine)	(None, 9)	0	dense_7[0][0] dense_8[0][0]
=====			
Total params: 1,689,258			
Trainable params: 1,689,258			
Non-trainable params: 0			

None

Testing for 100 episodes ...

Episode 1: reward: 480.000, steps: 8860
Episode 2: reward: 469.000, steps: 8882
Episode 3: reward: 494.000, steps: 8873
Episode 4: reward: 584.000, steps: 10000
Episode 5: reward: 578.000, steps: 10000
Episode 6: reward: 583.000, steps: 10000
Episode 7: reward: 564.000, steps: 10000
Episode 8: reward: 477.000, steps: 8848
Episode 9: reward: 599.000, steps: 10000
Episode 10: reward: 591.000, steps: 10000

Episode 11: reward: 464.000, steps: 8850
Episode 12: reward: 605.000, steps: 10000
Episode 13: reward: 585.000, steps: 10000
Episode 14: reward: 594.000, steps: 10000
Episode 15: reward: 571.000, steps: 10000
Episode 16: reward: 591.000, steps: 10000
Episode 17: reward: 557.000, steps: 10000
Episode 18: reward: 499.000, steps: 8863
Episode 19: reward: 479.000, steps: 8907
Episode 20: reward: 563.000, steps: 10000
Episode 21: reward: 588.000, steps: 10000
Episode 22: reward: 571.000, steps: 10000
Episode 23: reward: 546.000, steps: 10000
Episode 24: reward: 413.000, steps: 8842
Episode 25: reward: 645.000, steps: 10000
Episode 26: reward: 573.000, steps: 10000
Episode 27: reward: 563.000, steps: 10000
Episode 28: reward: 490.000, steps: 8858
Episode 29: reward: 582.000, steps: 10000
Episode 30: reward: 482.000, steps: 8884
Episode 31: reward: 522.000, steps: 10000
Episode 32: reward: 483.000, steps: 8900
Episode 33: reward: 599.000, steps: 10000
Episode 34: reward: 416.000, steps: 8836
Episode 35: reward: 580.000, steps: 10000
Episode 36: reward: 494.000, steps: 8866
Episode 37: reward: 575.000, steps: 10000
Episode 38: reward: 619.000, steps: 10000
Episode 39: reward: 412.000, steps: 8895
Episode 40: reward: 458.000, steps: 8910
Episode 41: reward: 581.000, steps: 10000
Episode 42: reward: 649.000, steps: 10000
Episode 43: reward: 591.000, steps: 10000
Episode 44: reward: 582.000, steps: 10000
Episode 45: reward: 603.000, steps: 10000
Episode 46: reward: 560.000, steps: 10000
Episode 47: reward: 544.000, steps: 10000
Episode 48: reward: 571.000, steps: 10000
Episode 49: reward: 575.000, steps: 10000
Episode 50: reward: 485.000, steps: 8810
Episode 51: reward: 588.000, steps: 10000

Episode 52: reward: 575.000, steps: 10000
Episode 53: reward: 550.000, steps: 10000
Episode 54: reward: 547.000, steps: 10000
Episode 55: reward: 483.000, steps: 8910
Episode 56: reward: 573.000, steps: 10000
Episode 57: reward: 451.000, steps: 8846
Episode 58: reward: 569.000, steps: 10000
Episode 59: reward: 534.000, steps: 10000
Episode 60: reward: 438.000, steps: 8883
Episode 61: reward: 608.000, steps: 10000
Episode 62: reward: 440.000, steps: 8916
Episode 63: reward: 584.000, steps: 10000
Episode 64: reward: 583.000, steps: 10000
Episode 65: reward: 629.000, steps: 10000
Episode 66: reward: 549.000, steps: 10000
Episode 67: reward: 587.000, steps: 10000
Episode 68: reward: 600.000, steps: 10000
Episode 69: reward: 601.000, steps: 10000
Episode 70: reward: 461.000, steps: 8881
Episode 71: reward: 590.000, steps: 10000
Episode 72: reward: 577.000, steps: 10000
Episode 73: reward: 584.000, steps: 10000
Episode 74: reward: 591.000, steps: 10000
Episode 75: reward: 533.000, steps: 10000
Episode 76: reward: 612.000, steps: 10000
Episode 77: reward: 475.000, steps: 8857
Episode 78: reward: 589.000, steps: 10000
Episode 79: reward: 574.000, steps: 10000
Episode 80: reward: 564.000, steps: 10000
Episode 81: reward: 563.000, steps: 10000
Episode 82: reward: 437.000, steps: 8897
Episode 83: reward: 558.000, steps: 10000
Episode 84: reward: 586.000, steps: 10000
Episode 85: reward: 578.000, steps: 10000
Episode 86: reward: 584.000, steps: 10000
Episode 87: reward: 485.000, steps: 8896
Episode 88: reward: 478.000, steps: 8943
Episode 89: reward: 560.000, steps: 10000
Episode 90: reward: 559.000, steps: 10000
Episode 91: reward: 614.000, steps: 10000
Episode 92: reward: 549.000, steps: 10000

```
Episode 93: reward: 602.000, steps: 10000
Episode 94: reward: 594.000, steps: 10000
Episode 95: reward: 567.000, steps: 10000
Episode 96: reward: 612.000, steps: 10000
Episode 97: reward: 605.000, steps: 10000
Episode 98: reward: 535.000, steps: 10000
Episode 99: reward: 557.000, steps: 10000
Episode 100: reward: 563.000, steps: 10000
dueling_dqn: media = 551.04 | % episodios  $\geq$  200 pts = 100.0%
Guardado 'test_results_dueling_dqn_20260728_142544.json'.
```

3. Justificación de los parámetros seleccionados y de los resultados obtenidos

Con el fin de realizar una comparación justa entre las distintas variantes de DQN, todos los experimentos se han llevado a cabo utilizando la misma configuración de entrenamiento. De esta forma, las diferencias observadas en el rendimiento pueden atribuirse principalmente a las modificaciones introducidas en cada algoritmo y no a cambios en los hiperparámetros.

Los principales hiperparámetros utilizados son los siguientes:

Hiperparámetro	Valor
Learning rate	0.00025
Gamma	0.99
Batch size	32
Replay memory	1.000.000
Target update	10.000 pasos
Epsilon inicial	1.0
Epsilon final	0.1
Epsilon decay	250.000 pasos
Window Length	4

- **Learning rate (0.00025):** se utiliza una tasa de aprendizaje reducida para favorecer un entrenamiento estable y evitar cambios demasiado bruscos en los pesos de la red neuronal.
- **Factor de descuento ($\gamma = 0.99$):** se emplea un valor elevado para dar importancia a las recompensas futuras, ya que en Enduro muchas decisiones tienen efectos que se reflejan varios pasos más adelante.
- **Tamaño del lote (batch size):** se utilizan minibatches de experiencias almacenadas en el replay buffer para aproximar el gradiente y mejorar la eficiencia del aprendizaje.
- **Replay Memory (1.000.000 de transiciones):** un buffer de gran tamaño permite almacenar experiencias diversas y romper la correlación entre muestras consecutivas, favoreciendo un entrenamiento más estable.
- **Actualización de la red objetivo (cada 10 000 pasos):** la red objetivo se actualiza periódicamente para proporcionar estimaciones más estables durante el aprendizaje y reducir las oscilaciones del entrenamiento.
- **Exploración ϵ -greedy:** la probabilidad de exploración disminuye progresivamente desde un valor inicial de 1 hasta un mínimo de 0.1. Esto permite que el agente explore el entorno durante las primeras etapas del entrenamiento y, posteriormente, aproveche el conocimiento adquirido.
- **Epsilon decay (250 000 pasos):** la reducción gradual de la exploración evita que el agente deje de explorar demasiado pronto y favorece un equilibrio entre exploración y explotación durante el entrenamiento.
- **Window Length (4):** procesar 4 *frames* consecutivos es vital en juegos de conducción. Un solo *frame* estático no revela la velocidad ni la trayectoria de los coches rivales; agruparlos permite a la red neuronal inferir el movimiento y la profundidad espacial para anticiparse y poder esquivarlos a tiempo.

Los resultados obtenidos con el modelo Baseline DQN muestran que el agente es capaz de aprender una estrategia eficaz para desenvolverse en el entorno Enduro. La evolución de las métricas durante el entrenamiento refleja una mejora progresiva del rendimiento y una recompensa media elevada durante la fase de evaluación. Estos resultados convierten al modelo Baseline en una referencia adecuada para comparar el comportamiento del resto de variantes implementadas.

En teoría, Double DQN suele obtener mejores resultados que el DQN clásico al reducir la sobreestimación de los valores Q pero en nuestros experimentos ha ocurrido lo contrario. El agente ha obtenido una recompensa media de aproximadamente 8 puntos, lo que indica que no ha conseguido aprender una estrategia adecuada para jugar a Enduro.

Planteamos como hipótesis que la red ha caído en un mínimo local, adoptando lo que se conoce como una "política subóptima perezosa". En las mecánicas de *Enduro*, acelerar conlleva un riesgo alto de colisionar con otros vehículos y perder puntos. Al parecer, si los pesos de evaluación no se sintonizan bien en las etapas tempranas, el agente descubre rápidamente que la estrategia más "segura" para no recibir penalizaciones es simplemente frenar constantemente o no acelerar nunca. De esta forma, el coche sobrevive sin chocar durante todo el episodio, pero apenas adelanta a los rivales, lo que justifica una puntuación tan baja (8.00) tras 100 episodios de prueba.

Este resultado puede deberse a que la configuración de hiperparámetros que se ha utilizado no era la más adecuada para esta variante o a que el tiempo de entrenamiento no ha sido suficiente para que el agente aprendiera correctamente y lograse salir de esa "zona de confort" inicial.

En el caso de Dueling DQN, los resultados muestran un comportamiento más sólido durante el entrenamiento y una mejora en la recompensa obtenida durante la evaluación respecto al modelo Baseline. Esto indica que la modificación de la arquitectura resulta beneficiosa para este entorno, permitiendo al agente aprender una estrategia más eficaz. En conjunto, esta variante es la que ofrece el mejor rendimiento de las tres evaluadas en este trabajo.

Gráficas

```
In [ ]: %pip install numpy>=1.20
        %pip install matplotlib
```

Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: matplotlib in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (3.7.5)

Requirement already satisfied: importlib-resources>=3.2.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (6.4.5)

Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (1.4.7)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (3.1.4)

Requirement already satisfied: packaging>=20.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (26.2)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (4.57.0)

Requirement already satisfied: pillow>=6.2.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (9.5.0)

Requirement already satisfied: cycler>=0.10 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (0.12.1)

Requirement already satisfied: numpy<2,>=1.20 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (1.24.4)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (1.1.1)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: zipp>=3.1.0 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from importlib-resources>=3.2.0->matplotlib) (3.20.2)

Requirement already satisfied: six>=1.5 in c:\users\rosai\onedrive\documentos\learning\viu - master ia\08miar - aprendizaje por refuerzo\proyecto\gym_env\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.15.0)

Note: you may need to restart the kernel to use updated packages.

```
In [ ]: import json
import matplotlib.pyplot as plt
```

```
run_timestamps = {
    'baseline_dqn': "20260725_164222",
    'double_dqn': "20260731_234157",
    'dueling_dqn': "20260728_142544",
}
```

```
def generate_training_graphs(name):
```

```
    with open(f'dqn_Enduro-v0_{name}_log_{run_timestamps[name]}.json') as f:
```

```

log_data = json.load(f)

print(log_data.keys())
print("Episodios registrados:", len(log_data['episode_reward']))

# Eje X: pasos acumulados de entrenamiento (más informativo que el índice de episodio)
steps = log_data['nb_steps']

fig, axes = plt.subplots(4, 1, figsize=(10, 14), sharex=True)

axes[0].plot(steps, log_data['episode_reward'])
axes[0].set_ylabel('Reward por episodio')
axes[0].set_title(f'Progresión del entrenamiento - {name}')
axes[0].grid(True)

axes[1].plot(steps, log_data['mean_q'], color='orange')
axes[1].set_ylabel('mean_q')
axes[1].grid(True)

axes[2].plot(steps, log_data['mae'], color='green')
axes[2].set_ylabel('mae')
axes[2].grid(True)

axes[3].plot(steps, log_data['loss'], color='red')
axes[3].set_ylabel('loss')
axes[3].set_xlabel('Pasos de entrenamiento')
axes[3].grid(True)

plt.tight_layout()
plt.show()

```

```

In [ ]: import json
import numpy as np
import matplotlib.pyplot as plt

def generate_test_graphs(name):
    with open(f'test_results_{name}_{run_timestamps[name]}.json') as f:
        test_results = json.load(f)

```

```

rewards_arr = np.array(test_results[name]['rewards'])
threshold = test_results[name]['threshold']

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Gráfica 1: recompensa por episodio de test, con la línea de umbral
axes[0].plot(range(1, len(rewards_arr) + 1), rewards_arr, marker='o', markersize=3)
axes[0].axhline(threshold, color='red', linestyle='--', label=f'Umbral día 1 ({threshold})')
axes[0].set_xlabel('Episodio de test')
axes[0].set_ylabel('Recompensa')
axes[0].set_title(f'{name} - Recompensa en 100 episodios de test')
axes[0].legend()
axes[0].grid(True)

# Gráfica 2: distribución (histograma)
axes[1].hist(rewards_arr, bins=15, edgecolor='black')
axes[1].axvline(threshold, color='red', linestyle='--', label=f'Umbral día 1 ({threshold})')
axes[1].set_xlabel('Recompensa')
axes[1].set_ylabel('Nº de episodios')
axes[1].set_title(f'{name} - Distribución de recompensas')
axes[1].legend()
axes[1].grid(True)

plt.tight_layout()
plt.savefig(f'test_{name}.png', dpi=150)
plt.show()

print(f"Media: {test_results[name]['mean']:.2f}")
print(f"% episodios ≥ {threshold}: {test_results[name]['pct_above_threshold']:.1f}%")

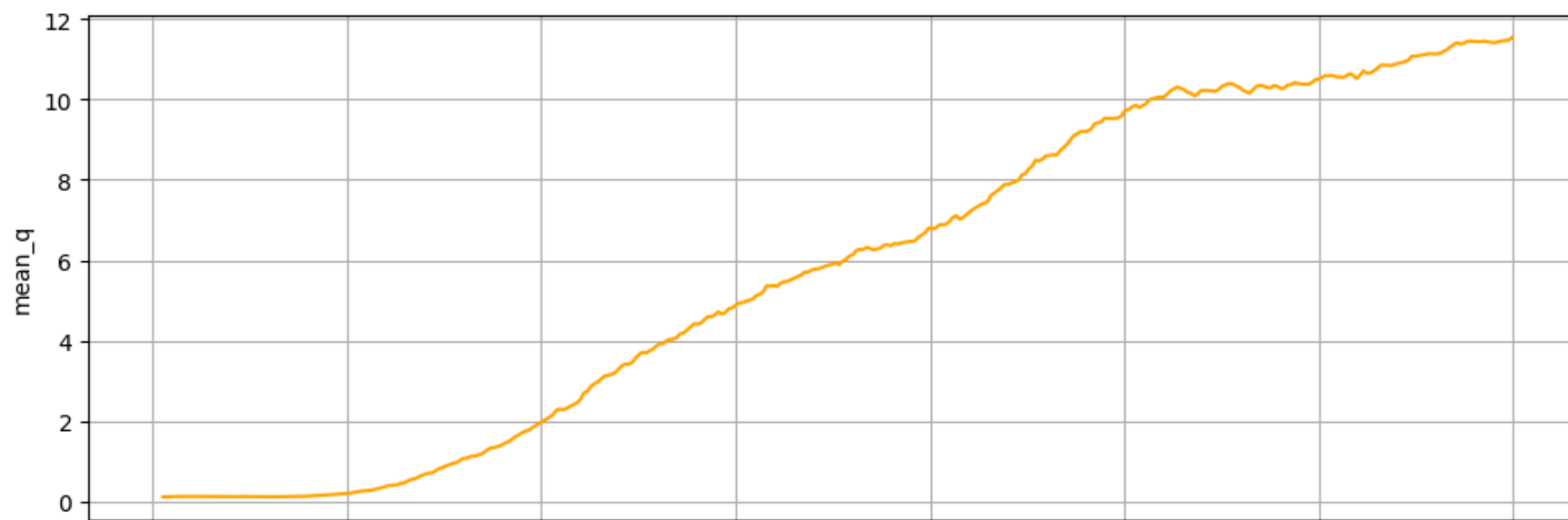
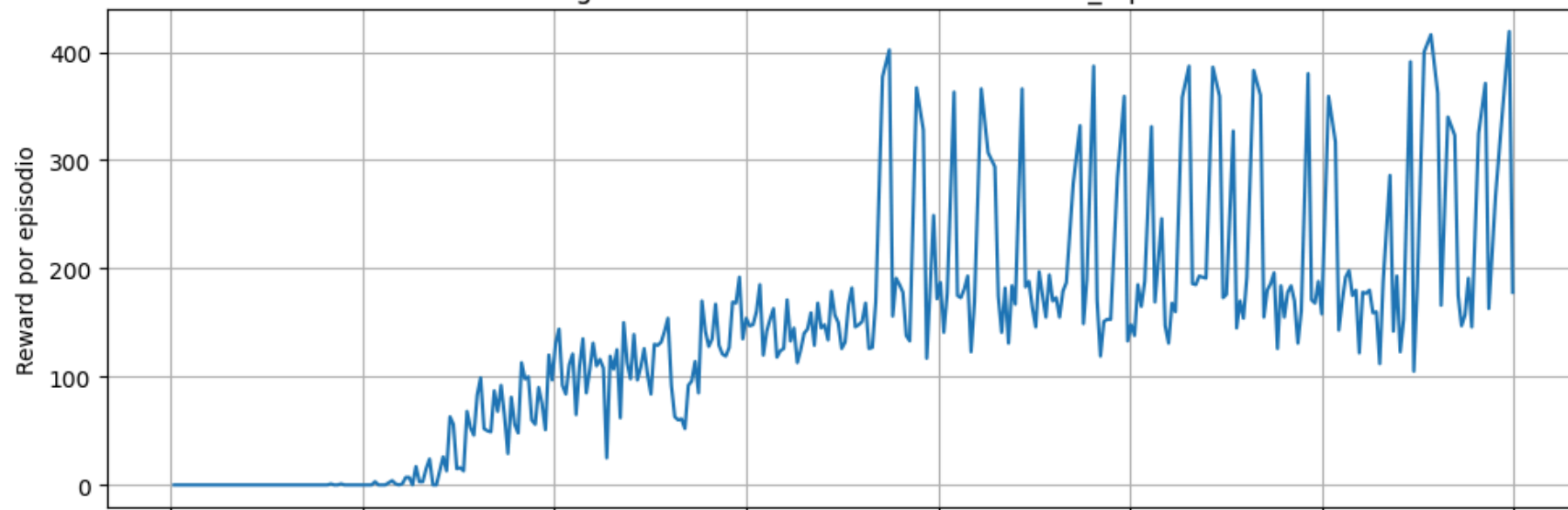
```

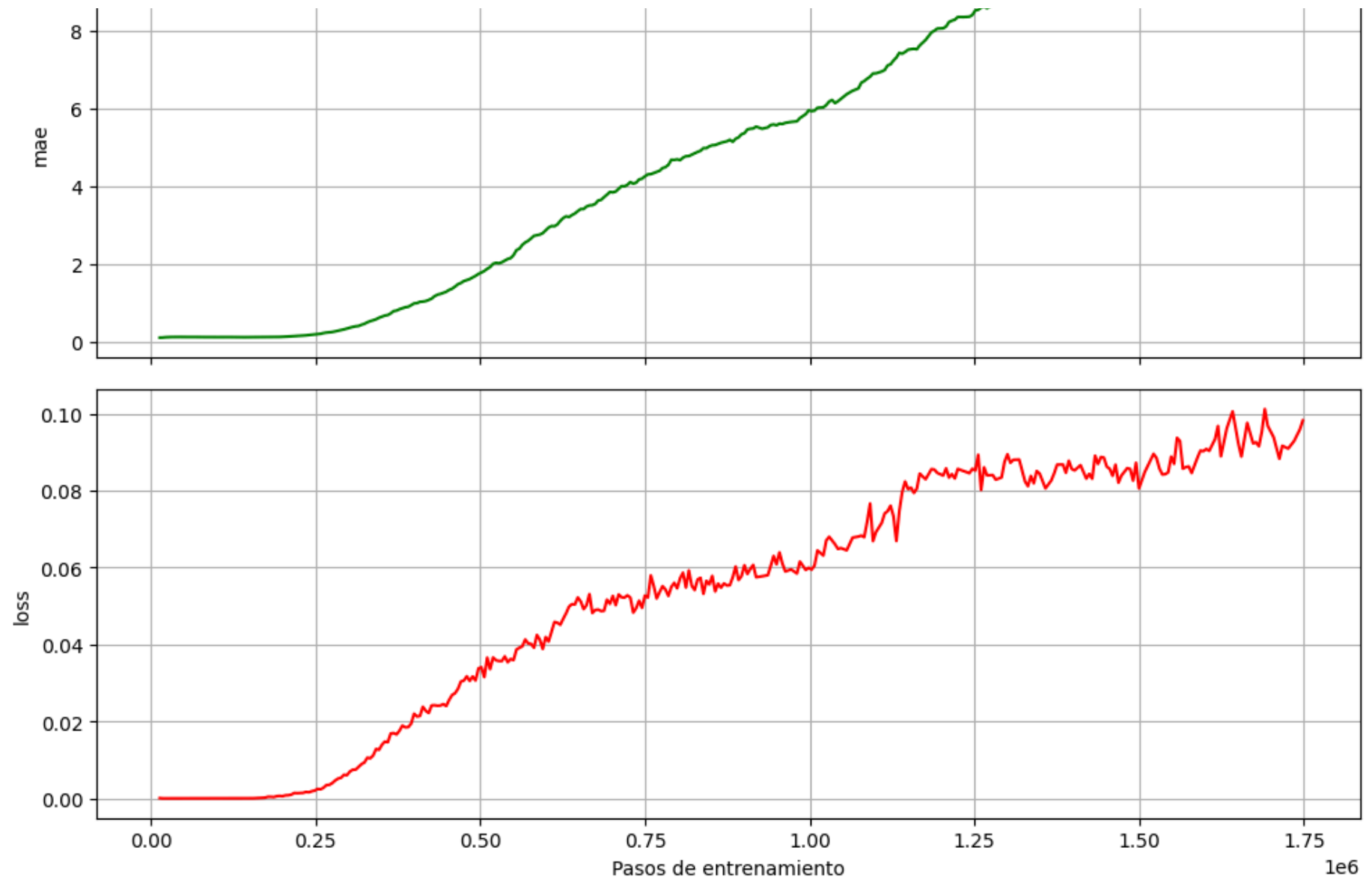
Baseline DQN

```
In [ ]: generate_training_graphs('baseline_dqn')
```

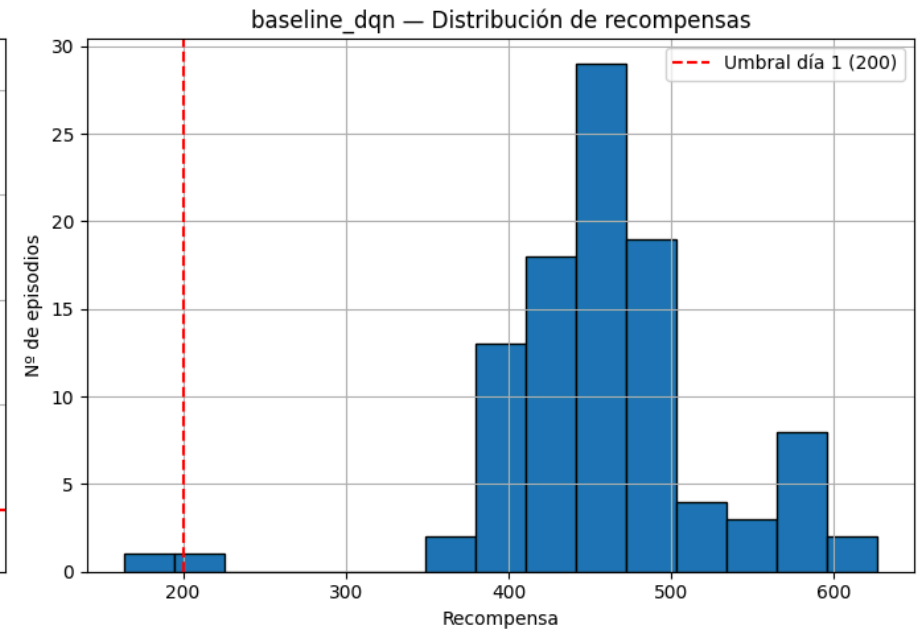
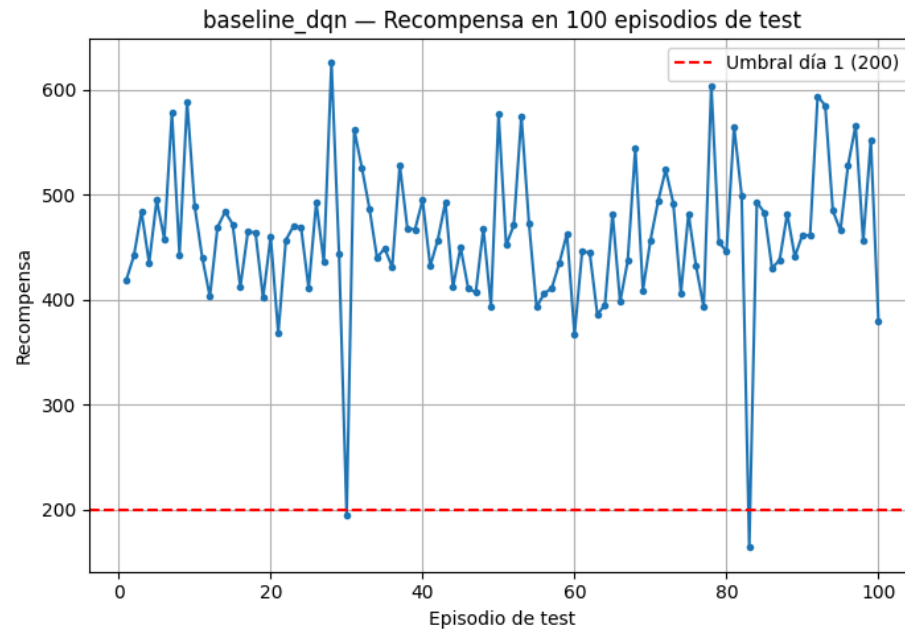
```
dict_keys(['loss', 'mae', 'mean_q', 'mean_eps', 'episode_reward', 'nb_episode_steps', 'nb_steps', 'episode', 'duration'])
Episodios registrados: 355
```


Progresión del entrenamiento — baseline_dqn





```
In [ ]: generate_test_graphs('baseline_dqn')
```



Media: 461.29
 % episodios ≥ 200 : 98.0%

Las gráficas correspondientes al modelo Baseline DQN muestran una evolución positiva a lo largo del entrenamiento. La recompensa obtenida aumenta de forma progresiva tras las primeras etapas de exploración, lo que indica que el agente va aprendiendo una estrategia cada vez más eficaz para desenvolverse en el entorno. Aunque tanto la función de pérdida como los valores Q presentan pequeñas oscilaciones, este comportamiento es habitual en los algoritmos DQN y no impide que el entrenamiento converja hacia una política estable.

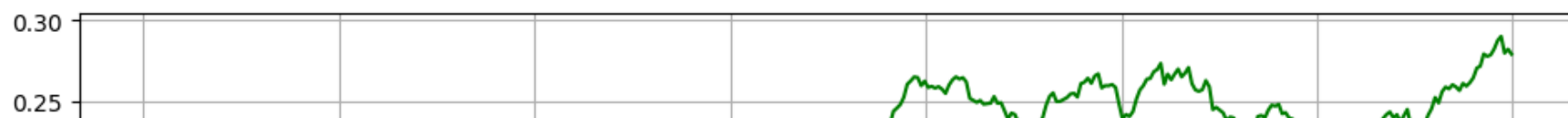
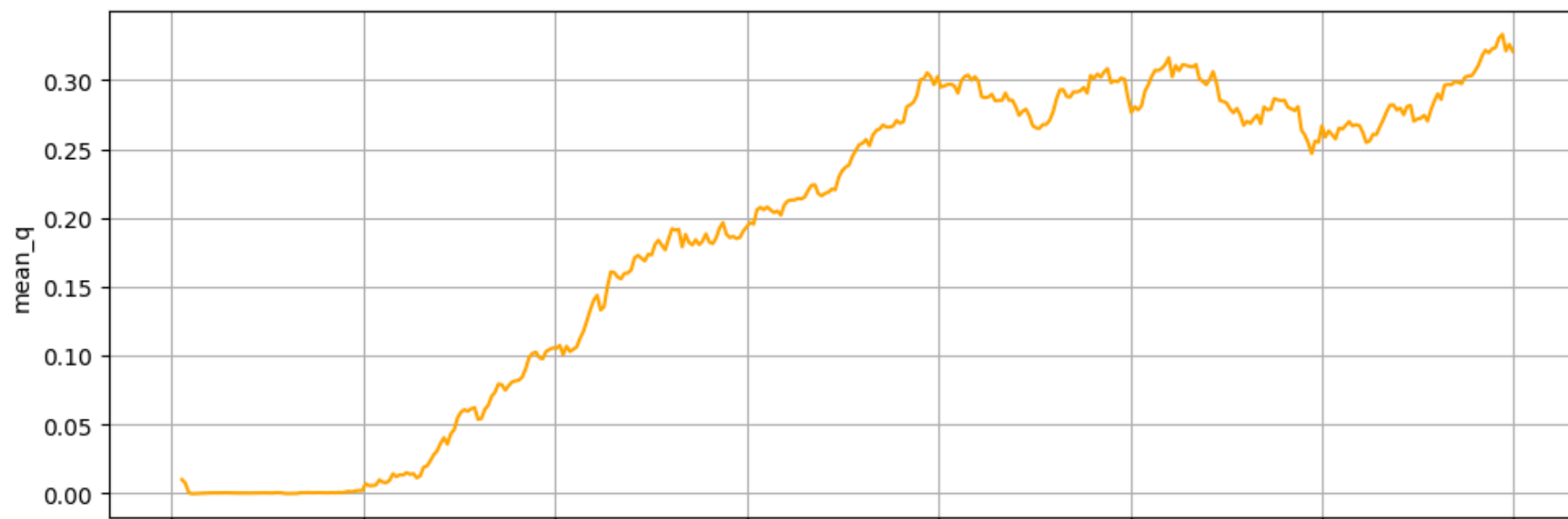
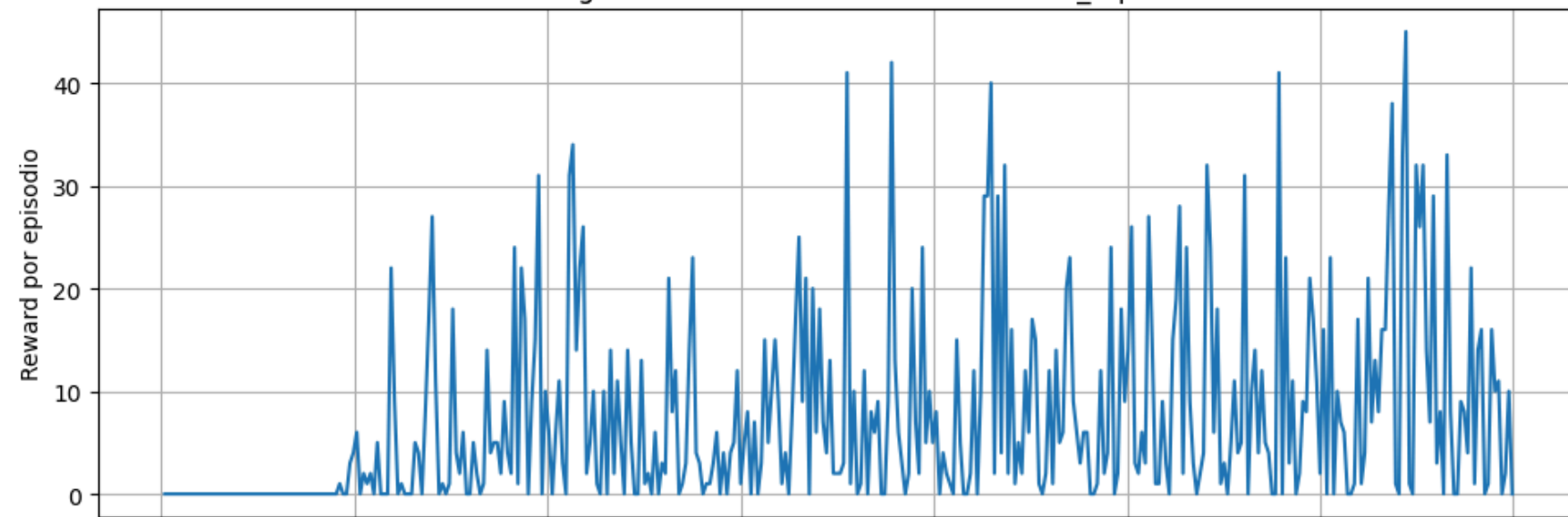
Los resultados de la evaluación confirman este comportamiento, alcanzando una recompensa media de aproximadamente 461 puntos, muy por encima del mínimo exigido para la práctica.

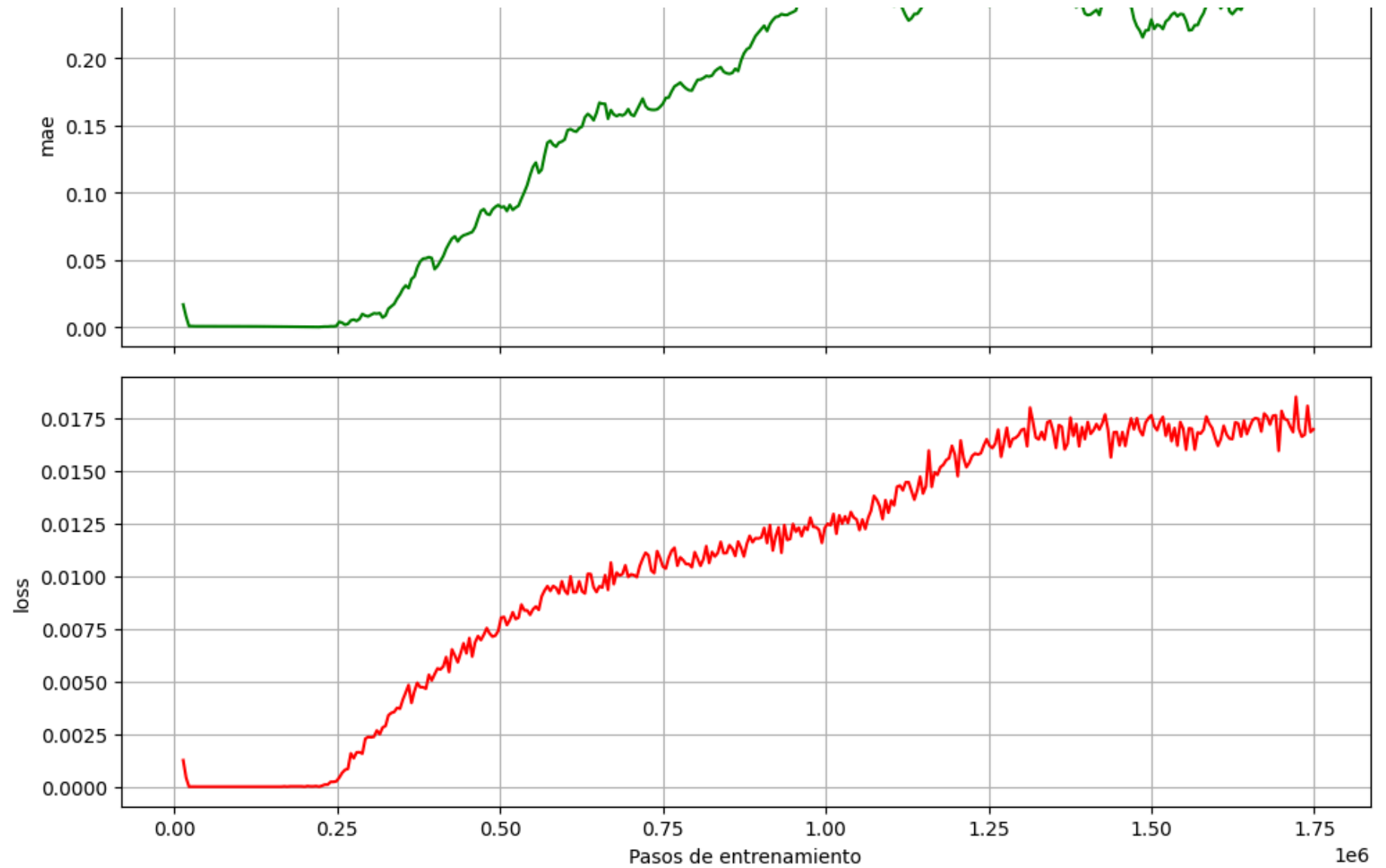
Double DQN

```
In [ ]: generate_training_graphs('double_dqn')
```

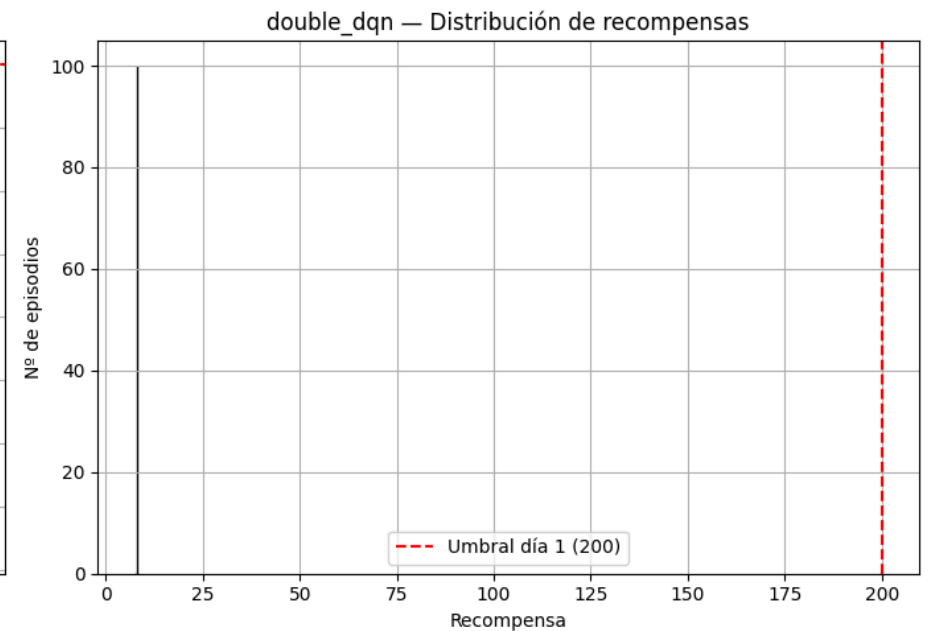
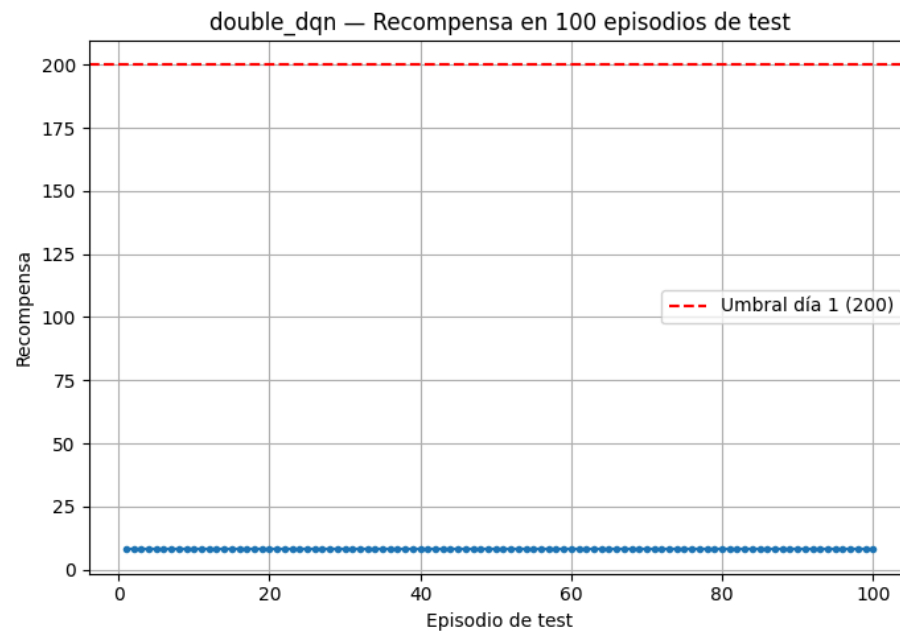
```
dict_keys(['loss', 'mae', 'mean_q', 'mean_eps', 'episode_reward', 'nb_episode_steps', 'nb_steps', 'episode', 'duration'])
Episodios registrados: 394
```

Progresión del entrenamiento — double_dqn





```
In [ ]: generate_test_graphs('double_dqn')
```



Media: 8.00
 % episodios ≥ 200 : 0.0%

En el caso de Double DQN, las gráficas muestran un comportamiento muy diferente al observado en el modelo Baseline. La recompensa apenas aumenta durante el entrenamiento y el agente no consigue mejorar de forma significativa su rendimiento. El modelo terminó el entrenamiento con un `mean_q` muy bajo, cercano a 0.32, en comparación con el Baseline DQN que alcanzó un valor de 11.46, lo que demuestra que la variante Double no ha logrado propagar correctamente las recompensas futuras durante el entrenamiento. Esto se refleja también en la evaluación final, donde obtiene una recompensa media cercana a 8 puntos.

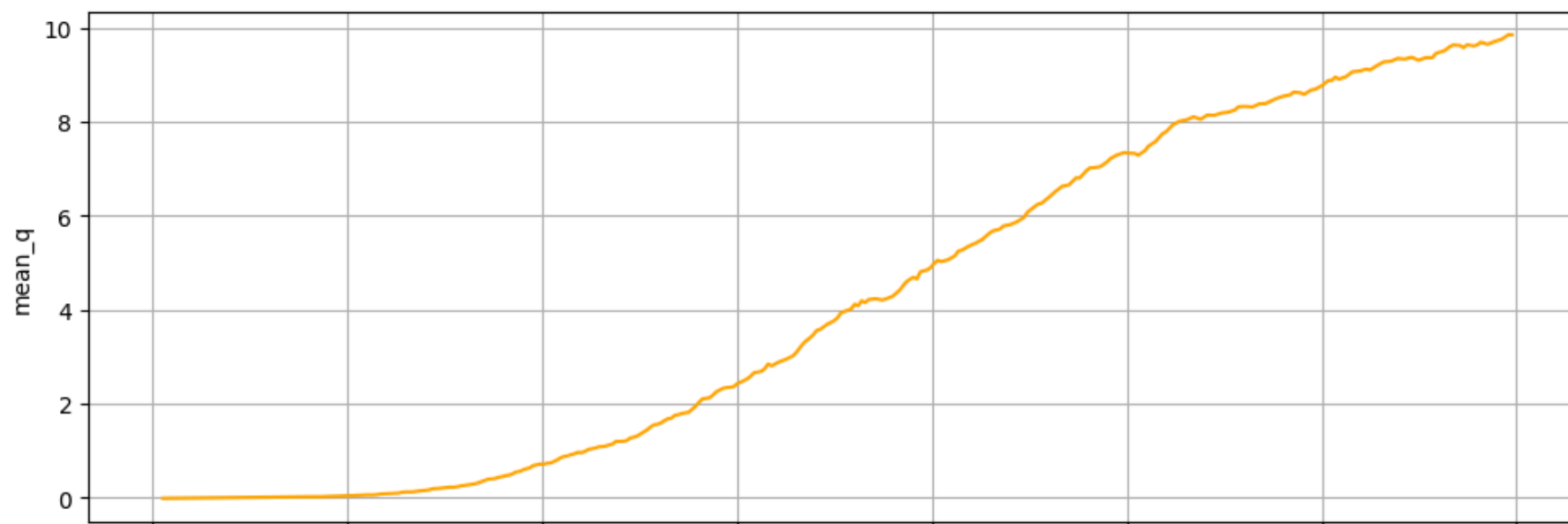
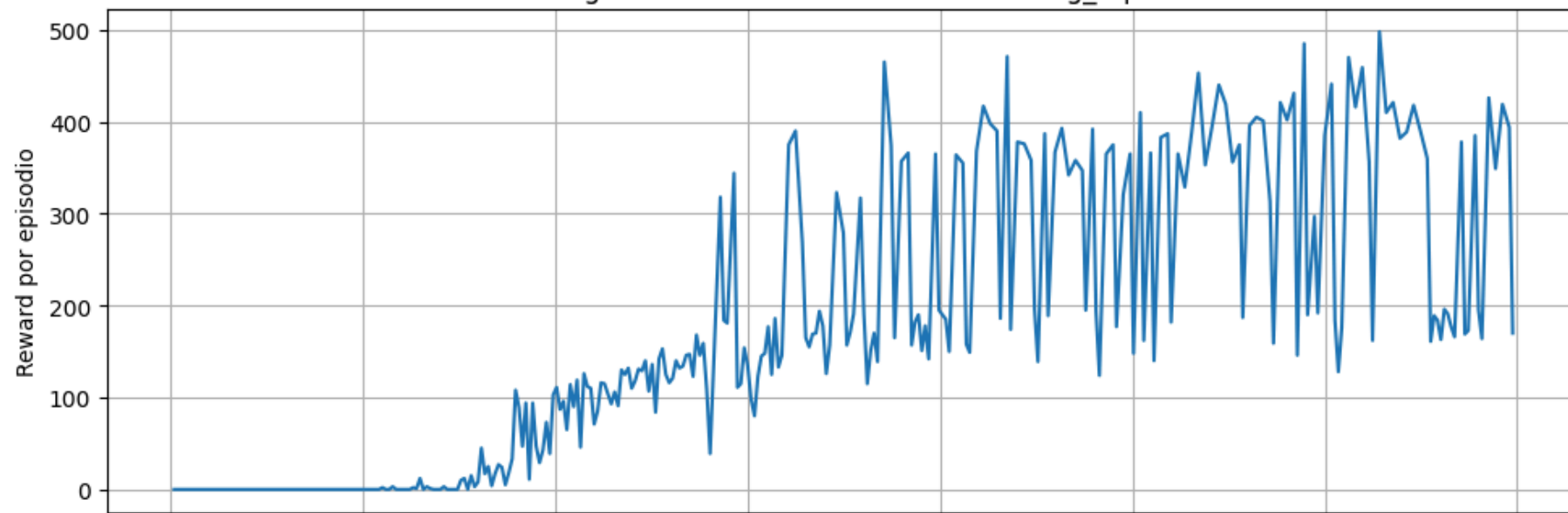
Aunque Double DQN está diseñado para reducir la sobreestimación de los valores Q, en este experimento la configuración utilizada no ha permitido que el agente aprenda una estrategia eficaz.

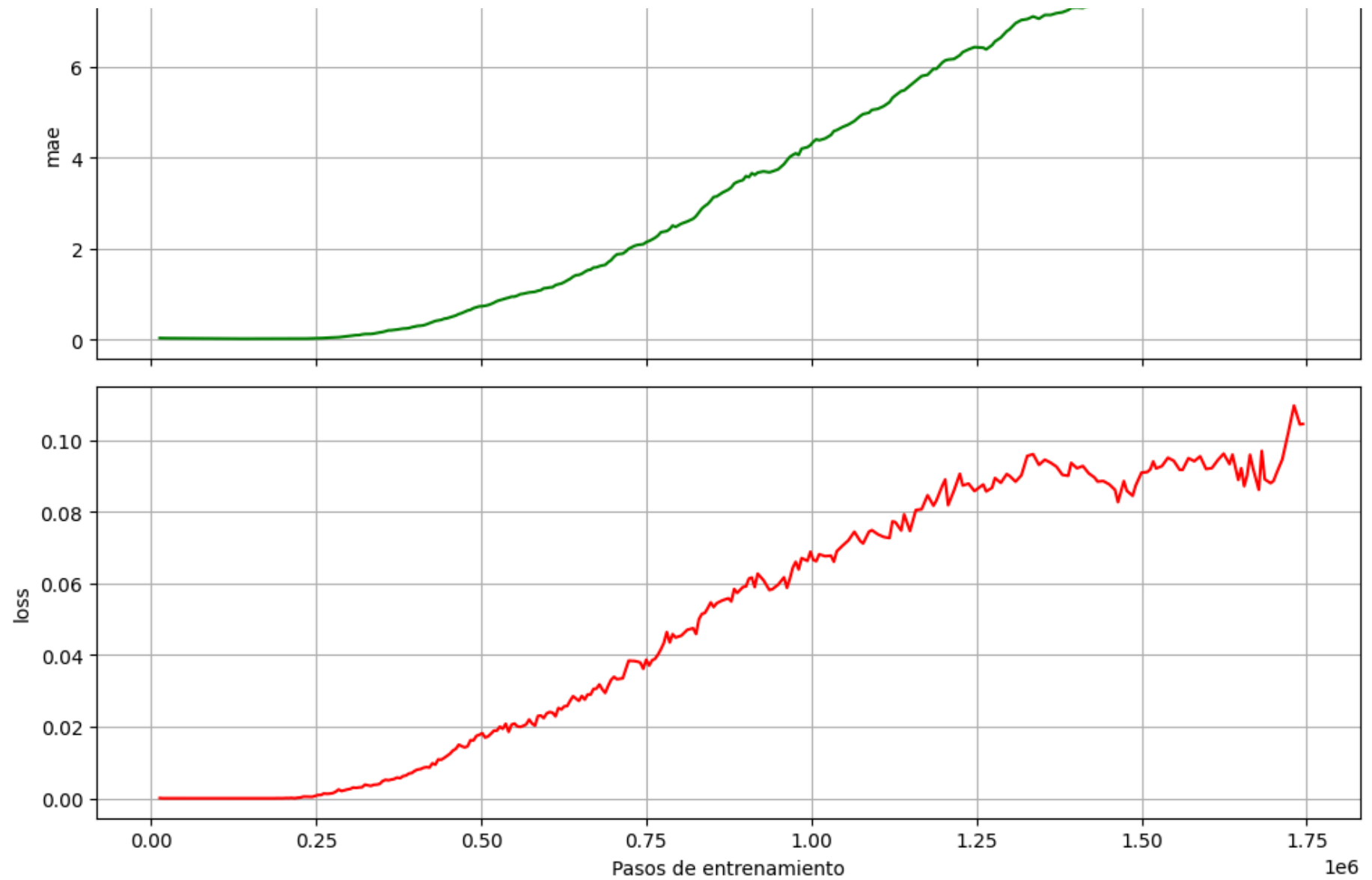
Dueling DQN

```
In [ ]: generate_training_graphs('dueling_dqn')
```

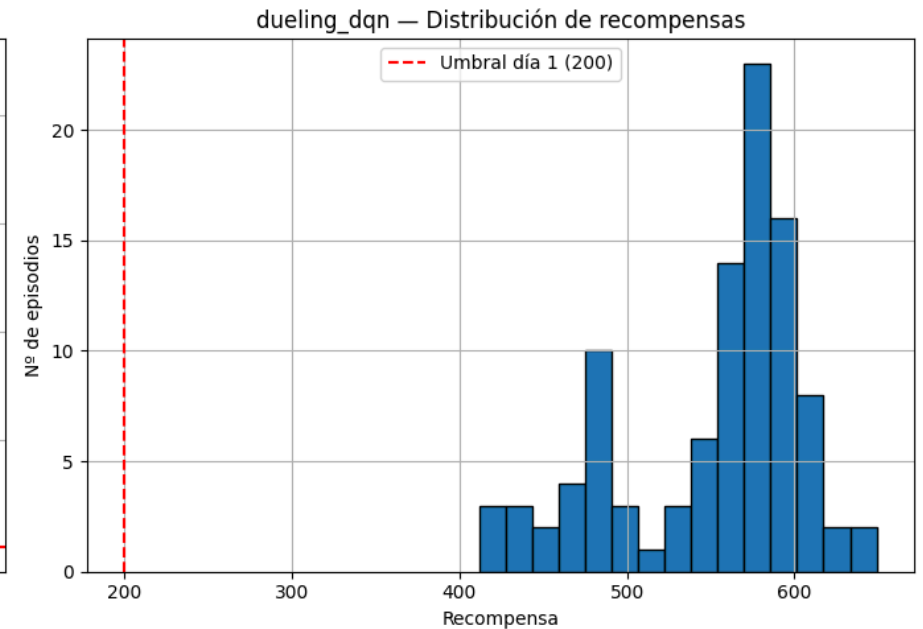
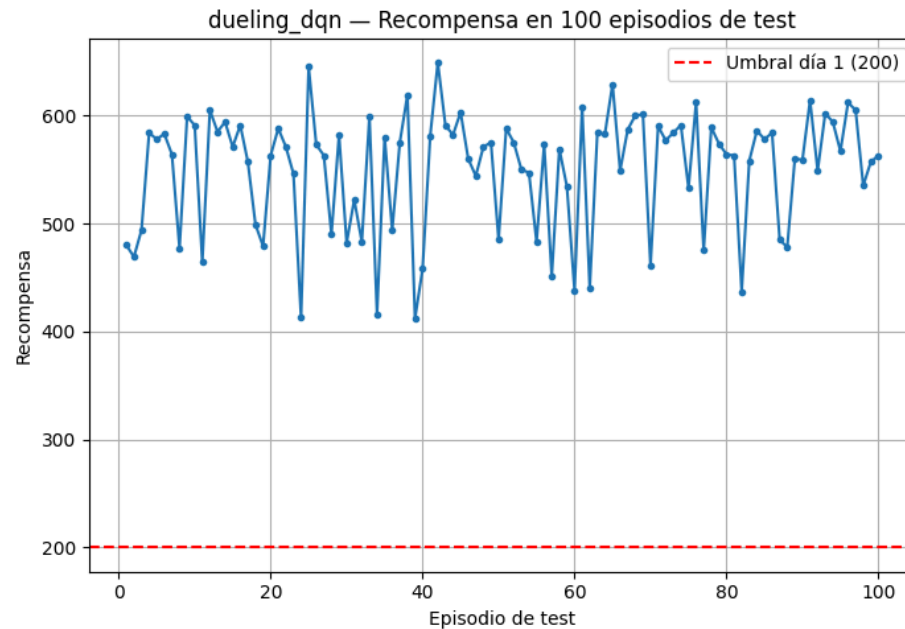
```
dict_keys(['loss', 'mae', 'mean_q', 'mean_eps', 'episode_reward', 'nb_episode_steps', 'nb_steps', 'episode', 'duration'])
Episodios registrados: 316
```

Progresión del entrenamiento — dueling_dqn





```
In [ ]: generate_test_graphs('dueling_dqn')
```

Media: 551.04
% episodios ≥ 200 : 100.0%

Las gráficas obtenidas para Dueling DQN muestran la evolución más favorable de las tres variantes evaluadas. La recompensa aumenta de manera más consistente durante el entrenamiento y presenta una mayor estabilidad en comparación con los modelos anteriores. Este comportamiento también se observa en la fase de evaluación, donde el agente obtiene las mejores recompensas medias del estudio.

Estos resultados sugieren que la arquitectura Dueling DQN se adapta especialmente bien al entorno Enduro, ya que permite aprender de forma más eficiente el valor de cada estado y diferenciar mejor qué acciones aportan una mayor ventaja en cada situación.

Gráficas comparativas

```
In [ ]: import matplotlib.pyplot as plt

# 1. Curva de aprendizaje (recompensa media móvil durante el entrenamiento)
histories = {
    'baseline_dqn': f'dqn_Enduro-v0_baseline_dqn_log_{run_timestamps["baseline_dqn"]}.json',
    'double_dqn': f'dqn_Enduro-v0_double_dqn_log_{run_timestamps["double_dqn"]}.json',
    'dueling_dqn': f'dqn_Enduro-v0_dueling_dqn_log_{run_timestamps["dueling_dqn"]}.json',
```

```

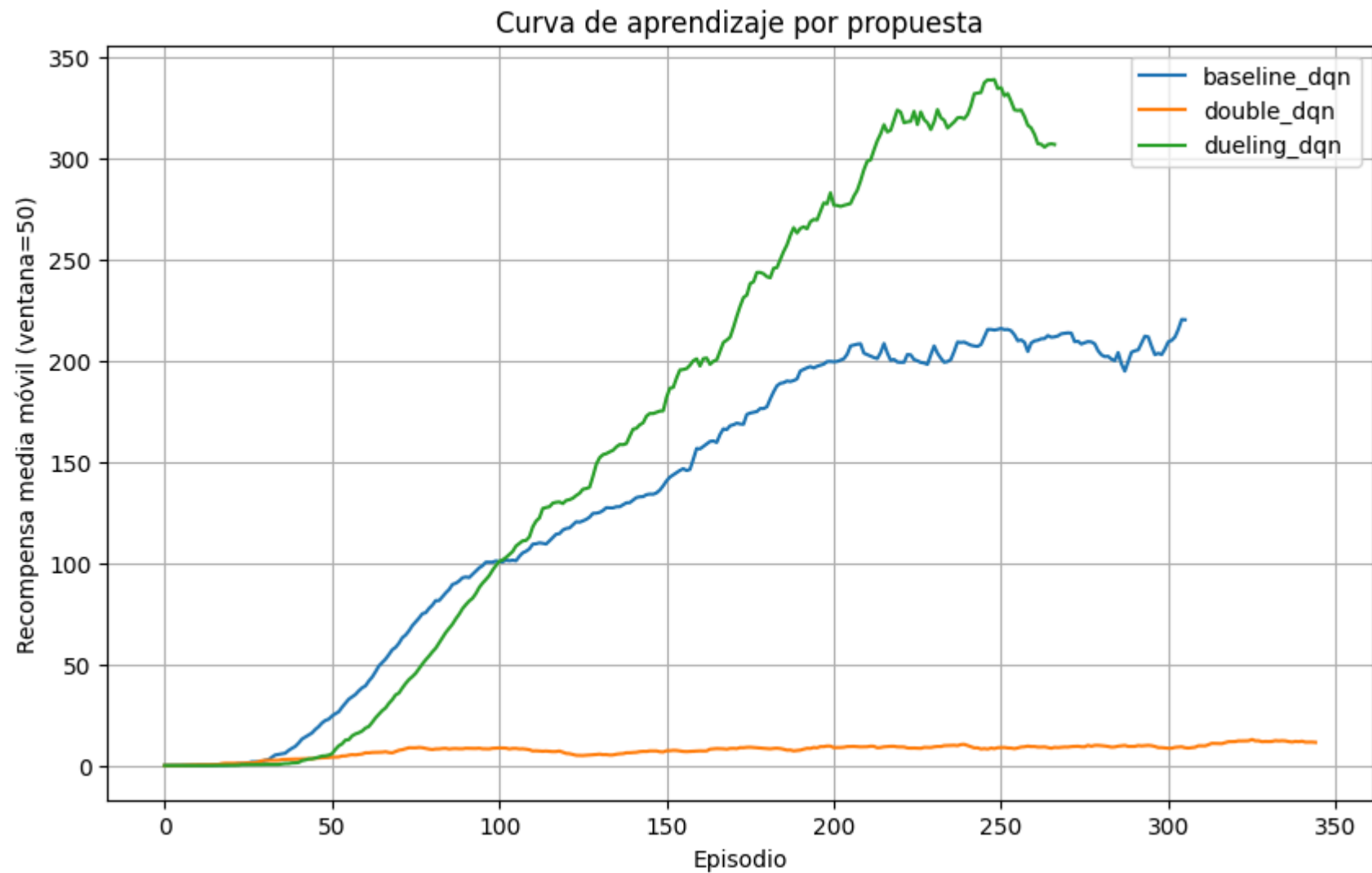
}
plt.figure(figsize=(10, 6))
for name, log_file in histories.items():
    with open(log_file) as f:
        log_data = json.load(f)
        rewards = log_data['episode_reward']
        window = 50
        moving_avg = np.convolve(rewards, np.ones(window)/window, mode='valid')
        plt.plot(moving_avg, label=name)
plt.xlabel('Episodio')
plt.ylabel('Recompensa media móvil (ventana=50)')
plt.title('Curva de aprendizaje por propuesta')
plt.legend()
plt.grid(True)
plt.show()

# 2. Distribución de recompensas en test (boxplot)
test_results = {
    'baseline_dqn': f'test_results_baseline_dqn_{run_timestamps["baseline_dqn"]}.json',
    'double_dqn': f'test_results_double_dqn_{run_timestamps["double_dqn"]}.json',
    'dueling_dqn': f'test_results_dueling_dqn_{run_timestamps["dueling_dqn"]}.json',
}
results = {}
for name, test_file in test_results.items():
    with open(test_file) as f:
        test_data = json.load(f)
        results[name] = test_data[name]['rewards']
        REWARD_THRESHOLD = test_data[name]['threshold']
plt.figure(figsize=(8, 6))
plt.boxplot([results[name] for name in results], labels=list(results.keys()))
plt.axhline(REWARD_THRESHOLD, color='red', linestyle='--', label='Objetivo día 1')
plt.ylabel('Recompensa por episodio (test)')
plt.title('Distribución de recompensas en 100 episodios de test')
plt.legend()
plt.grid(True)
plt.show()

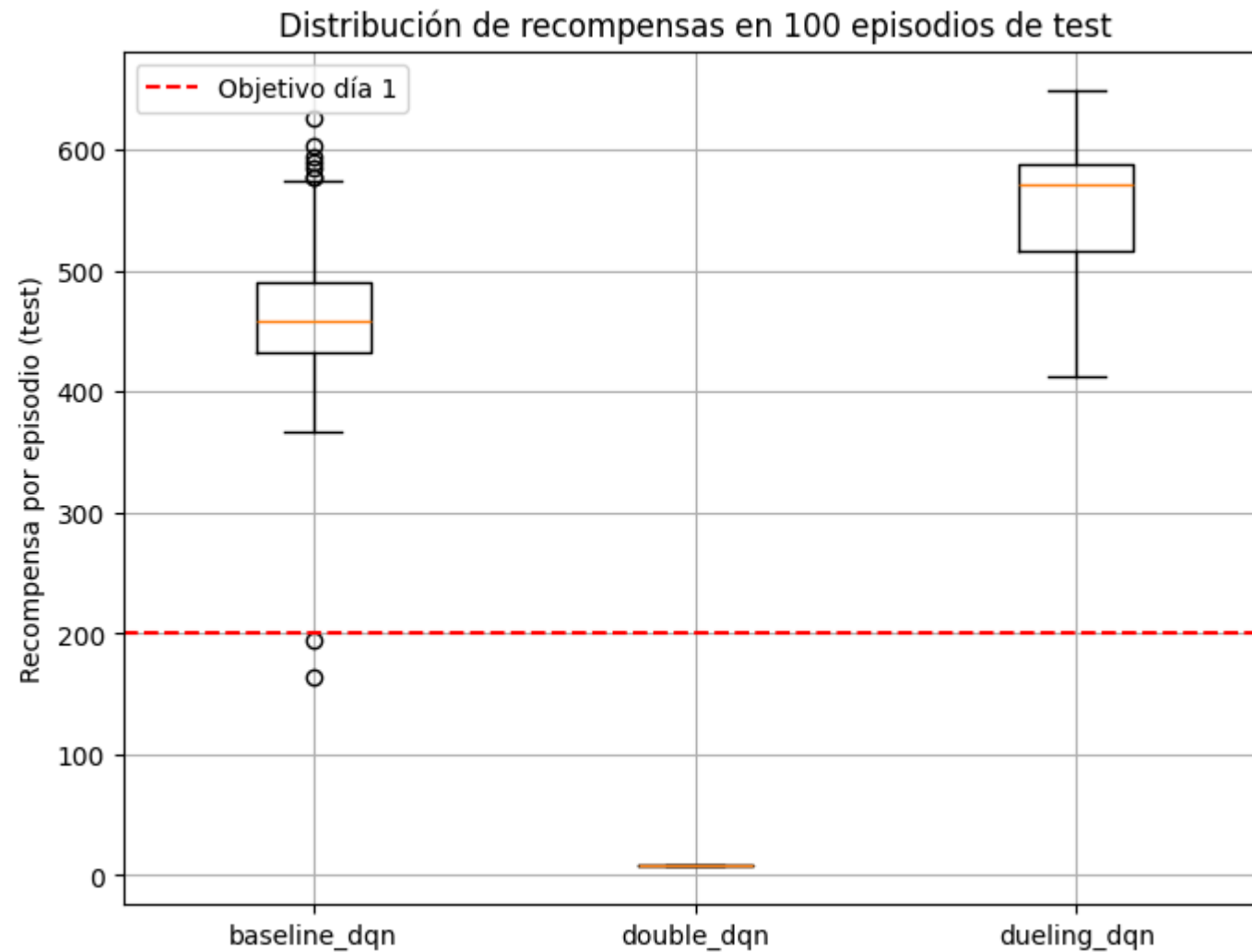
# 3. Recompensa por episodio de test (comparación directa)
plt.figure(figsize=(10, 6))
for name in results:
    plt.plot(results[name], label=name, alpha=0.7)

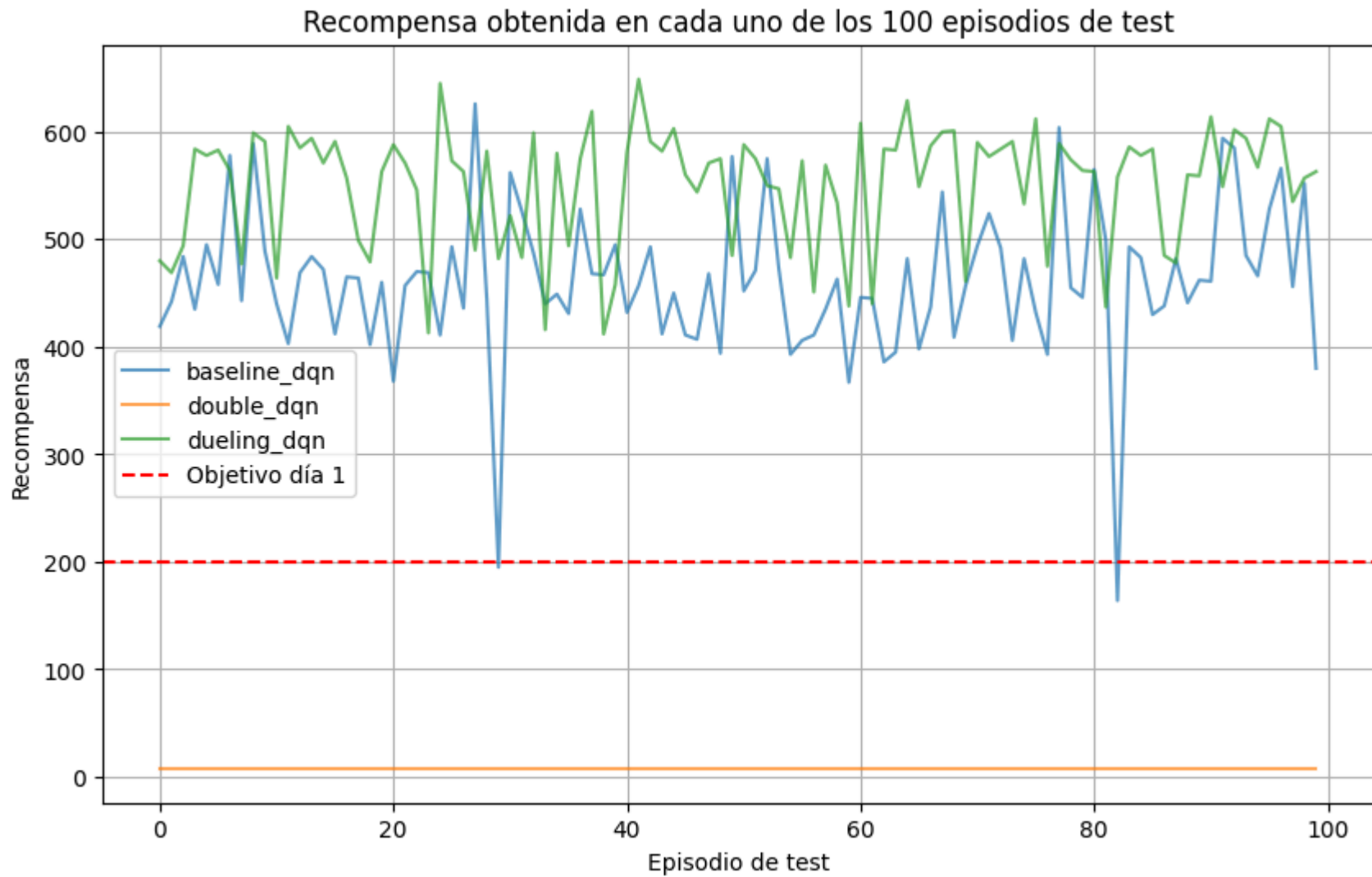
```

```
plt.axhline(REWARD_THRESHOLD, color='red', linestyle='--', label='Objetivo día 1')
plt.xlabel('Episodio de test')
plt.ylabel('Recompensa')
plt.title('Recompensa obtenida en cada uno de los 100 episodios de test')
plt.legend()
plt.grid(True)
plt.show()
```



C:\Users\rosai\AppData\Local\Temp\ipykernel_37544\2753323120.py:37: MatplotlibDeprecationWarning: The 'labels' parameter of box plot() has been renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
plt.boxplot([results[name] for name in results], labels=list(results.keys()))





En las gráficas comparativas se pueden ver claramente las diferencias de rendimiento entre las tres variantes implementadas.

Se observa que Baseline DQN y Dueling DQN consiguen aprender una estrategia eficaz para resolver el entorno, mientras que Double DQN no alcanza un rendimiento similar con la configuración utilizada. Entre las tres variantes, Dueling DQN obtiene los mejores resultados, mostrando una evolución más estable durante el entrenamiento y las mayores recompensas en la fase de evaluación.

En conjunto, estos resultados indican que la arquitectura Dueling DQN es la que mejor se adapta al entorno Enduro de las alternativas analizadas.

Conclusiones

En este proyecto se ha implementado y evaluado un agente de Aprendizaje por Refuerzo basado en DQN para el entorno Enduro-v0, comparando tres variantes de la arquitectura: DQN estándar, Double DQN y Dueling DQN.

Los resultados obtenidos muestran que las mejoras introducidas en un algoritmo no garantizan por sí solas un mejor rendimiento si no van acompañadas de un ajuste adecuado de los hiperparámetros y del proceso de entrenamiento. En nuestro caso, la versión Double DQN no consiguió aprender una política eficaz, obteniendo una recompensa media de únicamente 8 puntos en la fase de evaluación. Este comportamiento indica que el agente no llegó a converger correctamente y pone de manifiesto la importancia de ajustar parámetros como la exploración, la tasa de aprendizaje o el número de pasos de entrenamiento.

Por el contrario, tanto DQN estándar como Dueling DQN lograron superar ampliamente el umbral mínimo establecido para la práctica (200 puntos). El modelo DQN alcanzó una recompensa media cercana a los 461 puntos, demostrando una política estable y capaz de completar con éxito el primer día de carrera en la mayoría de los episodios de prueba. La arquitectura Dueling DQN mostró un rendimiento aún más sólido, obteniendo recompensas superiores de forma consistente durante la evaluación, lo que confirma la utilidad de separar la estimación del valor del estado y de la ventaja de cada acción en este tipo de entornos.

En conjunto, los resultados permiten concluir que la arquitectura Dueling DQN ha sido la alternativa más eficaz para resolver el problema planteado, ofreciendo un aprendizaje más robusto y un comportamiento más estable que las otras variantes evaluadas. Además, el desarrollo de este proyecto ha permitido comprender de forma práctica el funcionamiento de los algoritmos DQN y comprobar cómo pequeñas modificaciones en la arquitectura o en la estrategia de aprendizaje pueden tener un impacto significativo en el rendimiento final del agente.
